Existence and Being

Robert Johnston

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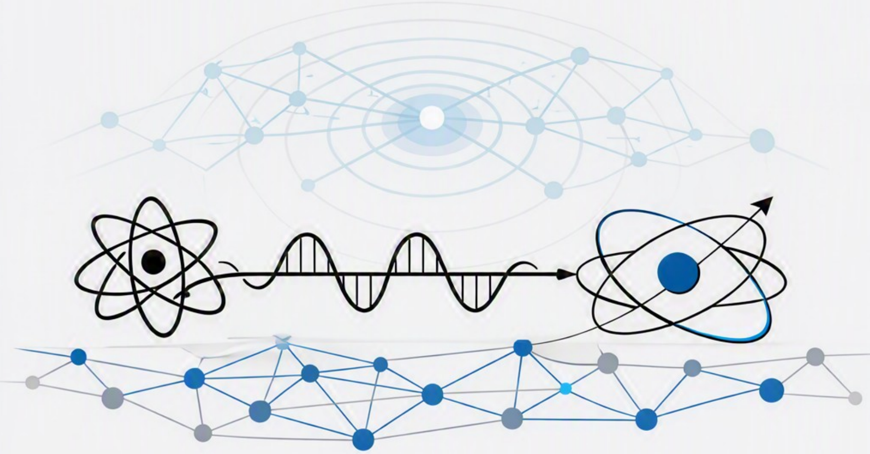
from physics to society



ROBERT JOHNSTON

Preface

The "being" of things is indifferent to whatever things might be "for someone". (Hartmann 1948)

The world as we find it is rich in things. This book will look at a series of ontological challenges and solutions ranging from physical foundations to societies, cultures and the entities that inhabit them. The book will focus on the ontological status of entities and their properties, and how they interact with each other. The book will also consider the ontological status of events, processes and relations.

Entity is the most general term that will be used to refer to things that exist. Entities can be physical, social, cultural, or mental. They can be concrete or abstract. They can be simple or complex. They can be individual or collective. They can be natural or artificial. They can be real, ideal or fictional. They can be actual or potential. They can be stable or dynamic. They can be independent or dependent. They can be autonomous or relational. The guiding principle in this book is that entities can have properties and interactions that are independent of whatever anyone or anything knows about them. For example, experiments are for finding out about the physical world not for instantiating it. That is, the physical world is there even when no one is looking. The discussion will engage with ontologies that deal principally with an autonomous world that is independent of considerations of what is known about the entities or who is interacting with them. This is not to neglect artifacts. Artifacts have properties that are a consequence of the intentions of their creator.

However, opposing views will also be considered in their strongest form. For example, there are quantum theories that focus on what can be known about a physical system rather than what the systems behaviour is *per se*. The original Copenhagen interpretation of quantum mechanics falls into this category, but approaches such as Quantum Bayesianism, commonly shortened to *QBism* (Fuchs and Schack 2013), interpret the quantum state as capturing a degree of personal belief. Such theories do not escape from ontological issues. There are questions about the ontological assumptions associated with the theory, with options ranging from a form of idealism to "participatory realism".

Karl Popper's three-world ontology (Popper 1972) can seem like a useful framework for investigating the ontological status of entities and their properties. Popper proposed three worlds:

1. The physical world of matter and energy, which is independent of human knowledge and perception.
2. The world of mental states and consciousness, which is dependent on human knowledge and perception.
3. The world of abstract objects and ideas, which is independent of human knowledge and perception.

This would have been my starting framework if I had not discovered Nicolai Hartmann's *New Ontology* (Hartmann 1949a). It provides a richer framework that allows for the semi-autonomy of mental states and ideas. Hartmann's ontology is more general and more flexible than Popper's. It allows for a more nuanced understanding of the relationships between entities and their properties.

The ontological framework for this book is therefore taken from Nicolai Hartmann's *new ontology programme* (Hartmann 1949a) that was developed in a number of very substantial publications in the first half of the 20th century. In my interpretation of it, it is the open and pluralistic nature of Hartmann's new ontology that is most attractive. It can include categories of substance, process, organism, consciousness and more. Its layer structure of the *Real Sphere* allows the discussion of emergence of multiple entities and supervenience. The openness and pluralism is more important than the layered ontology. This is shown by the interaction of the *Ideal Sphere* with the *Real Sphere*. Ideal Sphere entities, such as mathematical objects, can be examined and their ontological status clarified in relation to the *Real Sphere*.

Many chapters are specifically about the challenges posed by Quantum Mechanics (QM), which is an Ideal Sphere entity. Quantum mechanics was already making confirmed predictions when Hartmann was writing, and he provided some analysis, but, did not get much further than identifying that QM would provide some ontological challenges. In Hartmann's layered ontological model the foundation for the *Real Sphere* is the inorganic layer, with categories such as space, time, process and substance. It is because of the importance of this foundational role that so much space is devoted to the inorganic layer and QM in particular in this book. The inorganic layer is the foundation for the organic layer, which is the foundation for the psychical layer, which is the foundation for the spiritual layer. The ideal sphere is not a layer of the real sphere, but it interacts with it. It is largely activity in the spiritual layer that leads to the creation and discovery of ideal entities in domains such as mathematics and logic.

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Part I.

Ontological Foundations

1. Critical ontology and layers of reality

"Das Sein des Seienden ist indifferent gegen das, was das Seiende 'für jemanden' ist." (Hartmann 1948)

"The 'being' of things is indifferent to whatever things might be 'for someone'."

Nicolai Hartmann asked "How is a critical ontology at all possible?" (Hartmann 1924). He was writing at a time when metaphysics was under attack or being dismissed as irrelevant. Part of his answer was that ontology, properly structured, is more like a science of what exists than speculative metaphysics. Critical ontology is a way of investigating what exists and how it exists. The investigation of being must be considered before turning to how things behave. Hartmann's writings on ontology (Hartmann 1948), (Hartmann 1937), (Hartmann 1939), (Hartmann 1949b) developed this answer further. In addition to taking seriously the phenomena of normal life and natural science, he examined in detail the history of ontology. Hartmann's critical historical examination, ranging from the pre-socratics to his contemporaries such as Heidegger, leads him to classify many approaches as mistaken in not addressing existence as such but merely the appearance or knowledge of things. He also argued that assigning all entities to either the realm of ideas or physical substance is a fundamental mistake.

> **Aside:** *Critical ontology* is a phrase used elsewhere in philosophy studies (Richardson 2023), but in a different sense to that used here. For example, as the branch of philosophy

> that studies social ontology in order to critique specific ideologies or "end" to *social injustice*. This has given rise to critical studies in the social sciences and underpins identity politics that then itself tends to become ideological. The use of "critical" in the set of investigations presented here has nothing to do with critical studies or critical theory or any of its ideological derivatives. "Critical" in these investigations means a stance or approach that examines theories, claims and proposals for consistency and correspondence with the facts together with a recognition that there is always the possibility of error. This comprehensively fallibilist stance should apply to all invesitigations but will be explicitly recognised in this one.

In addition to critical historical investigation, ontology needs ideas, terminology and structure. Hartmann's concepts of *heterogeneous ontic spheres* and *layers*, introduced above provide a starting point for this set of investigations but they will not escape critique either.

1.1. Entities, spheres and layers

The phrase of Hartmann's *What is as such* will be taken as the point of departure for these investigations. It will be seen that there are many ways of being and the way things are depends on other things and the way they are. It is time to introduce some terminology.

The term "entity" will be used for *what is*. An entity - purely as such, whatever it may be in its particulars - exists indifferently to whether it is known or not known; whether it is knowable or unknowable. Without entities there would be nothing to have properties, to make appearances nor to induce perceptions.

The "object" is related to the entity and is used when the entity stands in relation to another entity. That is, it becomes an object through relationship with other objects.

The philosophical task of organising what exists into categories can be traced back to Aristotle, at least, and 'object' is a candidate for the category under which everything falls but so, perhaps, is "thing" or "entity". Nominal definitions can only provide an arbitrary distinction between the terms and so do not provide progress. Rather than just formally fixing technical definitions, use will be made of intuition from normal usage. This normal usage will provide guidance on the distinctions to be introduced above:

- "Entity" will be used to designate that which exists as such. *Entities will have their own particular way of being.*
- "Thing" can designate a physical entity. *Contrary to some physicalist ontologies not all entities are things.*
- An "object" is an entity that has properties, and these properties appear in relation to other entities.

In physical theories the entities of interest will be things that are objects. As we will see in our discussion of quantum physics, what makes an object physical will require development of the concept. Already it may seem that objects and things are the entities of interest and society can dispense with discussion of featureless entities. What is gained by insisting on the investigation of "entities" is the recognition that existence is not merely a matter of appearance.

Appearance only provides access to the object. This notion of appearance is interactional and reflexive. In particular, considering candidate things, quantum particles such as electrons, protons and positrons are objects and it is through their properties that they interact with other objects, and it is their properties that are observables in the sense of *standard quantum mechanics*. However, there are aspects to them being *things* that will mean extending the concept of physical from things in space and time.

The ontological structures introduced by Hartmann include:

- The ideal sphere of being

- The real sphere of being
- The layered structure of real being.
- The Dasein and Sosein relationship.

A major purpose of Hartmann's ontology is to clarify the mistakes and pitfalls in both idealism and realism. These two "isms" make the dogmatic claim that fundamentally everything is either immaterial (Ideal) or everything physical (Real). Taking a critical ontology approach makes clear that there is no need to follow either dogmatic path but claims it is wrong to identify the real with the material. The real sphere is much larger.

The ideal sphere includes entities such a logic, mathematics, scientific theories. It is also the sphere where ethical and aesthetic values reside. The ideal and real spheres are not separate forms of existence but different ways of being. This is a position going back to Plato who argued that ideal being is superior to material being and more real because eternal. In contrast Hartmann argues that the real (as set out below) is the ontological stronger sphere. The two spheres are distinguished in that everything real is individual, unique, destructible; whereas everything ideal is universal, returnable, always existing.

The real sphere consists of the natural world of common experience and its elaboration through scientific investigation. This sphere has four layers or strata (Schichten): inorganic, organic, psychic, and spiritual.

- **Inorganic stratum**, with categories such as space, time, process, substance, is organised through causality.
- **Organic stratum**, with categories such as metabolism, assimilation, self-regulation, self-reproduction and adaptation, is organically organised - the peculiarity of the organic network is unrecognizable for the time being.
- **Psychic stratum**, with categories such as consciousness, pleasure, act and content. Equally unknown is the form of determi-

nation of mental acts. That is, currently, we do not understand how mind works.

- **Spiritual stratum**, that includes categories of fear, hope, will, freedom, thought, personality, but also society, historicity, or intersubjectivity. Its organisation principle is, according to Hartmann, Finality (Hartmann 1949a, 58). In this layer and in the psychic we are all players and this changes what a method of scientific exploration of being is going to be. The work of Hayek will provide a starting point for such deliberations (Hayek 1952)

  Note: *Spirit* is employed here in the sense of the German term *Geist* that originates from the German idealist tradition influenced by Hegel among others. However it is also used in standard German. For example, in Geisteswissenschaft (Humanities)—and its meaning covers both individual mind as well as superindividual culture, which is why it lacks an adequate English translation. It need not be interpreted in any religious sense here. In addition, "psychic" relates to mind and mental - no interpretation in terms of occult "psychic phenomena" is intended.

In the stratified structure the organic layer builds on the inorganic and the psychic builds on the organic. It is only through the system of categories that apply to them that distinguish the levels. The relationship between the layers and their categories requires systemic treatment and will be covered in a separate section (the section titled 1.4).

1.2. "Dasein" and "Sosein"

"Dasein" and "Sosein" ("being there" and "being so" although the original German will be used in the following translation) are interacting ways of being in both spheres. This is illustrated by a quotation from Hartmann (1948) page 123:

> Das Dasein des Baumes an seiner Stelle “ist” selbst ein Sosein des Waldes, der Wald wäre anders ohne ihn; das Dasein des Astes am Baum “ist” ein Sosein des Baumes; das Dasein des Blattes am Aste “ist” ein Sosein des Astes; das Dasein der Rippe im Blatt “ist” ein Sosein des Blattes. Diese Reihe läßt sich nach beiden Seiten verlängern; immer ist das Dasein des einen zugleich Sosein des anderen. Aber sie läßt sich auch umkehren: das Sosein des Blattes “ist” das Dasein der Rippe, das Sosein des Astes ist das Dasein des Blattes usf. Daß es immer nur ein Bruchstück des Soseins ist, das im Dasein von etwas anderem besteht, daran wird man hierbei keinen Anstoß nehmen dürfen. Denn es handelt sich gar nicht um die Vollständigkeit des Soseins. Wohl aber läßt sich sagen, daß auch die übrigen Bruchstücke des Soseins auf dieselbe Weise im Dasein von immer wieder anderem und anderem bestehen.

1.2.1. Translation

> The Dasein of the tree in its place “is” itself a Sosein of the forest, the forest would be different without it; the Dasein of the branch on the tree “is” a Sosein of the tree; the Dasein of the leaf on the branch “is” a Sosein of the branch; the Dasein of the rib in the leaf “is” a Sosein of the leaf. This series can be extended to both sides; always the Dasein of one is at the same time Sosein of the other. But it can also be reversed: the Sosein of the leaf “is” the Dasein of the rib, the Sosein of the branch is the Dasein of the leaf, etc. That it is always only a fragment of the Soseins, which consists in the Dasein of something else, must not be taken negatively. Because it is not at all about the completeness of being. But it can be said that the other fragments of Sosein also exist in the repeating existence of always different and different things.

1.3. Critical ontology

For the purpose of these investigations Hartmann's work provides not just a philosophical system but a way of investigating *being* systematically. It is critically constructive and can be adopted as a way of proceeding without agreeing with Hartmann's specific conclusions.

Through step by step investigation we can gain greater insight into the constituents of the world. In these investigations topics will be tackled in physical and social science as well as philosophy where ambiguity over what exists or dogmatic positions risk hiding the nature of what there is and how it acts. For example, in physics, the status of space and time has been debated for millennia. Some claim that one or the other or both are illusory. In economics *homo economicus* is often declared not to exist but can be found implicitly and explicitly in many textbooks.

It is the current debate on the foundations of quantum mechanics on the status of the wavefunction and the particle that rekindled my interest in ontological questions. So quantum physics will have a number of chapters devoted to it and be referred to throughout these investigations.

Hartmann's philosophy will provide the initial structure for theses ontological investigations. A very useful and concise introduction to Hartmann's philosophy as whole is provided by Predrag Cicovacki in The Analysis of Wonder (Cicovacki 2014).

1.4. Layers of being

To discuss ontology in general and to compare competing ontological proposals some structure is useful. This structure will help to place *candidate beings* in relation to others, understand their way of being and, for example, avoid the category mistake of identifying elements of physical reality with their mathematical representation. Here we will explore, in general terms, the proposal that reality is indeed stratified and secondly what that stratification may be. In *critical ontology* it will be important to distinguish whether these levels are levels of reality *per se* or levels in a description of reality.

The starting point cannot just be comprehensive scepticism about whether anything exists or not. That will get us nowhere. We will start by taking the phenomena around us as the they appear. There appear to be things that take up space and are static. There appear to be things that take space and grow. There appear to be things that have some purpose and make choices. There appear to be things that are made up of other things. In addition, there appear to be beings with emotional states and awareness of making choices. There is also awareness of resistance or pushback from other things. There is awareness of being the capacity of some beings to change the state of other things. We may be mistaken about particular things and their relationship to each other but the appearances are the appearances. In talking about things and beings, we have implicitly used categories such a space, time, choice, emotion, awareness and substance. The categories will play an important role is characterising the levels in which beings exist. There will be separate chapter on categories.

Hartmann proposes a pluralistic view of reality, that covers all the things and beings listed above, that appear in four dynamically interrelated strata or layers (Chapter 1 and Cicovacki (2014)):

1. Inorganic
2. Organic
3. Psychic
4. Spiritual.

The layers are presented in the order of decreasing ontological strength. That is, in each layer the beings depend on those in stronger layers to exist. However, the beings in the less strong layers cannot be reduced to beings in stronger layers. They have their specific categories of existence. There are also multi strata beings as will be illustrated at the end of this chapter. The question of whether these levels exist and the nature of their existence will not be examined here other than to recognise them as structural proposals and so will be seen to at least exist at the *spiritual level*, as part of an ontological theory.

The name given to each level is not problematic in English except for the fourth. *Spiritual being* is a translation from the German *geistige Sein* and the use of the term *Geist* follows the tradition influenced by Hegel and Dilthey - as it is used, for example, in Geisteswissenschaft,and its meaning covers both individual mind as well as superindividual culture, which is why it lacks an adequate English translation. No religious sense of the term is intended here. Beings at the spiritual level cannot exist without the lower levels in this ontology.

Although the listing of the four strata suggests a hierarchy, to provide this thought with some substance something needs to be said about the relationship between the strata. The levels of reality are characterised (and therefore distinguished) by their categories. With regard to the levels of reality, it is obvious that the categories in question are ontological rather than epistemological. There are categories that pertain to reality as whole - such as time, whole, part. At the inorganic level the categories of physics are different from those of chemistry. Some examples follow.

1. Inorganic stratum, with categories such as space, time, process, substance, is organised through causality.

2. Organic stratum, with categories such as metabolism, assimilation, self-regulation, self-reproduction and adaptation, is organically organised - the details of the organising principles are unknown for the time being.

3. The psychic stratum, with categories such as consciousness, pleasure, act and content, has its own psychic network of relations. Equally unknown is the inner essence of the form of determination of mental acts.

4. Spirit includes categories of fear, hope, will, freedom, thought, personality, but also society, historicity, or intersubjectivity. It is organised as three modes of spirit;

    - Personal - thought, freedom, conscious action and moral will
    - Objective - superindividual but living as culture, collective or society
    - Objectivated - superindividually existing but non-living. As in cultural artifacts and institutions.

All categories at the inorganic level, such as space and time, recur at the organic level. This recurrence of categories is called *superinformation*. There is also autonomy off levels and this is known as *superposition* - building above. For example, *consciousness*, a category of the psyche, is not spatial itself, although it will depend on a physical infrastructure.

Hartmann's categories are polar, correlative, and non-reducible. That is, each pair expresses two moments of being that are mutually dependent yet ontologically distinct. They are not contradictions but are tensions built into the structure of reality. The catagory pairs are:

1. Principle – Concretum

- Principle: the universal, law-like, general aspect of being.
- Concretum: the individual, determinate, fully actualised entity.

    Every concretum embodies principles; principles exist only through concreta. Universality and individuality are inseparable.

2. Structure – Modus

- Structure: the stable, ordered pattern or configuration of a being.
- Modus: the variable, contingent way in which that structure is realised.

Structure is invariant; modus is its concrete manifestation. A being's structure persists through changing modi.

3. Form – Matter

- Form: the organising, delimiting, shaping aspect.
- Matter: the indeterminate substrate capable of receiving form.

Form actualises matter; matter individuates form. This is a universal ontological relation, not merely Aristotelian.

4. Inner – Outer

- Inner: intrinsic, essential, non-relational aspects.
- Outer: relational, expressive, externally oriented aspects.

The inner grounds the outer; the outer manifests the inner. Being has both an inward essence and outward expression.

5. Determination – Dependence

- Determination: what a being is in itself — its qualities, powers, limits.
- Dependence: what a being requires from others to be what it is.

Determinations imply dependencies; dependencies shape determinations. No being is self-sufficient.

6. Quality – Quantity

- Quality: the "what-ness" — character, nature, kind.
- Quantity: the "how-much" — magnitude, number, degree.

Qualities are intensified or diminished through quantities; quantities are meaningful only relative to qualities.

7. Unity – Manifoldness

- Unity: coherence, wholeness, integration.

- Manifoldness: plurality of parts, aspects, or moments.

  Unity integrates manifoldness; manifoldness articulates unity. Every whole is a structured multiplicity.

8. Harmony – Conflict

- Harmony: ordered, mutually supportive relations.
- Conflict: tension, opposition, struggle.

  Harmony presupposes resolved conflict; conflict presupposes differentiated elements capable of harmony. Reality is dynamic, not static.

9. Contrast – Dimension

- Contrast: difference, polarity, opposition.
- Dimension: the continuum or field within which contrasts occur.

  Contrasts articulate dimensions; dimensions provide the space for contrasts. Being is structured by fields of possible variation.

10. Discretion – Continuity

- Discretion: separateness, discreteness, distinctness.
- Continuity: connectedness, gradation, seamlessness.

  Discrete elements exist within continuous fields; continuity is structured by discrete nodes. Reality is neither purely discrete nor purely continuous.

11. Substratum – Relation

- Substratum: the underlying bearer of properties.
- Relation: the connections that situate the substratum within a network.

Substrata are always in relations; relations presuppose relata. Being is both self-standing and interconnected.

12. Element – Structure

- Element: basic constituent unit.
- Structure: ordered whole composed of elements.

Elements gain identity through structure; structure exists only through elements. This is the fundamental category of compositional ontology.

The above only provides a sketch. For more detail on the categories of real being see Cicovacki (2014) who provides a bridge to the detailed work by Hartmann.

The basic structure can be summarised in a diagram.

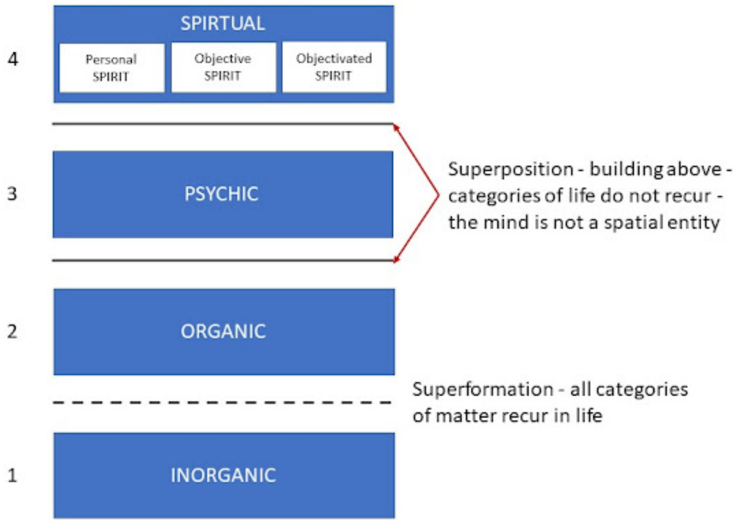


Figure 1.1.: Layers of being

Hartmann's model of reality includes mind and spirit superposed on an ontologically stronger foundation of life and matter. There is, however, a feedback loop that is not explicit in the diagram. The objectivated spirit gives rise to artifacts, such as books and architecture, that are also actualised as things at the material level.

For example, this content of this investitgation exists as objectivated spirit. It came into being because of my choice and was enabled by the infrastructure supplied by the web. Making and maintaining the documentation are choices made by me at the level of my personal spirit. The physical document is dependent on organised matter (inorganic level) the organisation of it is objectivated spirit because it was designed. The document and its chapters are captured and encoded in the infrastructure. The chapters may be instantiated on the screen of a reader's device. The readers use their sense organs and brain (organic level) to access the information that is processed through their psychic level and, hopefully, understood at the level of their personal spirit. The readers and I form a loose community that exists as objective spirit.

Simple beings such as electrons may only inhabit the inorganic level but the existence of complex beings may be across many or all strata. This complex existence is a process rather than a hierarchy.

2. Objects in a Field of Sense

Within the layers and spheres described in *Hartmann's architecture for critical ontology*, there is scope for introducing further structure. A **Field of Sense** (Gabriel 2015) is an organizational concept that can be employed within a layer or within a sphere.

While not all of Gabriel's consequences from his full ontology can be accepted within the critical approach adopted from Hartmann, a Field of Sense remains a potentially useful concept. It fits with Hartmann's distinction of item and object. An object is an item that appears in the context of other items. Some of these items become objects in their turn. Entities become objects in a Field of Sense.

2.1. Defining the Field of Sense

A Field of Sense is a domain in which objects appear in relationship to each other. The nature of these appearances depends on the properties of the objects involved.

- **Example:** A cube of ice melting on a table in a warm living room.
  - The **living room**, with all its other objects, is the Field of Sense.
  - The **table, air, and ice** are some of the objects appearing within it.
  - **Recursive Structure:** The living room can itself be an object within a larger Field of Sense, such as **the house**.

2.1.1. Gabriel's Original Ontology

For Gabriel, the mode of existence of an object depends entirely on the Field of Sense that grants it. In this view:

- **Existence:** To exist is to appear in a Field of Sense (belonging to a domain of appearance).
- **Sense:** Derived from Frege (Frege 1962), Gabriel generalizes "sense" from the grasping of a concept to the context that makes an appearance possible.
- **Example:** The "Morning Star" and "Evening Star" are two senses of the object Venus.
- **The Negative Result:** Gabriel argues that the "World" (the totality of all that exists) would have to be the Field of Sense of all Fields of Sense. Since it cannot appear within a larger field, Gabriel concludes that **the World does not exist.**

Note that in the example of Venus there are two Fields of Sense. The critical analysis would say that one object, Venus, has at least two modes of existence. Gabriel seems to be committed to two objects: the morning star and the evening star.

2.1.2. Proposed Modification: "The World is not an Object"

The following revised logical scheme is proposed to respond to the objection that existence is not strictly synonymous with appearance. While objects appear in fields that consist of other entities, appearance may not exhaust the meaning of "existing." Hartmann already reserves the term "object" for an entity that exists in relation to other entities.

2.1.3. The Revised Logical Scheme

1. **Appearance:** The properties of all objects appear in a Field of Sense.
2. **Sense:** A sense is the specific way in which an object appears.
3. **Recursion:** A Field of Sense can be an object in yet another Field of Sense.
4. **Relationality:** A Field of Sense must contain multiple objects; appearance is always a relationship.
5. **Totality:** The world cannot be an object in a Field of Sense (otherwise it would require a larger field to appear in).
6. **Definition of Objecthood:** To be an **object** is to appear in some Field of Sense.
7. **Conclusion:** *The World is not an object.*

   Note: This eliminates Gabriel's paradox. The World exists as a total system, but because it has no external entity to relate to, it cannot be categorized as an *object*.

2.2. Scope and Reference

- **Scope of Existence:** The physical universe, people, scientific theories, and fictional characters all exist in at least one Field of Sense.
- **Reference:** Fields of Sense are differentiated by reference (Frege 1962). Reference enables the recognition of the field in which an object appears, preventing "reference mistakes" (e.g., mistaking a TV play for one's actual family life).

2.2.1. Conclusion

By shifting the headline result from *"The World does not exist"* to ***"The World is not an object,"*** we maintain ontological realism.

This allows for investigation within and across **Hartmann's strata** (Inorganic, Organic, Psychic, and Spiritual) and both ontic spheres without the paradox of Gabriel's radical stance.

3. Process ontology

3.1. The Philosophical Landscape — Physical Layer

It was in *Taking Heisenberg's Potentia Seriously* (Kastner et al. 2018) that I first encountered Ruth Kastner's **Relativistic Transactional Interpretation (RTI)**. Michael Epperson, one of the co-authors, has written his own book (Epperson 2004) on quantum mechanics that makes extensive use of the concept of potentiality. This draws heavily on Whitehead's **Process Philosophy** but does not provide new physics, relying instead on a decoherence narrative used by many others. That narrative is not sufficient.

Kastner's transactions offer some novel physics in the relativistic formulation, involving an "offer and response" process. In addition, there is an interplay between **space-time events** and the quantum substratum *potentia*. Whitehead's process philosophy (which he misleadingly called a "philosophy of organism") may provide, or may be adapted to provide, a coherent cosmology encompassing quantum processes and events, but the physics of events still needs to be formulated.

3.2. Alternative Mechanisms

The Relativistic Transactional Interpretation is not the only approach to providing a mechanism for quantum state reduction—leading to events—that is not *ad hoc*. The **"Events, Trees, and Histories"**

(ETH) theory, as developed by Jürg Fröhlich and collaborators, also provides a mechanism. As with RTI, their approach requires interaction with massless modes (e.g., photons); however, retro-causality is not invoked.

The key concept in ETH is the **"Principle of Diminishing Potentialities."** This could be interpreted as adding what is missing from Kochen's reformulation of QM, but that is not a point that they make. To provide a rounded understanding of a physics that includes potentiality and processes leading to events, a cosmology is needed—an all-encompassing philosophical scheme.

3.3. The Whiteheadian Scheme

An all-encompassing philosophical scheme that includes events and processes at the microscopic, mesoscopic, and macroscopic scales can only be speculative, but speculation should not be arbitrary. Following Whitehead (2010):

> "…the philosophical scheme should be coherent, logical, and, in respect to its interpretation, applicable and adequate. Here 'applicable' means that some items of experience are thus interpretable, and 'adequate' means that there are no items incapable of such interpretation. 'Coherence,' as here employed, means that the fundamental ideas, in terms of which the scheme is developed, presuppose each other so that in isolation they are meaningless… The term 'logical' has its ordinary meaning, including 'logical' consistency, or lack of contradiction, the definition of constructs in logical terms, the exemplification of general logical notions in specific instances, and the principles of inference."

3.4. Key Terms in Process and Reality

Technical Term	Explanation	Relevance to Quantum Physics
Actual Entity / Occasion	The fundamental unit of reality; a single, concrete event or experience.	Quantum events or interactions. Each event is a discrete occurrence.
Prehension	The process by which an actual entity "grasps" or is influenced by other entities.	Particles interacting through fundamental forces; influencing one another's states.
Eternal Object	Abstract entities or pure potentials (similar to Platonic forms) that actual entities can realize.	Properties like position or momentum. Represented by self-adjoint operators, **not** the wavefunction.
Process	Reality as dynamic "becoming" rather than static "being."	The flux of particles and fields governed by probabilistic laws.
Creativity	The ultimate principle representing the continuous creation of new actual entities.	Inherent unpredictability and novelty; related to creation operators in QFT.
Singular Causality	Specific, individual causes contributing to a single event.	Specific interactions leading to the collapse of a quantum state.

Technical Term	Explanation	Relevance to Quantum Physics
Nomic Causality	General or universal causes (scientific laws).	General laws like the Heisenberg equation ($i\hbar\frac{d}{dt}A = [A, H]$).
Organism	The interconnected, interdependent nature of reality.	The holistic view of systems where the whole is more than the sum of its parts.
Concrescence	The process of an entity coming into being by integrating many prehensions.	How states evolve and integrate influences to result in a specific event.
Subjective Aim	The inherent goal or purpose an entity strives for in its "becoming."	Propensity theory; how an outcome is selected from a probability distribution.

3.4.1. Flow diagram of key terms

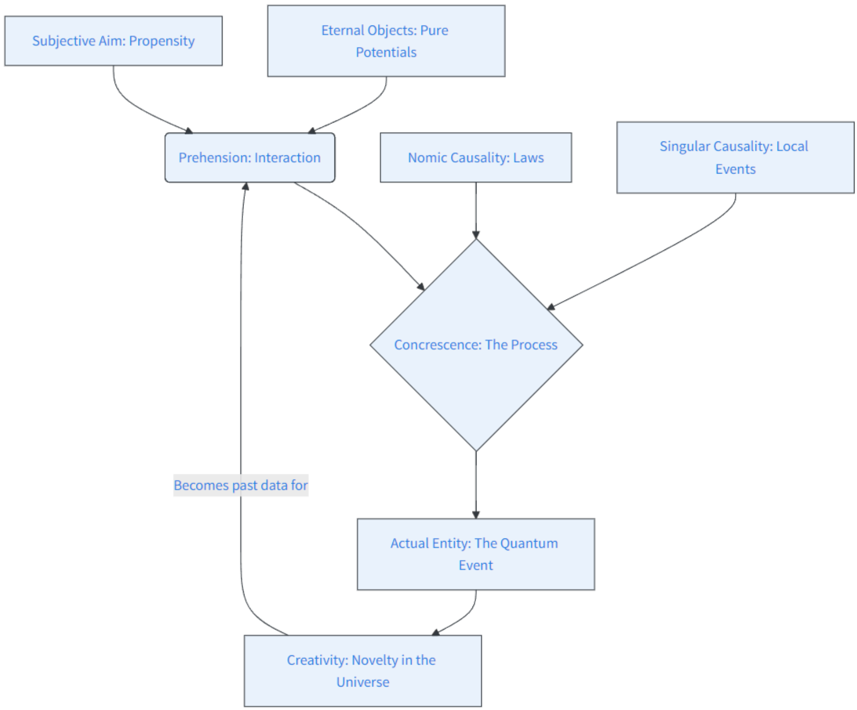


Figure 3.1.: Flow diagram of key terms

4. The Ontology of Dispositions and Objective Chance

The motivation for analysing the ontology of dispositions is to understand partial causes and objective chance. It is objective chance that is the stronger interest here, motivated by wanting to understand the behaviour of basic entities at the inorganic ontic level. That is, entities that fall within the remit of quantum physics.

Dispositional properties, as described by Anjum and Mumford (Anjum and Mumford 2018b), will be used to clarify how there can be properties of quantum entities that cause quantifiable property values to appear and influence other objects. This will develop into a dispositional understanding of probabilities in quantum mechanics.

While it is easy to appreciate dispositional properties intuitively, it can be harder to make the concept precise enough to be useful in constructing a scientific theory. To help, the following definitions will be used:

- **Definition 1:** A **disposition** is a property (such as solubility, fragility, elasticity) whose actualisation entails that the thing which has the property would change, or bring about some change, under certain conditions.
- **Definition 2:** A **dispositional power** is a physically existing dispositional property of an entity.

4.1. Intuitive vs. Quantum Dispositions

In everyday use, to say that some object is soluble is to say that it would dissolve if put in some liquid. To say that something is fragile is to say that it would break if, for instance, dropped in suitable circumstances; to say that something is elastic is to say that it would stretch when pulled. The fragility (solubility, elasticity) is a disposition; the breaking (dissolving, stretching) is the actualisation of the disposition.

These examples are intuitively uncontroversial, but position and momentum are not classical dispositional properties of a particle. Results such as that of Kochen and Specker (Kochen and Specker 1967) show that the quantum situation is more subtle than dispositional properties such as solubility and elasticity because they demonstrate a contradiction in assuming that it is possible to assign simultaneously values to all observable properties in all states. Their result, although called a paradox in their paper, indicates the need for an ontology that deals with this more subtle reality.

There is still ongoing debate about the nature of dispositions and key elements of this debate are outlined in Mumford (Mumford 2011). However, the application in physics makes the ontology of causal powers the more attractive interpretation of the nature of dispositions (Anjum and Mumford 2018a). These powers exist and can be independent of an observer.

4.2. Popper and Propensity

With the motivation to provide an objective interpretation of chance in quantum mechanics, Popper (Popper 1967), (Popper 1982) took a significant step towards a theory of dispositional properties. He took the position that quantum mechanics is a statistical description of the behaviour of physical entities and introduced dispositional properties through his propensity interpretation of probability:

> "...waves (even those of the second quantization) are mathematical representations of propensities, or of dispositional properties, of the physical situation (such as the experimental set-up), interpretable as propensities of the particles to take up certain states."

Popper proposed that the wave aspect of quantum mechanics determines the tendency of the particles to assume specific states under certain conditions, but with the strength of this disposition given by a probability. However, Popper developed his thinking prior to the demonstration that it is impossible, in general, to assign intrinsic values to all observable properties in all states, and he did not completely develop the ontological consequences. Popper remained committed to a physics in which particles have properties that take values as in classical physics but that the quantum theory provides a statistical description of that reality.

4.3. Properties and Fields of Sense

It is proposed here that properties of a physical entity need not all take values simultaneously for them to be physically existing properties. Dispositional properties are properties that govern how the entity tends to appear in various contexts. For example, a separate physical system may be of the form that couples to an electron's spin in a specific direction and instantiate a value for it. That would be how the electron spin appears to that system.

In the dispositional theory developed in this investitgation, there are properties—objective causal powers—and in addition, the probabilistic appearance of the properties as property values. The concepts required to clarify what is meant by "to exist" and "to have a property" are set out in *Why critical ontology?* (Chapter 1) and *Fields of Sense* (Chapter 2).

Dispositional properties cannot themselves appear in a **Field of Sense**, but their effects can. This is the point where the modification of the Field of Sense ontology is used that recognises that there are properties of objects that affect appearances but do not themselves appear. This modification means that Fields of Sense do not provide a complete ontology, contrary to what seems to be proposed by Markus Gabriel (Gabriel 2015).

A dispositional property such as fragility appears if the fragile object breaks—on falling from a table, for example. This is how it appears in a domestic Field of Sense. The fragile object, for example a porcelain cup, exists and appears in the domestic scene even if it does not break. It also has properties that are not dispositional; for example, it has a mass. So, an object can have non-dispositional properties that pin it down in a context. The electron has the properties of charge and (rest) mass that pin it down in the physical universe in which it has (as will be explained elsewhere) dispositional spin, position, and momentum.

4.3.1. Summary of Ontological Position

The position taken on the ontology of dispositions is:

1. Physical objects have at least one property that appears or causes a value to appear in some Field of Sense.
2. Dispositional powers cause property values to appear in Fields of Sense.
3. Physical dispositions are properties of some physical objects.

4.4. Concepts of causal powers

The concept of dispositions has played a key role in the formulation of objective quantum chance in this blog. However, there is ambiguity about what these are as powers. Ruth Porter Groff (Groff 2021) has helpfully addressed this issue and identified four senses of the term. She identifies dispositional power to be conceptualised as an:

- Activity
- Capacity
- Essence
- Necessitation.

While Groff indicates that there is a further task to work out which is correct in a given context. That is, depending on what is (what the world is like), there could be powers of distinct types. For example, at the various levels of reality (Hartmann) different concepts could play their part. The intention is that clarity on which concept of power applies will strengthen any theory of quantum chance.

These concepts of powers need to be contrasted with what is called the Humean view that there are no necessary connections or causes in nature. That is, there are no powers.

4.4.1. Activity

Consider a film in which each frame is static. Playing the film gives the impression of movement or activity. If activity in the world is like the film, then activity is an illusion or a metaphor. In which case activity is not an aspect of how things are, and the ontology can be called passive. If activity is not just a sequence of static configurations, then activity may not be an illusion. We can follow Groff and use the term anti-passivist to refer to the opinion that activity is a real and irreducible component of the world.

Activity is taken to cover a range of things. Movement, deliberation (moving away from the visual film example), inquiry and chemical reaction (to take an inorganic example). Any instance of causation is an activity.

The view that there is activity in the world has common sense on its side. For example, action as captured by verbs as part of the deep structure of language.

From this perspective, to say that things in the world have causal powers is to say that things engage in activity and are able to do. Reality is in this sense genuinely, irreducibly, non-metaphorically dynamic. In contrast Esfeld's (Esfeld and Deckert 2017) primitive ontology, that is in favour with some who defend the Bohmian version of Quantum Mechanics, is passive and at best kinematic. It is an example of an extreme (or to use Esfeld's own term Super) Humean ontology.

Real activity contrasts with the Humean view that inanimate matter is essentially passive and never intrinsically active. In real activity the action of things depends on their causal powers. Examples of activity, from Cartwright (2007):

1. The carburettor feeds gasoline and air to a car's engine ...
2. The pistons suck air in through the chamber...
3. The low-pressure air sucks gasoline out of a nozzle ...
4. The throttle valve allows air to flow through the nozzle ...
5. Pressing the pedal opens the throttle valve more, speeding the airflow and sucking in more gasoline...
6. ...

These examples indicate that mechanisms can undertake activity.

4.4.2. Capacity

A capacity is a way that something presently is, such that it could be a way that it presently is not. The phenomenon of a capacity is thus inherently modal, invoking possibility. That is capacity is the potential to be in a possible state for that thing. This type of potentiality is attached to the nature of a thing and is sometimes called real, or metaphysical possibility, in contrast to logical possibility.

A capacity may be engaged in activity but there is nothing about the concept of a real possibility that requires realism about activity to be either built into it or entailed by it. A power, as a capacity, is a property that need not be fully actualised in the present in order to exist.

Capacities are properties that include, as part of their identity in the present, non-actualized but nevertheless real potential manifestations. By contrast, for the Humean the only properties that exist are categorical properties; properties that are fully occurrent or actual.

4.4.3. Essence

Essentialism is when things (or some kinds of things) are such that they could not be, in part or in full, otherwise without ceasing to be what they are. Among dispositionalists, Bird Bird (2007) defines powers, or potencies, as fundamental properties whose identities are not just dispositional, but fixed. A power as an essence is to be a property whose identity is essential to it.

4.4.4. Necessitation

Necessitation is when one thing is the case, some other thing must be the case. A necessary connection is a power in this sense. Equating powers with necessary connections is proposed by Armstrong Armstrong (2005).

It is not clear that even those anti-passivists who are most focused on defending the reality of necessary connections (in the name of defending the notion of a law) do believe in metaphysical necessitation.

What is meant by the term ‘metaphysical necessitation’? To answer this, we need to know

- whether one who affirms it believes

- – that things of a given kind necessarily tend to behave in one way or another, or
  - – that they must behave in one way or another; and also

- whether or not one who affirms the existence of metaphysical necessitation holds that given behaviours necessarily bring about assigned outcomes.

Accepting the above would tend someone strongly towards metaphysical determinism and intuitively this would be a natural consequence of necessitation. However, it may be that a version of metaphysical necessity that commits only to the existence of necessary tendencies does not translate into a commitment to what may be called 'causal necessitarianism,' or even hard determinism.

4.5. Conclusion

It is conceivable that all the concepts described above have role to play in understanding the role of powers in the world. However, from the posts in this blog on quantum physics and in particular quantum chance it is capacity that seems to be the best fit. This is because the potential to have a physical property is attached to a physical object and possibilities are captured by the set of possible values that can be made actual. For example, an electron has the property of spin with the potential to take certain value. The possible values that can be actualised and with whichever probabilities depends on this potential and the context the particle find itself in.

The concept of activity also has causal force but seems better suited to powers of designed mechanism or psychological or social situations. If a theory can be developed of what gives rise to the occurrence of an actual event in Quantum Mechanics, then activity may be a valid concept for the power in question.

Essence plays a role in quantum chance in that the power that governs the tendency for an object, such as an electron, to take particular values of spin is an essential property of an electron. That is, an electron would not be an electron if spin and quantum chance were not aspects of its state.

Necessary connections have a role to play in a physical theory even in the presence of objective chance. The evolution of the wavefunction governed by the Schrödinger equation is deterministic making the state of the object necessarily connected to its state at an earlier time.

However, it is proposed that potentiality and possibility are the key concepts, equivalent to capacity, which play the key role in the developing theory of quantum chance. The most complete treatment to date of powers as potentiality and possibility is by Vetter (2015) whose classification of potentiality as a localised modality and possibility as a non-localised modality looks promising. We may then have the quantum state of the electron representing the chance potential to take certain spin values while the possible spin values that can be actualised will depend on the non-localised situation that constrains the quantum state.

5. Potentiality and Probability

As outlined in the previous chapter, Barbara Vetter (2015) developed the concept of potentiality in her theory of dispositional powers. In that theory, potentials are dispositions that are responsible for the manifestation of possibilities. The possibilities then tend to become actual events or states of affairs. The concept of "potential" in philosophy, in a sense close to that discussed here, goes back at least to Aristotle in *Metaphysics* Book Θ. (Aristotle, n.d.)

In contrast, probabilities are weightings summing to one that describe in what proportion the possibilities tend to appear. I propose that the potential underpins the actual appearance of the possibilities while probability shapes it. This will be discussed further in this chapter.

Barbara Vetter proposed a formal definition of possibility in terms of potentiality:

> **POSSIBILITY:** It is possible that $p =_{def}$ Something has an iterated potentiality for it to be the case that p.

So, it is further proposed that the probabilities are the weights that can be measured through this iteration using the frequency of appearances of each possibility. Note that this indicates how probabilities can be measured, but it is not a definition of probability.

In the field of disposition research, there is an unfortunate proliferation of terms meaning roughly the same thing. The concept of "power" brings out a disposition's causal role, but so does "potential." As technical terms in the field, both are dispositions. Now "tendency" will also be introduced, and it is often used as yet another flavor of disposition.

5.1. Tendencies

Barbara Vetter mentions tendencies in passing in her 2015 book on potentiality and, although she discusses graded dispositions, tendencies are not a major topic in that work. In "What Tends to Be: The Philosophy of Dispositional Modality," (Anjum and Mumford 2018b) Rani Lill Anjum and Stephen Mumford (2018) provide an examination of the relationship between dispositions and probabilities while developing a substantial theory of dispositional tendency.

In their treatment, powers are understood as disposing towards their manifestations, rather than necessitating them. This is consistent with Vetter's potentials. Tendencies are powers that do not necessitate manifestations, but nonetheless, the power will iteratively cause the possibility to be the case.

In common usage, a contingency is something that might possibly happen in the future. That is, it is a possibility. A more technical but still common view is that contingency is something that could be either true or false. This captures an aspect of possibility, but not completely because there is no role for **potentiality**; something responsible for the possibilities. There is also logical possibility in which anything that does not imply a contradiction is logically possible. This concept may be fine for logic, but in this discussion, it is possibilities that can appear in the world that are under consideration. Here, an actual possibility needs a potentiality to tend to produce it.

5.1.1. Example (adapted from Anjum and Mumford)

Struck matches tend to light. Although disposed to light when struck, we all know that there is no guarantee that they will light as there are many times a struck match fails to light. But there is an iterated potentiality for it to be the case that the match lights. The lighting of a struck match is more than a mere possibility or a logical possibility. There are many mere possibilities towards which the struck match has no disposition—that is, no potential in the match towards struck matches melting, for instance.

Iterated potentiality provides the tendency for possible outcomes to show some patterns in their manifestation. In very controlled cases, the number of cases of success in iterations of match striking could provide a measure of the strength of the power that is this potentiality. This would require a collection of matches that are essentially the same.

5.2. Initial Discussion of Probability

Anjum and Mumford introduce their discussion of probability through a simple example that builds on a familiar understanding of dispositional tendencies associated with fragility.

> "The fragility of a wine glass, for instance, might be understood to be a strong disposition towards breakage with as much as 0.8 probability, whereas the fragility of a car windscreen predicts its breaking to the lesser degree 0.3. Furthermore, it is open to a holder of such a theory to state that the probability of breakage can increase or decrease in the circumstances and, indeed, that the manifestation of the tendency occurs when and only when its probability reaches one."

This example is merely an introduction and needs further development, but already the claim that "the manifestation of the tendency occurs when and only when its probability reaches one" shows that it is not a model for objective probability. What is needed is a theory of dispositions that explains stable probability distributions. Of course, if the glass is broken, then the probability of it being broken is 1. However, this has nothing to do with the dispositional tendency to break. What is needed is a systemic understanding of the relationship between the strength of a dispositional tendency and the values or, in the continuous case, shape of a probability distribution.

In the quoted example above, each power is to be understood in terms of a probability of the occurrence of a certain effect, which is given a specific value. The fragility of a wine glass, for instance, might be understood to be a strong disposition towards breakage with as much as 0.8 probability, whereas the fragility of a car windscreen is less, and the probability of its breaking is a lesser degree 0.3. But given a wine glass or windscreen produced to certain norms and standards, it would be expected that the disposition towards breakage would be quite stable. A glass with a different disposition would be a different glass.

Anjum and Mumford claim that, in some understandings, the manifestation of a possibility occurs when and only when its probability reaches one (see Popper, *A World of Propensities*, (Popper 1990)). This is a misunderstanding of how probability works. Popper distinguished clearly between the mathematical probability measure and what he called the physical propensity, which is more like a force, but Popper does limit a propensity to have a strength of at most 1. As I will attempt to show below, Popper, in proposing the propensity interpretation of objective probabilities, oversimplifies the relationship between dispositions and probabilities. This confusion led Humphreys to draft a paper (Humphreys 1985), *The Philosophical Review*, to show that propensities cannot be probabilities. As indeed they are not. They are dispositions. That would leave open the proposition that probabilities are dispositional tendencies, but that also will turn out to be untenable.

The proposal by Anjum and Mumford that powers can **over-dispose** does seem to be sound. Over-disposing is where there is a stronger magnitude than what is minimally needed to bring about a particular possibility. This indicates that there is a difference between the notion of having a power to some degree and the probability of the power's manifestation occurring. Among other conclusions, this also shows that the dispositional tendency does not reduce to probability, preserving its status as a potential.

Anjum and Mumford continue the discussion using "propensity" as having a tendency to some degree, where degree is non-probabilistically defined. Anjum and Mumford use the notions of "power," "disposition," and "tendency" more or less interchangeably, whereas an object may have a power to a degree, there are powers that are simply properties. In what follows, I will try to eliminate the use of "propensity," except where commenting on the usage of others, and use "tendency" to qualify either "power," "potential," or "disposition" rather than let it stand on its own.

5.3. Probability vs. Necessity

A probability always takes a value within a bounded inclusive range between zero and one. If probability is 1, then probability theory stipulates that it is "almost certain" (occurs except for a set of cases of measure zero). In contrast to what Anjum and Mumford claim, it is not natural to interpret this as necessity because there are exceptions. For cases where there are only a finite set of possibilities, then probability 1 does mean that there are no exceptions. But as this is a special case in applied probability theory, there is no justification in equating it with logical or metaphysical necessity.

A power must be strong enough to reach the threshold to make the possibilities actual. Once the power is strong enough, then the probability distribution over the possibilities may be stable or affected by other aspects of the situation. So, instead of understanding powers and their degrees of strength as probabilistic, powers and their tendencies towards certain manifestations are the underpinning grounds of probabilities. Consider the example of tossing a coin.

A coin when tossed has the potential to fall either heads or tails. This tendency to fall either way can be made symmetric and then the coin is "fair." From which probability weightings of 1/2 for each outcome (taking account of the tossing mechanism) can be assumed and then confirmed by measuring the proportion of outcomes on iteration. The reason why the head and the tail events are equally probable statistically, when a fair coin is tossed, is that the coin is equally disposed towards those two outcomes due to its physical constitution. The probability weightings derive, in this example, from a symmetry in the potentiality, which in turn derives from the physical composition and detailed geometry of the coin.

5.3.1. The Rai Stone Example

Figure 5.1.: https://en.m.wikipedia.org/wiki/Rai_stones

Consider a society that uses very large stone discs as currency (e.g., Rai stones). On examination of the disc, it would be possible to conjecture that if it were tossed, then there would be two possible outcomes and that those outcomes are equally likely. But this disposition is not realized because of the effort required to construct the tossing mechanism, as such a stone may weigh several metric tons. The enabling disposition that would give rise to the iteration of possibilities would have been this missing tossing mechanism. It is not a property of the disc.

The manifestation of the dispositional tendency of the disc to come to lie in one of two states needs an external mechanism that is disposed through design to toss the coin in a certain way. If the mechanism is constructed, it may be too weak. It may tend to only flip the coin once, giving a sequence such as:

...$THTHTHHTHTHTHTHTHTH$...

That would give a frequency of T close to 1/2, but the sequence does not exhibit the potential for random outcomes to which the coin is disposed.

5.4. Probabilities and Chance

From the above: potential and possibility are more fundamental than (or prior to) probability. Both are needed to construct and explain objective probability. The alternative, subjective probability, is based on beliefs about possibilities, but that is not the same thing as what is actually possible and how things will appear independently of anyone's beliefs or judgments.

In this investigation, I have already referred to a dispositional tendency beginning to explain objective probabilities in quantum mechanics. The term "propensity" has been used to describe these probabilities. I now think that was wrong. Propensity should be reserved for the dispositional tendency that is responsible for the probabilities to avoid this term merging the underpinning dispositional elements and probability structure. Anjum and Mumford claim that they have made a key contribution to clarifying the relationship between dispositional tendencies and probability through their analysis of **over-disposition**.

Anjum and Mumford claim "information is lost in the putative conversion of propensities to probabilities," but only if the dispositional grounding of probabilities is forgotten. Their discussion is strongly influenced by their interest in application to medical evidence where a major goal is reduction of uncertainty. Anjum and Mumford propose two rules on how dispositions and probability relate:

1. The more something disposes towards an effect e, the more probable is e, *ceteris paribus*; and the more something over-disposes e, the closer we approach probability $P(e) = 1$.
2. There is a nonlinear "diminishing return" in over-disposing. E.g., if over-disposing e by a magnitude $x2$ produces a probability $P(e) = 0.98$, over-disposing $x3$ might increase that probability "only" to $P(e) = 0.99$, and over-disposing $x4$ "only" to $P(e) = 0.995$, and so on.

While these rules are fine as propositions, they miss the mark in explaining the relationship between dispositions and probability. In the coin tossing example, strengthening the mechanism is not about strengthening one outcome. Over-disposing does provide support for the distinction between the strength of the disposition and value of the probability, but the relationship between underpinning potentials, dispositional mechanisms, and the iterated outcomes needs to be made clear.

5.5. Radioactive Decay and Quantum Theory

Anjum and Mumford also discuss coin tossing and make substantially the same points as I made above. However, having clarified the distinction between propensity and probability, they revert to using the term propensity in a way that risks confusing the concepts of dispositional tendency and probabilities with random outcomes. They say "50% propensity" rather than "50% probability." They then introduce the term "chance" that they relate to outcomes in some specified situations.

Propensity is then reserved by them for potential probability while chance is the probability of an outcome in a situation. This is more confusing than helpful.

Anjum and Mumford go on to a discussion of radioactive decay that is known to be described by quantum theory. They make no mention of quantum theory (this will be corrected by them in Chapter 4) and strangely claim that radioactive decay is not probabilistic. The probability distributions derived from quantum mechanics unambiguously give the probability of decay per unit time. There are, per unit time, two possibilities: "decay" or "no decay." Their error is to claim "only one manifestation type" (decay) and from this that there is only one possibility. Ignoring quantum mechanics, they write:

> "The reason it is tempting to think of radioactive decay as probabilistic is that there is certainly a distinct tendency to decay that varies in strength for different kinds of particles, where that strength is specified in terms of a half-life (the time at which there is a 50/50 chance of decay having occurred)."

But no, the reason to think that radioactive decay is probabilistic is that our best theory of nuclear phenomena explains it in terms of probabilities. This misunderstanding leads them to introduce the concept of indeterministic propensities. Independently of how they arrived at the concept, it is left open as to whether there are non-probabilistic indeterminate powers, but radioactive decay is not an example.

5.6. Conclusion

The examples of the concept of chance provided by Anjum and Mumford can be derived in their examples from a correct application of probability theory. Chance is often used as a term for "objective probability," and I have done so in previous chapters. I will continue to follow that usage and exploit the clarification obtained from the analysis above shows that objective probability depends on the possibilities that are properties of an object. The manifestation of these possibilities may require an enabling mechanism. The statistical regularities displayed by these manifestations on iteration are due primarily to the physical properties of the object unless the enabling mechanism is badly designed.

The term "propensity" has given rise to much confusion in the literature. Now that we are reaching an explanation of objective probability, the term "propensity" might better be avoided.

I propose that the model of objective probability is that:

> **OBJECTIVE PROBABILITY:** An object has probabilistic properties if it is physically constituted so that it has a potential to manifest possibilities that show statistical regularities.

It is possible to describe statistical regularities without invoking the term "probability." Although some criticism of Anjum and Mumford is implied here, I recognize that their contribution has done much to disentangle considerations about the strength of dispositions that describe tendencies from a direct interpretation as probabilities. However, the value of the three distinctions they have identified is mixed:

1. **Chance and probability** are not fundamentally distinct and just require a correct application of probability theory.

2. **Probabilistic dispositional tendencies** are distinct from non-probabilistic dispositional tendencies: this is a real and fruitful distinction.
3. **Deterministic and indeterministic dispositional tendencies** also provide a useful distinction, but it remains to be seen whether there are fundamental non-probabilistic indeterministic dispositions.

5.7. The Kolmogorov probability axioms and objective chance"

Philosophy of Probability and Statistical Modelling by Mauricio Suárez (Suárez 2021) provides a historical overview of the philosophy of probability and argues for the significance of this philosophy for well-founded statistical modelling. Within this wider scope, the book discusses the themes: objective probability, propensities, and measurements. So, it has much in common with the themes of this investitgation.

Suárez defends a model of objective probability that disentangles propensity from single-case chance and the observed sequences of outputs that result in relative frequencies of outcomes. This is close to what I argued for in *Potentiality and Probability*, but there are differences that I will return to in a future chapter.

5.7.1. Subjective vs. Objective Probability

Suárez's main theme is objective probability, but he expends a lot of effort on examining subjective probabilities. I have no doubt that subjective probability has its place alongside objective probability but an *ad hoc* mixture of the two is to be avoided.

Subjective probability has been zealously defended by de Finetti and Jaynes to the point of them attempting to eliminate objective probability altogether. I believe, however, the chapters of this investitgation make a case for objective probability as an aspect of ontology. In this chapter I will focus on the Kolmogorov axioms of probability. Curiously, it is only in examining subjective probability that Suárez discusses the axioms of probability.

5.7.2. The Suárez Formulation

The axioms that Suárez states are as follows. Let $\{E_1, E_2, \dots, E_n\}$ be the set of events over which an agent's degrees of belief range; and let Ω be an event which occurs necessarily. The axioms of probability may be expressed as follows:

Axiom 1: $0 \leq P(E) \leq 1$, for any E: In other words, all probabilities lie in the real unit number interval.

Axiom 2: $P(\Omega) = 1$: The tautologous proposition, or the necessary event has probability one.

Axiom 3: If $\{E_1, E_2, \dots, E_n\}$ are exhaustive and exclusive events, then $P(E_1) + P(E_2) + \dots + P(E_n) = P(\Omega) = 1$: This is known as the addition law and is sometimes expressed equivalently as follows: If $\{E_1, E_2, \dots, E_n\}$ is a set of exclusive (but not necessarily exhaustive) events then: $P(E_1 \vee E_2 \vee \dots \vee E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$.

Axiom 4: $P(E_1 \wedge E_2) = P(E_1|E_2)P(E_2)$. This is sometimes known as the multiplication axiom, the axiom of conditional probability, or the ratio analysis of conditional probability since it expresses the conditional probability of E_1 given E_2.

According to Suárez, the Kolmogorov axioms are essentially equivalent to those above.

5.7.3. The Kolmogorov Formulation

Nathan Morrison has translated the Kolmogorov axioms (Kolmogorov 2018) as:

Let E be a collection of elements $\xi, \eta, \zeta, ...$, which we shall call elementary events, and $\mathfrak{F}$ a set of subsets of E; the elements of the set $\mathfrak{F}$ will be called random events.

> **I.** $\mathfrak{F}$ is a field of sets.
>
> **II.** $\mathfrak{F}$ contains the set E.
>
> **III.** To each set A in $\mathfrak{F}$ is assigned a non-negative real number $P(A)$. This number $P(A)$ is called the probability of the event A.
>
> **IV.** $P(E)$ equals 1.
>
> **V.** If A and B have no element in common, then $P(A \cup B) = P(A) + P(B)$.

Terminology has moved on and it would now be usual to identify the triple $(E, \mathfrak{F}, P)$ as the probability space with the term field reserved for the Borel field of sets $\mathfrak{F}$. In addition, it is preferable not to use the same symbol for the addition of numbers and the union of sets.

It is also important to note that $\xi, \eta, \zeta, ...$, indicating the elements of E does not constrain the set of elementary events to be a countable set. This is important for applications to statistical and quantum physics where, for example, particle positions and momentum can take a continuum of values.

It is a shorthand when Kolmogorov writes in Axiom III that the number $P(A)$ is the probability of the event A. The full interpretation is that $P(A)$ is the probability that the outcome will be an element in A. In the same Axiom III, the use of the term “assigned” is also deceptive. The probabilities are more properly assigned to the singleton set with each element of E as the sole member and from that the probability for each set in $\mathfrak{F}$ is constructed.

5.7.3.1. Mapping Logic to Set Theory

As Suárez is dealing with probabilities as credences (degrees of belief) in this section of his book, it would have been better to consistently employ the language and symbols of propositional logic.To maintain consistency, especially when dealing with credences, we can map the language of propositional logic to set theory:

Logical Category	Set Theoretical Category	Symbol Mapping
Elementary propositions	Elementary events	$\xi, \eta, ...$
Logical ‘And’	Set Intersection	$\wedge \to \cap$
Logical ‘Or’	Set Union	$\vee \to \cup$
Tautology	Universal Set	$\Omega \to E$

5.7.4. A Reformulated Neutral Axiom Set

I propose that this is a neutral set of axioms for a probability theory. It has one advantage that it can be applied to eventless situations such as in the quantum description of a free particle.

Let E be a collection of elements $\xi, \eta, \zeta, ...$, which we shall call **elementary possibilities** and $\mathfrak{F}$ a set of subsets of E.

I. $\mathfrak{F}$ is a field of sets.

II. $\mathfrak{F}$ contains the set E.

III. To each set A in $\mathfrak{F}$ is assigned a non-negative real number $P(A)$. This number $P(A)$ is called the probability of A, an element of $\mathfrak{F}$.

IV. $P(E)$ equals 1.

V. If A and B in $\mathfrak{F}$ have no element in common, then $P(A \cup B) = P(A) + P(B)$.

VI. For any A and B in $\mathfrak{F}$, $P(A|B) = \frac{P(A \cap B)}{P(B)}$ is the conditional probability.

Such a system of sets $\mathfrak{F}, E$, together with a definite assignment of numbers $P(A)$ for all $A \in \mathfrak{F}$, satisfying Axioms I-V, is called a **probability space**.

5.7.5. Applications: The Die and the Electron

In adopting these axioms for objective chance, the collection of elements can again be called elementary events or possible events, and P is a numerical assignment furnished by a theory or estimated through experiment.

The Die Example: A simple physical example is provided by a die with six sides. These six possibilities are the elementary events E. These outcomes are not in $\mathfrak{F}$, the set of subsets of E that Kolmogorov calls events. For example, the outcome "face 5 faces upwards" is not in the set of random events but the subset {"face 5 faces upwards"} is.

$P(\{\text{face 5 faces upwards}\})$ can be estimated as the proportion of times it occurs in a long run of repetitions. I emphasize that this is not a relative frequency interpretation; here the relative frequencies are an estimate of the numerical probability assignment. $P(\{\text{face 5 faces upwards}\})$ itself is the relative strength of the tendency for that event to occur, otherwise known as the **single-case chance**.

The Quantum Example: An example is the case of an observation of an electron governed by the Schrödinger equation. According to quantum mechanics the probability of it being observed at any predesignated spot is zero. In this example, all the elements of the set of elementary events have numerical probability assignment zero. This is where the field of events $\mathfrak{F}$ is useful in providing sets with non-zero probability in which the position of the electron may be observed.

5.7.6. Conclusion

The subjectivity enters through interpreting P as credence or degree of belief. In applications with objective probability, $P(A)$ is a numerical assignment of the strength of the single-chance tendency for an elementary event to appear in set A. In objective probability P is **ontological**, but in subjective probability it is **epistemic**.

5.8. Conditional probability: Renyi versus Kolmogorov

It is timely to Renyi's less well know but rival to Kolmogorov's axioms. In addition, Suarez (whose latest book was also discussed in a previous chapter(Suárez 2021)) appears to endorse the Renyi axioms (Rényi 1969) over those of Kolmogorov (Kolmogorov 2018) although only in a footnote. Stephen Mumford, Rani Lill Anjum and Johan Arnt Myrstad in the book What Tends to Be, Chapter 6 (Anjum and Mumford 2018b) also follow their analysis of conditional probability in the Kolmogorov axiomisation by taking the view that conditional probabilities should not be reducible to absolute probabilities.

In his Foundations of Probability (1969) Renyi provided an alternative axiomisation to that of Kolmogorov that takes conditional probability as the fundamental notion, otherwise he stays as close as possible to

Kolmogorov. As with the Kolmogorov axioms, I shall replace reference to events with possibilities.

Renyi's conditional probability space $(\Omega, \mathfrak{F}(\mathfrak{G}, P(F|G))$ is defined as follows.

The set Ω is the space of elementary possibilities and $\mathfrak{F}$ is a σ-field of subsets of Ω and $\mathfrak{G}$, a subset of $\mathfrak{F}$ (called the set of admissible conditions) having the properties):

(a) $G_1, G_2 \in \mathfrak{G} \Rightarrow G_1 \cup G_2 \in \mathfrak{G}$,

(b) $\exists\{G_n\}$, a sequence in $\mathfrak{G}$, such that $\cup_{n=1}^{\infty} G_n = \Omega$,

(c) $\emptyset \notin \mathfrak{G}$,

P is the conditional probability function satisfying the following four axioms:

- R0. $P : \mathfrak{F} \times \mathfrak{G} \to [0,1]$,
- R1. $\forall G \in \mathfrak{G}, P(G|G) = 1$
- R2. $\forall G \in \mathfrak{G}, P(\centerdot|G)$, is a countably additive measure on $\mathfrak{F}$.
- R3. If $\forall G_1, G_2 \in \mathfrak{G}, G_2 \subseteq G_1 \Rightarrow P(G_2|G_1) > 0$,

$$(\forall F \in \mathfrak{F}) P(F|G_2) = \frac{P(F \cap G_2|G_1)}{P(G_2|G_1)}.$$

Several problems have been examined by Stephen Mumford, Rani Lill Anjum and Johan Arnt Myrstad in What Tends to Be, Chapter 6, as part of a critique of the applicability of Kolmogorov's definition of conditional probability to the ontology of dispositions that tend to cause or hinder events. These have been analysed by them using Kolmogorov's absolute probabilities, but without a careful construction of the probability space appropriate for the application. These same examples will be analysed here using both Kolmogorov's and Renyi's formulation.

The first example that indicates a problem with absolute probability (absolute probability will be denoted by μ below to avoid confusion with Renyi's function P, the σ-field, $\mathfrak{F}$ is the same for both).

P1. For $A, B \in \mathfrak{F}$, let $\mu(A) = 1$ then $\mu(A|B) = 1$, μ is Kolmogorov's absolute probability.

Strictly this holds only for sets with measure greater than one. We can calculate this result from Kolmogorov's conditional probability as follows:

$$\mu(A \cap B) = \mu(B),$$

$$\mu(A|B) = \mu(A \cap B)/\mu(B) = \mu(B)/\mu(B) = 1.$$

This is not problematic within the mathematics but Mumford et al consider it to be if $\mu(A|B)$ is to be understood as a degree of implication. They claim that there must exist a condition under which the probability of A decreases. They justify this through an example: Say we strike a match and it lights. The match is lit and thus the (unconditional) probability of it being lit is one. Still, this does not entail that the match lit given that there was a strong wind. A number of conditions could counteract the match lighting, thus lowering the probability of this outcome. The match might be struck in water, the flammable tip could have been knocked off, the match could have been sucked into a black hole, and so on. Let us analyse this more closely. Let $A =$ "the match lights". Then, "The match is lit and thus the (unconditional) probability of it being lit is one." is equivalent to $\mu(A|A) = 1$. This is not controversial. They go on to bring other considerations into play and, intuitively, it seems evident that whether a match is lit or not will depend on the existing situation. For example, on whether it is wet or dry, windy, or not, and whether the match is stuck effectively. But this enlarges the space the elementary possibilities. In this enlarged probability space, the set A labelled "the match lights" is

$$A = \{(\text{"the match is lit", "it is windy", "it is dry", "match is struck"}\} \cup$$

$$\{(\text{"the match is lit", "it is not windy", "it is dry", "match is struck"}\} \cup$$

$$\{(\text{"the match is lit", "it is windy", "it is not dry", "match is struck"}\}......$$

where ...... indicates all the other subsets (elementary possibilities) that make up A.

Each elementary possibility is now a 4-tuple and A is the union of all sets consisting of a single 4-tuple in which the first item is "the match is lit". Similarly, a set can be constructed to represent $B =$ "match stuck". The probability function over the probability space is constructed from the probabilities assigned to each elementary possibility. An assignment can be made such that

$$\mu(A|B) = 1 \text{ or } \mu(A|B^C) = 0$$

where B^C ("match not struck") is the complement of B . It would not be a physically or causally feasible allocation of probabilities to have $ (A| B^C) =1 $ whereas $ (A| B^C) =0 $ is. Indeed, a physically valid allocation of probabilities should give $\mu(A \cap B^C) = \mu(\emptyset) = 0$. All Kolmogorov probability assignments of elementary possibilities with "the match is lit" and "match not struck" in the 4-tuples of elementary possibilities should be zero. The Kolmogorov ratio formula for the conditional probability would apply in the case when all the conditions are accommodated in the set of elementary possibilities. Therefore, P1 is not a problem for the Kolmogorov axioms if the probability space is appropriately modelled.

Appropriate modelling is just as relevant when using the Renyi axioms. I addition, as we are working in the context of conditions influencing outcomes, we will not allow outcomes that cannot be influences to be in the set of admissible conditions $\mathfrak{G}$. This has no effect on the analysis of P1 but, as will be discussed below, is important for modelling causal conditions.

A further problematic consequence of Kolmogorov's conditional probability, according to Mumford et al, is when A and B are probabilistically independent P2. $\mu(A \cap B) = \mu(A)\mu(B)$ implies $\mu(A|B) = \mu(A)$. This is indeed a consequence of Kolmogorov's definition. Renyi's formulation does not allow this analysis to be carried out, unless $\mathfrak{G} = \mathfrak{F}$. Mumford et al illustrate there concern through an example The dispositional properties of solubility and sweetness are not generally thought to be probabilistically dependent. Whatever is generally thought, the mathematical analysis will depend on the probability model. If two properties are probabilistically independent, then that should be captured in the model. However the objections of Mumford et al are combined with a criticism of Adam's Thesis Assert (if B then A) $=$ P(if B then A) $=P$(A given B) $= P(A|B)$ where $P(A|B)$ is given by the Kolmogorov ratio formula. However, it should be remembered that the Kolmogorov ratio formula can be simply showing correlation and not that B causes A or that B implies A to any degree. I do not want to get into defending or challenging this thesis here but within the Renyi axiomisation the Kolmogorov conditional probability formula only holds under special conditions, see R3. Independence, in Renyi's scheme, is only defined with reference to some conditioning set, C say. In which case probabilistic independence is described by the condition

$$P(A \cap B|C) = P(A|C)P(B|C)$$

and as a consequence, it is only if $B \subseteq C$ that

$$P(A|B) = \frac{P(A \cap B|C)}{P(B|C)} = P(A|C)$$

This means that if a set D in $\mathfrak{G}$ that is not a subset of C is used to condition $A \cap B$ then, in general,

$$P(A \cap B|D) \neq P(A|D)P(B|D)$$

even if

$$P(A \cap B|C) = P(A|C)P(B|C).$$

This shows that in the Renyi axiomisation statistical independence is not an absolute property of two sets.

The third objection is that regardless of the probabilistic relation between A and B, a third consequence of the Kolmogorov conditional probability definition is that whenever the probability of A and B is high $\mu(A|B)$ is high and so is $\mu(B|A)$: P3. $(\mu(A \cap B) \sim 1) \Rightarrow ((\mu(A|B) \sim 1) \wedge \mu(B|A) \sim 1))$. If $\mu(A \cap B) \sim 1$ then $A \equiv B$ but for a set of measure zero. Then $\mu(A) \sim 1$ and $\mu(B) \sim 1$ that implies the statement in P3. Mumford et al object The probability of the conjunction of 'the sun will rise tomorrow' and 'snow is white' is high. But this doesn't necessarily imply that the sun rising tomorrow is conditional upon snow being white, or vice versa. That may the case but the correlation between situations where both are the case is high. Once again, the problem is the identification of conditional probability with a degree of implication in Adam's Thesis. But it is well known that conditional probability may simply capture correlation. If we want to separate conditioning sets from other sets that are consequences in the σ-algebra generated by all elementary possibilities, then Renyi's axioms allow this.

The Renyi equivalent of P3 is

$$P(A \cap B|C) \sim 1 \Rightarrow (P(A|B) \sim 1, B \subseteq C) \wedge (P(B|A) \sim 1, A \subseteq C$$

It does holds, when both A and B are subsets of C but that is then a reasonable conclusion for the case of both A and B included in C. However, if one of the sets is not a subset of C then it will not hold in general.

When $\mathfrak{G}$ is a smaller set than $\mathfrak{F}$ it becomes useful for causal conditioning. We can exclude sets from $\mathfrak{G}$ that are outcomes and include sets that are causes. If we are interested in causes of snow be white, we will condition in facts of crystallography and local conditions that may turn snow yellow, as pointed out by Frank Zappa.

For the earlier example above the set A, "the match lights" would not be included in $\mathfrak{G}$. So for $C \in \mathfrak{G}, P(A|C)$ is a defined probability measure but $P(C|A)$ is not.

The Kolmogorov axioms are good for modelling situations where measurable sets represent events of the same status. If there are reasons to believe that some sets have the status of causal conditions for other sets then they should be modelled with Renyi's axiomisation (or some similar axiomisation) as subsets of the set of admissible conditions.

The next question is whether adopting, and modelling fully, with the Renyi scheme allows a counter to objections such as those of Humphreys (1985) to using conditional probabilities to represent dispositional probabilities.

6. An Ontology of Complex Systems

** Place holder for ontology of complex systems **

An ontology of complex systems studies the fundamental entities, properties, structures, and relations that define systems characterized by non-linearity, emergence, and feedback loops. Unlike reductionist frameworks that analyze isolated parts, this branch of metaphysics focuses on how wholes relate to their parts. [1, 2, 3, 4, 5]

6.1. Key Foundational Concepts

- Emergence: Ontological status given to macro-level properties that cannot be reduced to, or predicted by, the isolated components of a system. [6, 7]
- Non-linearity: Causal relationships where outputs are disproportionate to inputs, meaning small changes can trigger massive, unpredictable systemic shifts. [8, 9]
- Downward Causation: The controversial concept that the emergent macro-structure of a system can actively constrain or influence the behavior of its micro-parts. [10, 11, 12]
- System Boundaries: The problematic definition of where a complex system ends and its environment begins, as boundaries are often fluid, open, and dynamic. [13, 14, 15, 16]

6.2. Major Ontological Debates

6.2.1. 1. Substance vs. Process Ontology

- Substance view: Traditional ontologies view the world as a collection of static things (substances) that possess changing properties.
- Process view: Complex systems theorists often argue that processes are more fundamental than things. A complex system is not a static object, but a continuous stabilization of energy and information flow. [17, 18, 19, 20, 21]

6.2.2. 2. Reductionism vs. Emergentism

- Ontological Reductionism: The belief that a complex system is "nothing but" the sum of its fundamental physical parts.
- Ontological Emergentism: The view that complex systems introduce genuinely new layers of reality with unique causal powers that cannot be captured by the laws of physics alone. [22, 23]

6.2.3. 3. Realism of Macro-Structures

- Epistemic Realism: The stance that concepts like "the economy," "an ecosystem," or "the mind" are merely useful shorthand descriptions for human convenience.
- Ontological Realism: The assertion that these complex macro-structures exist objectively in reality, possessing independent causal power. [24, 25]

6.3. Applications to Artificial Intelligence

In the context of Barry Smith and Jobst Landgrebe's work (Why Machines Will Never Rule the World), this ontology is used to argue against the possibility of Artificial General Intelligence (AGI). They contend that because the human mind-brain is an open, non-linear complex system, its states cannot be formalized into the static, computable data structures required by digital computers. [26]

[1] https://pmc.ncbi.nlm.nih.gov

[2] https://peer.asee.org

[3] https://www.cambridge.org

[4] https://www.resilience.org

[5] https://pressbooks.openeducationalberta.ca

[6] https://medium.com

[7] https://pmc.ncbi.nlm.nih.gov

[8] https://academic.oup.com

[9] https://www.intechopen.com

[10] https://www.tandfonline.com

[11] https://www.self-organization.org

[12] https://www.facebook.com

[13] https://sebokwiki.org

[14] https://dokumen.pub

[15] https://www.tandfonline.com

[16] https://onlinelibrary.wiley.com

[17] https://www.christopherroosen.com

[18] https://link.springer.com

[19] https://www.tandfonline.com

[20] https://www.mdpi.com

[21] https://sk.sagepub.com

[22] https://dl.acm.org

[23] https://www.iit.wiki

[24] https://methods.sagepub.com

[25] https://bridgeresearchconsulting.com

[26] https://dataskeptic.com

Part II.

Physical Layer

7. A Relational Reconstruction of Quantum Mechanics

The mathematical formulation to be discussed here is a modification of Kochen's Reconstruction of Quantum Mechanics (Kochen 2015). It concentrates on a mathematical formulation using a union of σ-algebras that he calls a **complex**. In Kochen's reformulation, it is through the interaction of a quantum system under consideration with another system that properties gain physical values. Such properties are said by Kochen to be **relational** or **extrinsic**, as opposed to fixed intrinsic properties such as the charge of an electron.

While the mathematics depends heavily on the reconstruction by Kochen, it makes use of **von Neumann $*$-algebras** (see Thirring (Thirring 1979)) rather than the Boolean algebras used by Kochen. This choice is made to avoid any hint of a logical interpretation and will be useful when we move on to discussing relativistic quantum mechanics and field theory. This formulation also differs from Kochen's in the treatment and assignment of both intrinsic and extrinsic properties. The extrinsic properties will be treated here as an actualization of potential, dispositional properties of the quantum entity that are shaped by its context.

In the mathematical formulation of quantum mechanics, a physical system, **S**, is characterized by a list of properties (called observables in standard quantum mechanics) that can be represented by abstract self-adjoint operators:

$$\mathcal{O}_{\mathbf{S}} = \{\hat{X}_i = \hat{X}_i^* \mid i \in \mathcal{I}_{\mathbf{S}}\},$$

with $\mathcal{I}_{\mathbf{S}}$ a set of indices depending on $\mathbf{S}$, where every operator $\hat{X} \in \mathcal{O}_{\mathbf{S}}$ represents a physical property characteristic of $\mathbf{S}$, such as the total momentum, energy, or spin of all particles localized in some bounded region of physical space and belonging to an ensemble of (possibly infinitely many) particles constituting the system $\mathbf{S}$.

At every time t, there is a representation of $\mathcal{O}_{\mathbf{S}}$ by self-adjoint operators acting on a separable Hilbert space $\mathcal{H}_{\mathbf{S}}$:

$$\mathcal{O}_{\mathbf{S}} \ni \hat{X}_i \mapsto X_i = X_i^* \in \mathcal{A}(\mathcal{H}_{\mathbf{S}}),$$

where $\mathcal{A}(\mathcal{H}_{\mathbf{S}})$ is the $*$-algebra of all bounded operators acting on $\mathcal{H}_{\mathbf{S}}$.

In general, the operators in the $*$-algebra do not all commute. This has deep physical consequences:

- Properties that cannot appear together are represented in quantum mechanics by self-adjoint operators that do not all commute.
- The sets of operators that do commute provide a set of properties for a measurable mathematical structure which is a **σ-algebra**.

The σ-algebra is essential to constructing mathematical probability models using the Kolmogorov axiomization (Kolmogorov 2018). Therefore, each collection of commuting observables of the object has an associated σ-algebra. So, a collection of σ-algebras that together capture all the property values associated with the object covers all possible events for that object in itself. This structure is the minimal one which contains all the σ-algebras arising from all the properties of the particle.

Each dispositional property of an object, such as spin or position, has a $\sigma - algebra$ of possible values that appear in interaction with other objects. All these σ-algebras form a complex property structure for the object under consideration. The formal definition of this notion is as follows:

> **Definition:** Let F be a family of σ-algebras. The σ-complex Q_F based on F is the union, $\cup B$, of all σ-algebras B lying in F:

$$Q_F = \bigcup_{B \in F} B.$$

Generally, the family F is left implicit, and reference is simply made to a σ-complex Q; however, the family F does tie the complex to an object that has properties that take values depending on the set of possible interactions that it can have with other objects. Usually, σ-complexes that are closed under the formation of sub-σ-algebras are discussed. It is possible, in any case, to always close a σ-complex by adding all its sub-σ-algebras. This complex will include all the possible values that the dispositional properties of a particle can take in all possible interaction contexts. B denotes the σ-algebra relating to the property values of a particular (commuting) set of observables of the particle or system.

In the mathematical formulation of quantum mechanics, $\mathcal{H}$ is a Hilbert space. For the quantum object under consideration, all properties are represented by self-adjoint operators and possible events are represented by projection operators. For pairwise commuting projection operators, closure under the operation of orthogonal complement and countable product $\prod_i \pi_i$ forms a σ-algebra. The family of all such σ-algebras forms a σ-complex, $Q(\mathcal{H})$, for the object in its possible interaction contexts.

7.1. Proposition

In quantum mechanics, the operators representing the possible values of the properties of a system form the σ-complex $Q(\mathcal{H})$ of projections of the Hilbert space $\mathcal{H}$ of the system.

It has been shown (Kochen 2015) that this proposition holds and the mathematical formalism using $Q(\mathcal{H})$ is equivalent to the standard quantum formulation of the theory on $\mathcal{H}$. However, the interpretation of the formalism here is that while a quantum object has a complete set of properties, they appear only when the situation invokes the σ-algebra in the complex that allows that property to take a specific set of values.

7.2. The Measurement Problem

{#sec-measurement-problem} This treatment is adapted from Kochen's reconstruction of quantum mechanics (Kochen 2015). It is a mathematical formulation of the theory that is equivalent to the standard formulation, but it is not a physical theory. The physical theory is the interpretation of the mathematics. The interpretation here is that the properties of a quantum object are relational and appear only in interaction with other objects. The measurement problem arises because the standard formulation of quantum mechanics assumes that an isolated system evolves unitarily according to Schrödinger's equation, but when a measurement occurs, an eigenvalue of the operator representing the observable being measured is randomly selected as the result of the measurement. This additional mechanism is not explained by the unitary evolution and leads to a contradiction between the predicted state and the observed state after measurement.

The Measurement Problem refers to a postulate in standard quantum mechanics, which assumes that an isolated system undergoes unitary evolution via Schrödinger's equation and then an eigenvalue of the operator representing the observable being measured (an observable is a property of the system that the experimental setup is designed to measure) is randomly selected as the result of the measurement, as presented by Bohm (2001), for example. However, if a property $\hat{A}$ of a system S is measured by an apparatus T, the total system $S + T$, if assumed to be isolated, then undergoes unitary evolution. The random selection of an eigenvalue is an additional mechanism.

7.3. Ideal Measurement

The mathematical formulation of an ideal measurement, in standard quantum mechanics, is as follows for system (S) in a pure state (_k):

Take the spectral decomposition of an operator representing an observable to be

$$A = \sum_i a_i \pi_i.$$

Each π_i is a one-dimensional projection with eigenstate ϕ_i and $\{a_i\}_i$ is the set of eigenvalues. The apparatus T is assumed to be sensitive to the different eigenstates of A.

Hence, if the initial state of S is ϕ_k and the apparatus T is in a neutral state ψ_0, so that the state of $S + T$ is

$$\phi_k \otimes \psi_0,$$

the system evolves into the state

$$\phi_k \otimes \psi_k,$$

where the $\{\psi_i\}_i$ are the states of the apparatus operator corresponding to the states $\{\phi_i\}_i$ of the system (S).

T and its interaction with S will have been chosen to achieve this: a perfectly designed measurement apparatus to be in ψ_l whenever S is in ϕ_l for all l.

But now, for the case of a more general initial state

$$\phi = \sum_i c_i \phi_i,$$

linearity gives:

$$\Gamma = \sum_i c_i \phi_i \otimes \psi_i.$$

A problem with this for standard quantum mechanics is that the completed measurement gives a particular apparatus state ψ_k, say, indicating that the state of S is ϕ_k, so that the state of the total system is $\phi_k \otimes \psi_k$, in contradiction to the derived evolved state

$$\sum_i c_i \phi_i \otimes \psi_i.$$

This evolution does not describe what happens in an experiment.

7.4. Conditioning and Reduction

In contrast, the reduction can also be considered from the viewpoint of the conditioning of the states. If the state p of $S + T$ just prior to measurement is ρ_Γ, corresponding to

$$\Gamma = \sum_i c_i \phi_i \otimes \psi_i,$$

then after the measurement it is in the conditioned state, by equation (**) in the post *Quantum chance*:

$$
\begin{aligned}
p(\cdot \mid (\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})) &= \frac{(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\rho_\Gamma(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})}{\mathbf{tr}((\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\rho_\Gamma)} \\
&= \frac{(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\left(\sum_i c_i\phi_i \otimes \psi_i\right)(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})}{\mathbf{tr}((\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\left(\sum_i c_i\phi_i \otimes \psi_i\right))} \\
&= \pi_{\phi_k \otimes \psi_k}.
\end{aligned}
$$

Hence, the new conditioned state of $S+T$ is the reduced state $\phi_k \otimes \psi_k$. This is not surprising as it is conditioned on being in just the (k)th state of (_) and so it projects that element out. This is not a resolution of the measurement problem but merely makes use of the probabilistic formulation of the theory to show conditioning forces the state reduction.

7.5. Beyond Unitary Evolution

Whereas standard quantum mechanics must add a means to reconcile the unitary evolution of $S+T$ with the measured reduced states of S and T, the interpretation argued for in this paper takes the opposite approach to the orthodox interpretation. The point of departure is not the unitary development of an isolated system, but rather the result of an interaction.

It is the conditions under which dynamical evolution occurs that must be further investigated, rather than the additional reduced state mechanism. Therefore, it should not be taken for granted, as assumed in standard quantum mechanics, that an isolated system evolves unitarily.

The question to be addressed is whether in a measurement the σ-complex structure of $S+T$ undergoes a symmetry transformation at separate times of the process. This is formalised as the condition for the existence of a representation

$$\alpha : \mathbb{R} \to \mathrm{Aut}(Q).$$

The outcome of a measurement cannot be given by a unitary process.

7.6. Failure of Automorphism Description

A completed measurement or a state preparation has two distinct elements of

$$Q(\mathcal{H}) = Q(\mathcal{H}_S \otimes \mathcal{H}_T)$$

at initial time 0 which end up being mapped to the same element at a later time (t).

One such element is an initial state (_0) resulting in a state (_k _k), for some (k). However, a second such element (_k _0) also results in the state (_k _k).

If the state ϕ is chosen to be distinct from ϕ_k, then the two elements

$$\pi_{\phi\otimes\psi_0}, \qquad \pi_{\phi_k\otimes\psi_0}$$

of $Q(\mathcal{H})$ both map to the same element

$$\pi_{\phi_k\otimes\psi_k}.$$

However, any automorphism α_t is a one-to-one map on Q, so the measurement process cannot be described by a representation

$$\alpha : \mathbb{R} \to \mathrm{Aut}(Q),$$

and hence **a unitary evolution cannot explain what is observed**.

7.7. State Reduction in Dynamics

The Measurement Problem must be resolved by a theory that includes state reduction in its dynamics in addition to periods of unitary evolution. The GRW theory provides an example of a partial mechanism for this—partial because it only reduces the wavefunction to one that is more localised rather than full transition from possibility to actuality.

Bohmian mechanics avoids this by proposing a particle trajectory dynamics that requires no more state reduction than in classical probability. In Bohmian mechanics the particle always has an actual position.

7.8. Internal Interactions

For a composite system it should not only be outside forces that can break symmetry, but internal interactions. In the state

$$\Gamma = \sum_i c_i \phi_i \otimes \psi_i$$

introduced above, the total, but still isolated, system (S+T) has a set of (i) property values associated with the states ({ _i   _i}_i).

However, the interacting object (T) as part of the system (S+T) will have the state (_k) of (S) appear with probability (|c_k|^2). This provides a matrix mechanics interpretation of reduction as a physical transition probability for the system (S) in the presence of the apparatus (T).

State reduction does take place in isolated compound systems with internal interactions and the reduction of the state is due to the combined system's properties but traceable to the dispositional power to take specific property values associated with (S).

7.9. Timing of Experimental Records

In an experiment the results are recorded at the time of the experiment. This experimental recording is not part of the formal theory. The theory provides transition probabilities but nothing to time the transition.

8. Heisenberg, Potentiality, and Quantum Ontology

Heisenberg, in one of his excursions into philosophy, *Physics and Philosophy*, (Heisenberg 1958), discusses epistemology, logic, and ontology. In developing his ideas on how the innovations of quantum mechanics impact philosophy, he introduces (or rather reintroduces, with reference to Aristotle) the concepts of **potentiality**.

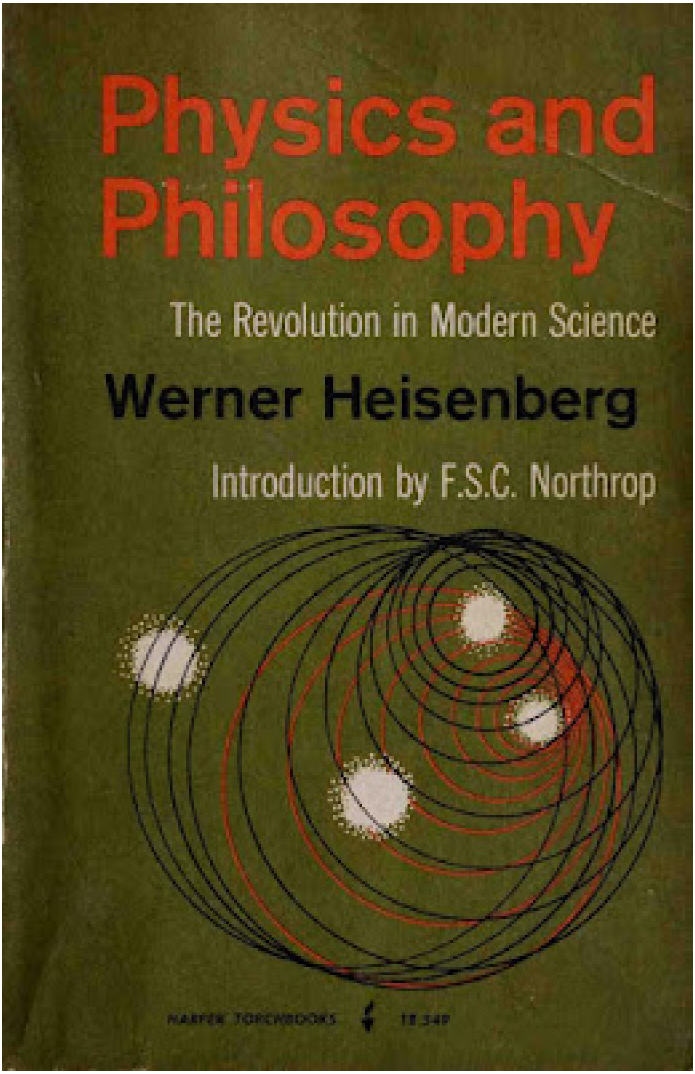


Figure 8.1.: Physics and Philosophy

He comes to potentiality through considering the logic required to refer to things that quantum mechanics describes. That is, it

> "… concerns the ontology that underlies the modified logical patterns. If the pair of complex numbers represents a 'statement' in the sense just described, there should exist a 'state' or a 'situation' in nature in which the statement is correct. We will use the word 'state' in this connection. The 'states' corresponding to complementary statements are then called 'coexistent states' by Weizsäcker. This term 'coexistent' describes the situation correctly; it would in fact be difficult to call them 'different states,' since every state contains to some extent also the other 'coexistent states.' This concept of 'state' would then form a first definition concerning the ontology of quantum theory. One sees at once that this use of the word 'state,' especially the term 'coexistent state,' is so different from the usual materialistic ontology that one may doubt whether one is using a convenient terminology. On the other hand, if one considers the word 'state' as describing some potentiality rather than a reality—one may even simply replace the term 'state' by the term 'potentiality'—then the concept of 'coexistent potentialities' is quite plausible, since one potentiality may involve or overlap other potentialities."

In the version of quantum mechanics that I have been developing, these *coexistent potentialities* should be understood in relation to the σ-complex. Despite his hints, Heisenberg does not succeed in including potentiality in the required logic. What is needed is a **modal logic** that includes a **POTENTIALITY operator**. Barbara Vetter constructs such a modal logic and defends it with modal metaphysical arguments in her book *Potentiality: From Dispositions to Modality* (Vetter 2015).

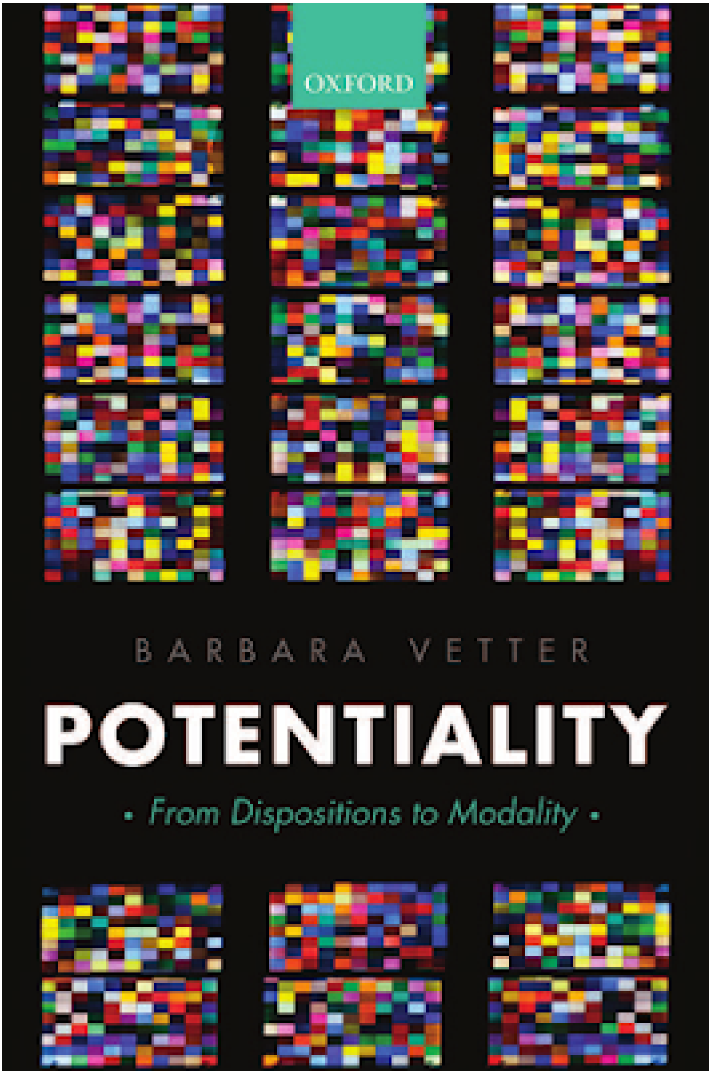


Figure 8.2.: Potentiality: From Dispositions to Modality

There is a need for a theory going beyond the naïve position that there are objects that simply have the potential to have certain properties.

In quantum mechanics in particular it is not sufficient to say that an electron has the potential to be in some place. If this is elaborated to saying that it has some position with a certain probability, we get a deceptive oversimplification. The theory of potentiality developed by Vetter gives potentialities the status of **properties of objects**. Contrary to Heisenberg's view that "one considers the word 'state' as describing some potentiality rather than a reality," Vetter's theory allows us to enlarge the domain of what is real to include potentialities.

8.1. Possibility and Iterated Potentiality

In standard quantum mechanics, observables can take possible values. The possibility is described by the theory, and the values are represented by the spectrum of the self-adjoint operator that represents the observable.

Vetter proposes, in general, to define possibility as follows:

> **POSSIBILITY:**
> *It is possible that $p =_{\mathrm{df}}$ Something has an iterated potentiality for it to be the case that p.*

We have now moved on to potentiality and, more generally, its **iteration**. I have discussed dispositions in previous posts and now explore Vetter's proposal that we call those properties which form the metaphysical background for disposition ascriptions **potentialities**. In the definition, the *something* also needs explanation. Something can be an object or any collection of objects.

Potentialities are individuated not by a pair of stimulus and manifestation and a corresponding conditional, but by their **manifestation alone**. The manifestation of a potentiality is the property that the potentiality's possessor would have if the potentiality were to be manifested. A manifestation can be a property of an object (or collection of objects) or a relation that the object (or collection) has with something else.

8.2. Joint Potentiality

The relation between a potentiality and its manifestation—the relation "… is a potentiality to …"—allows a classification of joint potentialities:

- **Type 1 joint potentialities**: the manifestation is a relation between, or a plural property of, *all* its possessors.
- **Type 2 joint potentialities**: the manifestation consists in a property or relation of *only some*, but not all, of its possessors.

Examples of Type 1 include the door key's joint potentiality with the door for the former to unlock the latter.

Examples of Type 2 include:

- a uranium pile (with a potential to go unstable) plus the rods and fail-safe mechanism

- a glass (with a potential to break) but protected by styrofoam

- a city (with a potential to be destroyed) and its defence system

In all three, only one of the objects manifests the disposition or potentiality, but they jointly have the potentiality to a certain degree.

8.3. Extrinsic Potentiality

An object has an **extrinsic potentiality** if it has a joint potentiality whose co-possessors are the *dependees* of the extrinsic potentiality. In the key–door example, the door is part of the manifestation (unlocking the door). In general, where a potentiality's manifestation consists in a relation to a particular other object—such as opening *this* particular door—the potentiality will be extrinsic.

8.4. Iterated Potentiality

Potentialities themselves are properties. So objects can have:

- potentialities to have potentialities
- potentialities to have potentialities to have potentialities
- and so on, indefinitely.

These are **iterated potentialities**.

8.5. Formalising Potentiality

In Vetter's theory, potentialities include dispositions and abilities as properties of individual objects or collections of objects. POTENTIALITY cannot be defined, because it is metaphysically fundamental, but it can be **formalised**.

Just as modal operators act on propositions, the modal potentiality operator is:

$$\mathrm{POT}$$

POT is a **predicate-forming operator**. Where Φ is a predicate:

$$\mathrm{POT}[\Phi](t)$$

ascribes to t the potentiality to Φ.

To allow POT to operate on sentences, we introduce the λ-operator:

$$\mathrm{POT}[\Phi](t) \to \mathrm{POT}[\lambda x.\, \phi](t)$$

where $\lambda x.\phi$ means “is such that ϕ”.

A simple iterated potentiality is:

$$\mathrm{POT}[\lambda x.\, \mathrm{POT}[F](x)](a)$$

which ascribes to a the potentiality to have the potentiality to be F.

8.6. The Electron

The electron, as described by quantum mechanics, demonstrates the ontological strangeness of quantum objects:

- it has the potential to be somewhere
- to have a momentum
- to have charge and mass

But quantum mechanics gives more structure: the electron appears somewhere with a **probability**, and these probability models are combined in a **σ-complex**, corresponding (possibly) to Weizsäcker’s “coexistent states”.

The electron has a **joint potentiality** with other objects to manifest one of the probability distributions from the complex. That probability distribution weights the appearance of specific values of the property with a numerical probability.

In an experiment the electron does not appear directly but only through observable effects on the objects with which it is in a relationship. The electron never appears in the way that a macroscopic object (such as a book) appears, with phonomonological directness. The electron is never in space-time but it does influence other objects that are in space-time to show the effects of its presence as a potentiality.

Thus:

- the probability that the electron appears in a region of space

- in certain circumstances **and** on in relation to other objects.

is itself an **iterated potentiality**.

This example indicates how the concept of potentiality, as formalised, can be used to describe quantum processes. It remains to be seen how useful this will be when we introduce considerations of spin, entanglement, interference, etc. This will be explored in further chapters.

9. The double slit experiment

Having discussed the issues with quantum measurement in general, and shown that standard interpretation of quantum mechanics is incomplete, Young's double slit experiment with electrons will be discussed in two variants (for background see Chapter 1 of Hall (2013)): The standard configuration, to be described below, Figure (1). A configuration with a pointer that acts behind the slits to point in the direction of the passing electron, Figure (2).

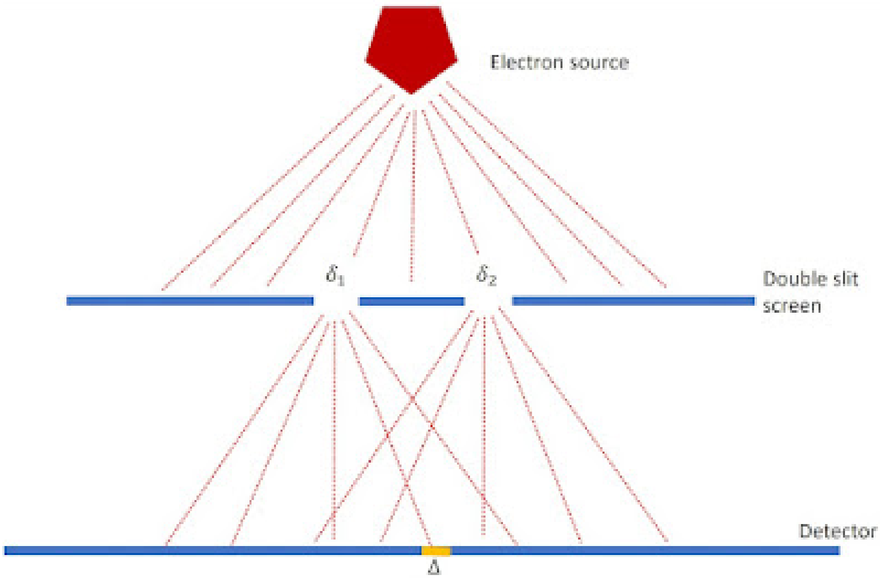


Figure 9.1.: Figure (1) The setup for the double slit experiment is shown. An electron source sends one particle at a time toward the screen with the slits. The slits are marked by δ_1 and δ_2 and a sample region Δ is shown on the detector screen.

9.1. Standard configuration

The experiment, Figure (1), is as follows:

There is a source of electrons that move towards a screen with two slits.
The intensity of the beam is low and only one electron is moving towards the detector at any time.

The slits are marked δ_1 and δ_2.
Δ be some arbitrary region on the electron detector.

Let Y_1 and Y_2 be the projection operators for position in the regions of the two slits δ_1 and δ_2. Then $Y_1 + Y_2$ is the projection of position for the union $\delta_1 \cup \delta_2$. Let X be the operator for position in a local region Δ on the detection screen. Assume the electron is only constrained to pass through the slits without being constrained as to which, then under those conditions the conditional probability is given by the Law of Alternatives:

$$p(X|Y_1+Y_2) = p(X|Y_1)p(Y_1|Y_1+Y_2) + p(X|Y_2)p(Y_2|Y_1+Y_2)$$
$$+\frac{\mathrm{tr}(Y_1\rho Y_2 X) + \mathrm{tr}(Y_2\rho Y_1 X)}{\mathrm{tr}(\rho(Y_1+Y_2))}.$$

This can be written more compactly as

$$p(X|Y_1+Y_2) = p(X|Y_1)p(Y_1|Y_1+Y_2)$$
$$+p(X|Y_2)p(Y_2|Y_1+Y_2) + p(X|Y_1+Y_2)_I$$

where $p(X|Y_1+Y_2)_I$ is the interference term.

Note that if X commutes with either Y_1 or Y_2, this interference term vanishes, because $Y_iY_j = 0$ for $i \neq j$. This can be observed if the detector is right next to the two-slit screen because X then coincides with either Y_1 or Y_2. If the detector is a distance from the two-slit screen (e.g. $X = \Delta$), then the state of the electron evolves unitarily via α_t, so $\alpha_t(Y_i) = u_t Y_i u_t^{-1}$ no longer commutes with $\alpha(X)$, giving rise to the non-zero interference term.

The standard explanation of the interference effect is that the state of the particle is, or acts as, a coherent pair of waves emanating from the slits, which exhibit constructive and destructive interference effects. This was, of course, the explanation for Young's original experiment with light. For individual quantum particles, however, there is the unexplained local event observed at the detection screen. This is a problem for the theory. While the Born interpretation of the wavefunction provides a probability distribution for the particle position it requires the detection screen, operating outside what is described by the mathematical theory, to act as a sampling mechanism for that distribution.

The explanation given in this blog is that the two-slit screen functions as a preparation of the state for the particle, by which the state is conditioned, or reduced, to pass through the region $\delta_1 \cup \delta_2$. This reduction is not a position measurement, since $\delta_1 \cup \delta_2$ is not a localised region (as it would be for a single-slit screen). Once the particle reaches a detection screen then, in interaction with the screen, it appears in a random local region Δ and its position takes a value. Just as in the standard Born interpretation of the wavefunction, it is not explained in the theory how the electron takes the value that the detector detects other than invoking random sampling of the possible values.

So, the interference pattern on the detector is built up over time as more electrons arrive and are sampled by the detector.

9.2. The introduction of an interaction with a pointer

This section is adapted from Bricmont (2017), Appendix 5.A and Maudlin (2019).

The experiment, illustrated in Figure (2), is now as follows:

There is again a source of electrons that move towards a screen with two slits.
The intensity of the beam is low and only one electron is moving towards the detector at any time.

The slits are marked δ_1 and δ_2.
Δ be some arbitrary region on the electron detector.

A pointer P is introduced. It is a quantum object with three states: neutral P_0, points to slit 1 (P_1), and points to slit 2 (P_2). The interaction with the electron causes the pointer to move towards it.

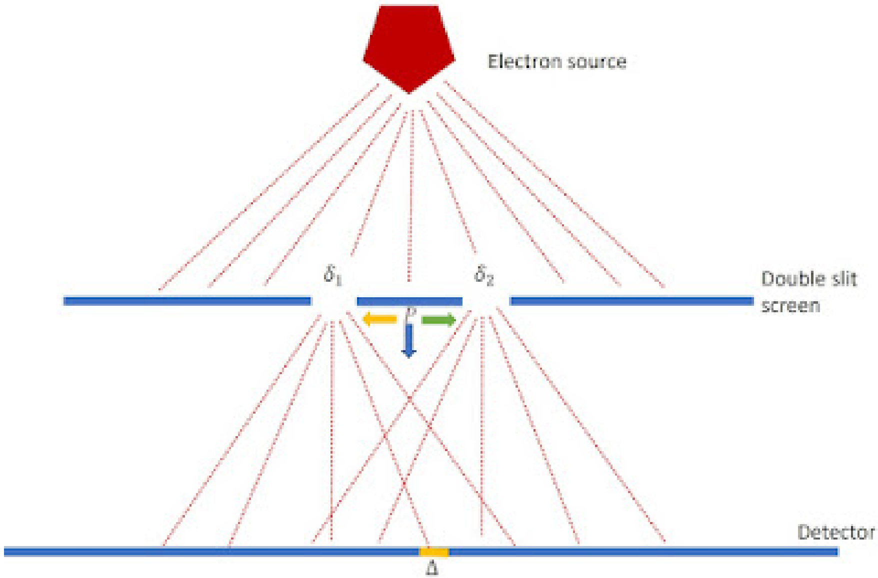


Figure 9.2.: Figure (2) The setup for the double slit experiment is as in Figure (1) but for the addition of a three state pointer that interacts with the electron as it passes through slit δ_1 or δ_2.

Again let Y_1 and Y_2 be the projection operators for position in the regions of the two slits δ_1 and δ_2. Then $Y_1 + Y_2$ is the projection of position for the union $\delta_1 \cup \delta_2$. Let X be the operator for position in a local region Δ on the detection screen. The operator representing the pointer has three eigenstates and therefore a three-dimensional Hilbert space $\mathcal{H}_P$. Without the pointer the Hilbert space is $\mathcal{H}_0$. The Hilbert space of the total system is $\mathcal{H}_P \otimes \mathcal{H}_0$.

The possible constituent states are:

- ϕ_1 pointer pointing to slit δ_1
- ϕ_2 pointer pointing to slit δ_2
- ϕ_0 pointer neutral
- ψ_1 electron through slit δ_1
- ψ_2 electron through slit δ_2
- Ψ_0 electron with pointer in neutral state P_0

Assuming the pointer starts in its neutral state, the initial wave function is

$$\begin{aligned} \Psi_0 &= \phi_0 \otimes (\psi_1 + \psi_2) \\ &= \phi_0 \otimes \psi_1 + \phi_0 \otimes \psi_2 \end{aligned}$$

9.2.1. Treatment I: Pointer reacts to which slit the electron passes through

Time evolution gives:

$$\Psi = \phi_1 \otimes \psi_1 + \phi_2 \otimes \psi_2.$$

The interference term becomes:

$$p(X|Y_1 + Y_2)_I = \frac{\mathrm{tr}(Y_1 \rho_\Psi Y_2 X) + \mathrm{tr}(Y_2 \rho_\Psi Y_1 X)}{\mathrm{tr}(\rho_\Psi (Y_1 + Y_2))}$$

Using orthogonality of pointer states:

$$p(X|Y_1 + Y_2)_I = 0$$

Interference disappears.

9.2.2. Treatment II: Pointer does *not* react to which slit the electron passes through

General evolution:

$$\Psi = \sum_{i \in \{0,1,2\}} a_i \phi_i \otimes \psi_1 + \sum_{i \in \{0,1,2\}} b_i \phi_i \otimes \psi_2$$

with

$$a_0 = b_0, \quad a_1 = b_2, \quad a_2 = b_1.$$

The detection pattern is:

$$(\Psi, P \otimes X\,\Psi) = \sum_i |a_i|^2 (\psi_1, X\psi_1)$$

$$+ \sum_i |a_i|^2 (\psi_2, X\psi_2) + \sum_i a_1^* a_2 (\psi_2, X\psi_1) + \sum_i a_2^* a_1 (\psi_1, X\psi_2)$$

Let $C = \sum_i |a_i|^2$:

$$(\Psi, P \otimes X\,\Psi) = C\big[(\psi_1, X\psi_1) + (\psi_2, X\psi_2)\big] + \mathrm{Re}\{2a_1^* a_2(\psi_2, X\psi_1)\}.$$

Interference persists.

10. Locality, Microcausality, and Entanglement

Until now, the discussion has tried, as far as possible, to free quantum mechanics from reference to measurement but quantum mechanics also has a problem with locality. First of all it is worth remembering that classical mechanics also had a locality problem and it was tolerated for two centuries. Newton's theory of gravity was published in 1687 and structurally similar Coulomb's law of electric charge attraction and repulsion was published in 1785. In both cases any local change in mass or charge, whether magnitude or position, had an instantaneous effect everywhere. There was no mechanism in the physics for propagation of the effect of a change on mass (gravity) or charge (electricity). A local change had an effect across the universe. The solution to this was found first for electricity in combination with magnetism. Faraday proposed the existence of a field. The mathematical formulation of this concept by Maxwell led to the classical electromagnetic theory (1865) and provided a propagation mechanism.

The success of electromagnetic theory brought to the fore the locality problem with classical dynamics. The space and time translation invariance in classical Newtonian dynamics did not follow the same transformation rules as in the electromagnetic theory and there was still no mechanism proposed for the propagation of gravitational effect. As is well known, Einstein, inspired by Maxwell, solved both anomalies with first the special theory , which indicated that causal effects can not propagate faster than the speed of light in empty space. Then the general theory of relativity provided a propagation mechanism by making space-time an active physical entity supporting gravitational interaction and waves.

By the time the general theory of relativity was formulated it was evident that classical physical theory had a further deep problem; it could not explain atomic and other micro phenomena. To tackle this problem solutions were found for specific situations. Max Plank introduce his constant $\hbar$ to resolve the problem of the ultraviolet singularity in the black body radiation spectrum by proposing energy quantisation (Bohm 1951). This same constant came to be fundamental in explaining atomic energy levels, the photoelectric effect role and more generally the quantisation of action.

Quantum theory took shape is the 1920's with the rival formations, by Heisenberg and Schrödinger (with much help from others), shown to be formally equivalent in terms of experimental consequences. The space time translation symmetry of special relativity was also built into an equation for the electron proposed by Dirac that in turn implied the existence of anti-matter. But a fully relativistic quantum mechanics remains a research topic (Haag 1996).

To combine particle theory with electromagnetism quantum electrodynamics was developed. This theory was remarkably successful in its empirical confirmation but relied on some dubious mathematical manipulations. To deal with this the mathematical foundations of quantum field theory were examined. It is at this point that the first type of locality that we are going to consider appears in quantum theory in mathematically precise form (Haag 1996).

10.1. Causal Locality

A basic characteristic of physics in the context of special relativity and general relativity is that causal influences on a Lorentzian manifold spacetime propagate in timelike or light-like directions but not space-like. Space-like separated points in space-time lie outside each other's light cone, which means that no influence can pass from one to the other.

A further way of considering causality is that influences only propagate into the future in time-like and light-like directions, but this is not simple to dealt with in either classical special relativity or standard quantum mechanics because of their time reversal symmetry. One approach would be to treat irreversible processes through coarse-grained entropy in statistical physics. But this seems more like a mathematical trick that treats irreversibility as due to a lack of access to the detailed microscopic reversable dynamics. That is, irreversability is an illusion. A more fundamental approach is to develop a new physics, as is being attempted by Fröhlich (2021) and here in this book.

To return to Einstein causality, any two space-like-separated regions of spacetime should behave like independent subsystems. This causal locality is, with a slightly stronger technical definition, Einstein causality. This concept of locality when adopted in relativistic quantum theory (algebraic theory) implies that space-like separated local self-adjoint operators commute. This is sometimes known as microcausality. Microcausality is causal locality at the atomic level and below.

In quantum theory, where operators represent physical quantities, the microcausality condition requires that any operators commute that pertain to two points of space-time if these points cannot be linked by light propagation. This commutation means, as in standard quantum mechanics, that the physical quantities to which these operators correspond can be precisely determined locally, independently, and simultaneously. However, the operators in standard quantum theories and the non-relativistic alternatives discussed so far in this book don't have a natural definition of an operator that is local in space-time. For example, the position operator is not at any point in space. The points in space are held as potential values in the quantum state that is represented mathematically by the density matrix. How these potential values become actual is dealt with in standard quantum mechanics by the Born criterion, which is, however, tied to measurement situations. To remove this dependence on measurement situations is a major aim of this book and we will see that measurement only need be invoked when discussing how various form of locality and non-locality are known about.

As the introduction of classical fields cured Newtonian dynamics of action at a distance and eventually modified then replaced Newton's theory of gravity with General Relativity, the development of quantum field theory could perhaps cure standard quantum mechanics of its causal locality problem. Local quantum theory as set out in the book by Haag (1996) tackles this challenge. The technical details involved will be simplified here.

Although dealing with these questions coherently within non-relativistic quantum theory is not mathematically valid, it is instructuive to explore specific examples with an emphasis on physical intuition rather than mathematical rigour. Following Fröhlich (2023), it is natural to consider the spin of the particle to be local to that particle. Therefore, the spin operators, whether represented by Pauli matrices or by projection operators that project states associated with some subsets of the spin spectrum, can be assigned unambiguously to one particle or another.

In a situation where two particles are prepared so that they propagated in opposite directions their local interactions with other entities will eventually be space-like separated. The spin operators of one commute with the spin operators of the other. The local interaction of one cannot then be influence by the local interaction of the other. This is a specific example of microcausality.

But what if the preparation of the two particles entangles their quantum states? This entanglement may persist over any subsequent separation, if the particle does not first undergo any interaction with other particles or fields.

We note that entanglement is a state property whereas microcausality is an operator property and proceed to a discussion of entanglement and its consequences in a developed version of the two-particle example.

10.2. Entanglement and Non-Locality

The two-particle example we have been discussing only needs the introduction of a local spin measurement mechanism for each particle for it to become the version of the Einstein Podolsky, Rosen thought experiment formulated by Bohm (1951) . This post will follow Bohm's mathematical treatment closely but will avoid as far as possible invoking the results of measurements. Bohm's discussion follows the Copenhagen interpretation but also uses the concept of potentiality as developed by Heisenberg (1958).

The system in this example consists of experimental setups (described below) for two atoms (1 and 2) with spin 1/2 (up/down or $\pm\hbar$). The z direction spin aspect of the state of the total system consists of four basic wavefunctions:

$$|a\rangle = |+, z, 1\rangle|+, z, 2\rangle$$

$$|b\rangle = |-, z, 1\rangle|-, z, 2\rangle$$

$$|c\rangle = |+, z, 1\rangle|-, z, 2\rangle$$

$$|d\rangle = |-, z, 1\rangle|+, z, 2\rangle$$

it will be shown below that although the choice of the z direction is convenient the results of the analysis do not depend on it.

If the total system is prepared in a zero-spin state, then it can be represented by the linear combination:

$$|0\rangle = |c\rangle - |d\rangle \tag{1}$$

This correlation of the spin states of the particles is an example of quantum entanglement.

Each particle also has associated with its spin state a wavefunction that describes its motion and position. Theses space wavefunctions will not be shown explicitly here but are important conceptually because the particles aways move away from each other. The description of the thought experiment is completed by each particle undergoing a Stern-Gerlach experimental interaction at space-like separated regions of space-time, as shown below.

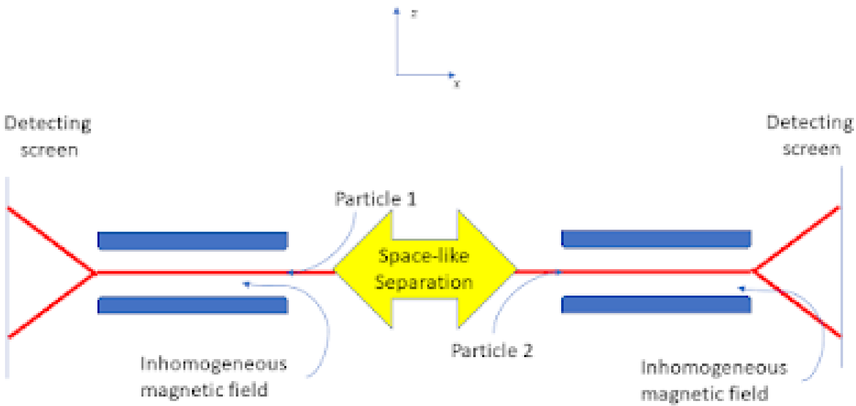


Figure 10.1.: Two space-like separated Stern-Gerlach interaction situations.

The detecting screen is a part of an experimental setup that is need for confirming the predictions of the theory but not the physics of the effects. Here we are primarily concerned with the interaction of the particles with the magnetic field $\mathfrak{H}$. The component of the system

Hamiltonian for the interaction of the particle spin with the magnetic field is, from Bohm (1951),

$$\mathcal{H} = \mu(\mathfrak{H}_0 + z_1\mathfrak{H}_0')\sigma_{1,z} + \mu(\mathfrak{H}_0 + z_2\mathfrak{H}_0')\sigma_{2,z}$$

where $\mu = \frac{e\hbar}{2mc}$, $\mathfrak{H}_0 = (\mathfrak{H}_z)_{z=0}$ and $\mathfrak{H}_0' = (\frac{\partial\mathfrak{H}_z}{\partial z})_{z=0}$. m and e are the electron mass and charge. c is the speed of light in vacuum. We also assume the magnetic fields have the same strength and spatial form in both regions but this not essential. It is also assumed that each particle interacts with its own local magnetic field at the same time. This is not a limiting assumption, but it is essential to assume that the time of the interaction is short enough for the local space-time regions to remain space-like separated.

The Schrödinger equation can now be solved for a wavefunction of the form

$$|\psi\rangle = f_c|c\rangle + f_d|d\rangle$$

with initial conditions given by equation (1). The result is, once the particles have passed through the region with non-zero magnetic field strength:

$$f_c = \frac{1}{\sqrt{2}}e^{-i\frac{\mu\mathfrak{H}_0'}{\hbar}(z_1-z_2)\Delta t}$$

and

$$f_d = -\frac{1}{\sqrt{2}}e^{i\frac{\mu\mathfrak{H}_0'}{\hbar}(z_1-z_2)\Delta t}$$

Where Δt is the time it takes for the particles to pass through the magnetic field.

Inserting the above results into the equation for $|\psi\rangle$ gives the post interaction wavefunction:

$$|\psi\rangle = \frac{1}{\sqrt{2}} e^{-i\frac{\mu \mathfrak{H}'_0}{\hbar}(z_1 - z_2)\Delta t}|c\rangle - \frac{1}{\sqrt{2}} e^{i\frac{\mu \mathfrak{H}'_0}{\hbar}(z_1 - z_2)\Delta t}|d\rangle$$

Therefore, for a system prepared with total spin zero undergoing local interactions in space-like separated regions, as shown in the figure, there is equal probability for each particle to be deflected either up or down. However, because of the correlation when one is deflected up the other is deflected down.

This may seem unsurprising because the total spin is prepared to be zero. No more surprising than taking a green card and a blue card, putting them in identical envelopes, shuffling them and then giving one to a friend to take far away. Opening the envelope you kept and seeing a green card means that the distant envelope contains a blue card. This, clearly, is not a non-local influence.

However, this is not the end of the story. As mentioned above, there is nothing special about the z direction of spin. The same analysis can be carried with x direction states, as follows:

$$\begin{aligned}
|a'\rangle &= |+, x, 1\rangle|+, x, 2\rangle \\
|b'\rangle &= |-, x, 1\rangle|-, x, 2\rangle \\
|c'\rangle &= |+, x, 1\rangle|-, x, 2\rangle \\
|d'\rangle &= |-, x, 1\rangle|+, x, 2\rangle
\end{aligned}$$

and again, the zero total spin state is:

$$|0'\rangle = |c'\rangle - |d'\rangle \tag{2}$$

Using the standard spin state relations (valid for both particles one and two, by introducing the appropriate tags (1 or 2), see Bohm (1951):

$$|+, x\rangle = \frac{1}{\sqrt{2}}(|+, z\rangle + |-, z\rangle) \quad \text{and} \quad |-, x\rangle = \frac{1}{\sqrt{2}}(|+, z\rangle - |-, z\rangle)$$

Inserting into equation (2), with some algebra, it can be shown that:

$$|0'\rangle = |0\rangle$$

Therefore, if the Stern-Gerlach setup is rotated to measure the x component of spin, exactly the same analysis can be carried out as for the z component giving the same anti-correlation effect. It must be stressed that we are discussing physical effects and not the results of experiments or the experimenter's knowledge of events at this point.

In general, there is no reason for the two space-like separated setups to be chosen in the same direction. If the choice is effectively random then when the direction of interaction does not coincide there will be no correlation between the outcomes but if they happen to be in the same direction, then there will be the $\pm$ anti-correlation. Locally the spin operators for the x, y and z do not commute. Their values are potential rather than actual and remain non-actual after the interactions. The situation is not like the classical coloured cards in envelopes example. There is no direction of spin fixed by the initial state preparation. Indeed, that would be inconsistent with a total spin zero state preparation. What the interaction does is chose a σ-algebra from the local σ-complex but the spin state of the system remains entangled.

As far as local effects are concerned, each particle behaves as expected for a spin 1/2 particle. This is causal locality. It is only if someone gets access to a sequence of measurements from both regions (here is the only place where detection enters this description of the physics of this situation) that the anti-correlation effect can be confirmed.

The effect depends on the preparation of the initial total system state. There is persistent correlation across any distance just as in the green and blue card example, but it is mysterious because the initial state does not hold an actual value of each spin component for each particle, unlike the actuality green and blue card example. There is no way for the one particle to be influenced by the choice of direction of measurement at the region where the other particle is, but a correlation of potentiality persists that depends on the details of the total quantum state.

It is perhaps too early to simply accept that there are non-causal, non-classical correlations of potentialities between two space-like separated regions. That would be a quantum generalisation of the blue and green card example. What the theory does predict is that the effect due to entanglement is not just epistemic but physical once potentiality is accepted as an aspect of the ontology.

11. Bell's Theorem

Bell's theorem has an untidy history. It is even debatable whether it merits being called a theorem. However, it is extremely important, as we show below, and we endeavour to shape it more clearly into the form of a theorem. This chapter is an adaptation making extensive use of the presentation by Myrvold et al. (2020), which summarises the original work of Bell (1964), John F. Clauser et al. (1969), Bell (1971), Clauser and Horne (1974), Aspect (1983) and Mermin (1986). The presentation here is intended to be self-contained, but the interested reader may consult the original sources.

We present the proof in two parts, as in Myrvold et al. (2020):

1. formulate a measurement framework and then derive a version of the inequality;
2. demonstrate that quantum mechanics predicts violations of the inequality.

This will show a conflict between a statistical analysis of possible experiments that assumes a locality concept that we call Bell locality (BL), and textbook quantum mechanics (QM). Further discussion is then devoted to the challenge of deciding experimentally between BL and QM.

The framework is a generalisation of the one used in Bell (1964). Bell's 1964 derivation assumed an experiment involving perfect anticorrelation of the results of aligned Stern-Gerlach experiments on a pair of entangled spin-$\frac{1}{2}$ particles (the section titled 10.2). This is not realistic, and experimental tests need this assumption to be relaxed because of the practical difficulties of maintaining quantum coherence and guaranteeing spacelike separation of the measurements. An experimental setup with polarised photons is more appropriate. Papers that took the steps from Bell's 1964 demonstration to an experimentally testable form include John F. Clauser et al. (1969), Bell (1971), Clauser and Horne (1974), Aspect (1983) and Mermin (1986).

The initial part of the proof will use a framework describing the general experimental setup and the assumptions sufficient to ensure satisfaction of the Bell inequalities. The second part will show that quantum mechanics predicts violations of the inequalities. The conceptual framework postulates an ensemble of configurations of two correlated subsystems, with each entangled system labelled 1 and 2. Each entangled system is characterised by a configuration state z that contains the entirety of its properties, including the properties of the individual sub-systems and any correlations between them. The state z is taken to be a complete specification of the properties of the configuration at the moment of generation and future evolution, other than future experimental decisions.

The state space Z is the totality of all states z, and we assume that the subsets of Z are measurable subsets, forming a measurable space to which probabilistic considerations may be applied. The probability distribution over Z is denoted by ρ. In addition, it is assumed that the mode of generation of the pairs establishes a probability distribution π that is independent of the histories of each of the two systems after they separate. Temporal evolution of the properties of the two systems after separation is possible, but it is assumed that the state z sets the probabilities for subsequent events (including any temporal evolution), and therefore the probabilities for outcomes of experiments to be performed on the sub-systems separately.

Different experiments may be performed on each subsystem. We use $a \in \mathcal{A}$ as variables ranging over possible experiment settings $\mathcal{A}$ on subsystem 1, and $b \in \mathcal{B}$ as variables ranging over experiment settings $\mathcal{B}$ on subsystem 2. It is not assumed that these parameters capture the complete state of the experimental apparatus, which might have a range of microstates corresponding to each experimental setting; consider them as labels. The following assumption concerns experimental procedures and the independence of the preparation of the entangled system from the choice of experiments to be performed.

> **Measurement Independence Assumption**: the preparation probability distribution ρ is independent of the processes by which the choice of experiments to be performed is made.

The results of an experiment with setting a on subsystem 1 are labelled by a real parameter s, taking values from a discrete set S_a of real numbers in the interval $[-1, 1]$. Equivalently, the result of an experiment on subsystem 2 is labelled by a parameter τ, taking values from a discrete set T_b in $[-1, 1]$. The sets of possible outcomes depend on the choice of experimental settings. The restriction of the outcome labels to the interval $[-1, 1]$ is a choice made only for convenience. It is essentially a measurement calibration, and it is important that the calibration be consistent across all experiments. The restriction to discrete outcome sets is not essential, but it is convenient for the proof of the Bell inequalities.

We assume that, for each pair of settings a, b, and every $z \in Z$, there is a probability function $\pi_{a,b}(s, \tau \mid z)$, which obviously sums to unity over all $s \in S_a$ and $\tau \in T_b$. We can use these probabilities to define marginal probabilities:

$$\pi^1_{a,b}(s \mid z) \equiv \sum_{\tau \in T_b} \pi_{a,b}(s, \tau \mid z), \tag{11.1}$$

$$\pi^2_{a,b}(\tau \mid z) \equiv \sum_{s \in S_a} \pi_{a,b}(s, \tau \mid z), \tag{11.2}$$

Define $A_z(a,b)$ and $B_z(a,b)$ as the expectation values, for complete state z, of the outcomes of experiments on subsystems 1 and 2, respectively, when the settings are a, b.

$$A_z(a,b) \equiv \sum_{s \in S_a} s\, \pi^1_{a,b}(s \mid z) \tag{11.3}$$

$$B_z(a,b) \equiv \sum_{\tau \in T_b} \tau\, \pi^2_{a,b}(\tau \mid z) \tag{11.4}$$

Now define the expectation value of the product $s \cdot \tau$ of outcomes:

$$E_z(a,b) \equiv \sum_{s \in S_a,\, \tau \in T_b} (s \cdot \tau)\, \pi_{a,b}(s, \tau \mid z). \tag{11.5}$$

Bell-type inequalities follow from considerations of locality and causality that have been called *factorisability* or *Bell locality* (BL). This is formulated as the following factorisation condition on the response probability distribution of outcomes, conditional on the complete state z of the configuration.

(BL) For any a, b, z, there exist probability functions $\pi^1_a(s \mid z)$ and $\pi^2_b(\tau \mid z)$ such that

$$\pi_{a,b}(s, \tau \mid z) = \pi^1_a(s \mid z)\, \pi^2_b(\tau \mid z).$$

Note that, even though the probability factors into independent subsystem-1 and subsystem-2 probabilities conditional on the total system state z, the outcome in subsystem 1 does not depend on the setting of the experiment in subsystem 2, and vice versa. The condition (BL) is a consequence of the conjunction of two conditions, which have been called *parameter independence* and *outcome independence* (see the section titled 11.1, below).

If the factorisability condition (BL) is satisfied, then $A_z(a,b) = A_z(a)$ and $B_z(a,b) = B_z(b)$, so

$$E_z(a,b) = A_z(a)B_z(b). \tag{11.6}$$

The above definitions are valid for experiments with any number of discrete outcomes. The simplest case is that in which each experiment has only two distinct outcomes, which we may label by ± 1.

To derive the inequalities, at least two separate experimental settings per subsystem must be considered, a, a', b, b'. Now consider the quantities

$$S_z(a,a',b,b') = |E_z(a,b) + E_z(a,b')| + |E_z(a',b) - E_z(a',b')|. \tag{11.7}$$

We now introduce a quantity related to $S_z(a,a',b,b')$, namely S_ρ, where the expectation values in Equation 11.7 are averaged over ρ, the distribution over states $z \in Z$.

$$S_\rho(a,a',b,b') = |\langle E_z(a,b)\rangle_\rho + \langle E_z(a,b')\rangle_\rho| + |\langle E_z(a',b)\rangle_\rho - \langle E_z(a',b')\rangle_\rho|. \tag{11.8}$$

Note that, in taking the expectation values of the quantities appearing on the right-hand side of this definition with respect to the same distribution ρ, we are invoking the Measurement Independence Assumption. Under this assumption and the BL condition, we shall prove the following:

Bell's theorem — part 1: For $S_\rho(a,a',b,b')$ given by Equation 11.8, the following inequality holds:

$$S_\rho(a,a',b,b') \leq 2. \tag{BT1}$$

This is due to J. F. Clauser et al. (1969) and is known as the *CHSH (Clauser–Horne–Shimony–Holt) inequality.*

Proof

From Equation 11.7, we have

$$\langle S_z(a,a',b,b')\rangle_\rho = \langle |E_z(a,b)+E_z(a,b')|\rangle_\rho + \langle |E_z(a',b)-E_z(a',b')|\rangle_\rho.$$

Since the absolute value of the average of any random variable cannot be greater than the average of its absolute value, it is clear that

$$S_\rho(a,a',b,b') \leq \langle S_z(a,a',b,b')\rangle_\rho. \tag{11.9}$$

If BL is satisfied, then, using Equation 11.5 and the fact that $A_z(a)$ and $A_z(a')$ lie in the interval $[-1,1]$,

$$S_z(a,a',b,b') \tag{11.10}$$

$$= |A_z(a)(B_z(b)+B_z(b'))| + |A_z(a')(B_z(b)-B_z(b'))|$$

$$\leq |B_z(b)+B_z(b')| + |B_z(b)-B_z(b')|.$$

It is easy to check that

$$|B_z(b)+B_z(b')| + |B_z(b)-B_z(b')| = 2\max(|B_z(b)|,\, |B_z(b')|).$$

Since $B_z(b)$ and $B_z(b')$ also lie in the interval $[-1,1]$, from the previous step we conclude that, for every z,

$$S_z(a, a', b, b') \leq 2. \tag{11.11}$$

Since this bound holds for every value of z, it must also hold for the expectation value of S_z.

$$\langle S_z(a, a', b, b') \rangle_\rho \leq 2. \tag{11.12}$$

This, together with Equation 11.9, yields the CHSH inequality and completes the proof of BT1. □

The final step of the proof of our Bell theorem is to show that a system described by a quantum mechanical state violates BT1, the CHSH inequality.

Bell's theorem — part 2: There is a quantum mechanical state and a set of experiments for which the CHSH inequality is violated.

$$\langle S_z(a, a', b, b') \rangle_\rho \nleq 2. \tag{BT2}$$

The set of experiments used to test the CHSH inequality will be referred to as CHSH experiments. Here, we consider one example involving polarisation-entangled photons, Figure 11.1.

Consider a pair of photons 1 and 2 propagating in the ζ-direction. Let $|\chi\rangle$ and $|\upsilon\rangle$ represent states in which photon j ($j = 1, 2$) is linearly polarised in the χ- and υ-directions, respectively. Neglecting the part of the state that describes movement in the ζ-direction, consider the following part of the state vector that describes the polarisation of the photons:

$$|\Phi\rangle = \frac{1}{\sqrt{2}} \left(|\chi\rangle_1 |\chi\rangle_2 + |\upsilon\rangle_1 |\upsilon\rangle_2 \right), \tag{11.13}$$

which is invariant under rotation of the χ and υ axes in the plane perpendicular to ζ. The total quantum state of the pair of photons 1 and 2 is invariant under exchange of the two photons, as required by the fact that photons are integral-spin particles. Suppose now that photons 1 and 2 impinge respectively on the faces of birefringent crystal polarisation analyzers I and II, with the entrance face of each analyzer perpendicular to ζ. Each analyzer has the property of separating light incident upon its face into two outgoing non-parallel rays, the *ordinary ray* and the *extraordinary ray*. The transmission axis of the analyzer is a direction with the property that a photon polarised along it will emerge in the ordinary ray (with certainty if the crystals are assumed to be ideal), while a photon polarised in a direction perpendicular to ζ and to the transmission axis will emerge in the extraordinary ray. See Figure 11.1 for a schematic of the experimental setup.

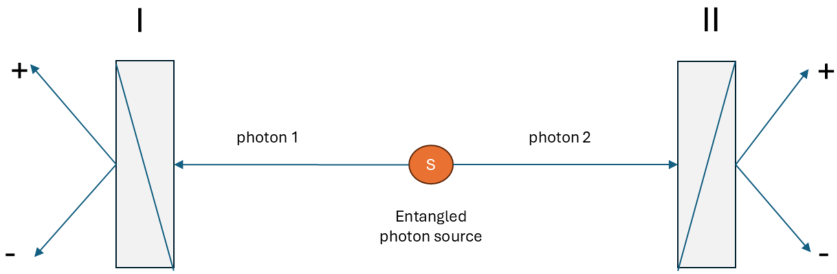


Figure 11.1.: Experimental Set-up

Photon pairs are emitted from the source, each pair quantum mechanically described by $|\Phi\rangle$ of Equation 11.13. I and II are polarisation analyzers, with outcomes $s = 1$ and $\tau = 1$ designating emergence in the ordinary ray, while $s = -1$ and $\tau = -1$ designate emergence in the extraordinary ray. The crystals are also idealised by assuming that no incident photon is absorbed.

The expectation value, in state $|\Phi\rangle$, of the product of s and τ, is

$$E_{\Phi}(\mathbf{a},\mathbf{b}) = \cos^2(\theta_{\mathbf{ab}}) - \sin^2(\theta_{\mathbf{ab}}) \tag{11.14}$$

$$= \cos(2\theta_{\mathbf{ab}}).$$

The details are calculated in the section titled 11.2, below.

Here $\theta_{\mathbf{ab}} = \alpha - \beta$ is the angle between the transmission axes of the two analyzers. The expectation value $E_{\Phi}(\mathbf{a},\mathbf{b})$ is a correlation function, and it is a function of the angle between the transmission axes of the two analyzers. The correlation function is maximal when the transmission axes are aligned, and minimal when they are perpendicular.

An important insight of Bell's was that it is important to consider the less-than-perfect correlations obtained when the device axes are not aligned. Bell's theorem shows that no model satisfying the condition BL can reproduce the quantum correlations for all angles. This can be seen by considering the quantum predictions (Equation 11.13, Equation 11.14) and plugging them into the expression for S. For the case of polarisation-entangled photons, we have

$$S_{\Phi}(\phi) = |2\cos(2\phi)| + |\cos(2\phi) - \cos(6\phi)|. \tag{11.15}$$

This takes on its maximum at $\phi = \pi/8$.

$$S_{\Phi}(\pi/8) = 2\sqrt{2} \approx 2.828. \tag{11.16}$$

11.1. Locality and causality assumptions

The key condition giving rise to the Bell inequality (BT1) is the factorisability condition (BL). This condition is motivated by considerations of locality and causality. It is also a purely mathematical relationship between probabilities. For the proof, it is only necessary to find one system described by a quantum state and a set of experiments for which the CHSH inequality is violated. However, for experimental tests it is important to understand the physical assumptions that motivate the condition (BL) and ensure that these are met. In this section we discuss these assumptions.

11.1.1. Bell's Principle of Local Causality

Bell's Principle of Local Causality has evolved since the publication of Bell (1964). We will not be concerned with the details of this evolution, but rather with the core idea that the result of a measurement on one system be independent of the setting of a distant measurement, provided the two measurements were not coordinated in their joint past.

In his final contribution (Bell and Aspect (2004)), Bell derives the factorisability condition (BL) from the *Principle of Local Causality*. He bases his analysis on the following:

- relativity prohibits superluminal causation;
- the cause-effect relation is asymmetric in time, with causes temporally preceding their effects;
- in a relativistic spacetime, events at spacelike separation are taken to have no temporal order.

Bell's final statement of the Principle of Local Causality is as follows (see Figure 11.2):

> **Principle of Local Causality** (PLC): A theory will be said to be locally causal if the probabilities attached to values of local beables in a spacetime region 1 are unaltered by specification of values of local beables in a spacelike separated region 2, when what happens in the backward light cone of 1 is already sufficiently specified, for example by a full specification of local beables in a spacetime region 3.

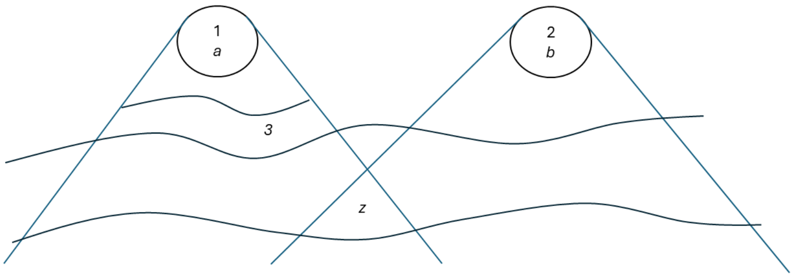


Figure 11.2.: Principle of Local Causality

Bell's term *beable* means any element of a physical theory that is taken to correspond to something physically real; therefore a beable is an entity in the Real Sphere as described in (Chapter 1), and *local beables* pertaining to a spacetime region are those contained within that region. The term was, of course, introduced to generalise beyond the term *observable* in textbook QM.

Note that the preparation state z is a complete specification of the properties of the entangled system at the moment of generation, and hence is in the backward light cone of both region 1 and region 2.

The formulation of the local causality condition is motivated by Bell (1976) with the remark,

> Now my intuitive notion of local causality is that events in **spacetime region 2** should not be "causes" of events in **spacelike separated region 1**, and vice versa. But this does not mean that the two sets of events should be uncorrelated, for they could have common causes in the overlap of their backward light cones.

The system state z is a common cause of correlations between the two subsystems. Correlations between two variables that are not in a cause-effect relation to each other can be explained by some common cause.

Bell then proceeds to formulate the condition of local causality, which is that a full specification of beables in the overlap of the backward light cones of the spacetime regions 1 and 2 should screen off correlations between them. Implicit in this is that correlations between two variables be susceptible to causal explanation, either via a causal connection between the variables, or via a common cause.

The Principle of Local Causality follows from the conjunction of three assumptions, all of which were explicitly formulated by Bell:

- (Temporal Asymmetry of Causality) Any cause of an event lies in its temporal past, and not in its temporal future.
- (Lorentz Invariance) The relation of temporal precedence is invariant under Lorentz boosts.
- Common cause.

11.1.2. Parameter independence and outcome independence

The condition BL of factorisability is the application, to the particular setup of CHSH experiments, of Bell's Principle of Local Causality. As we have seen, it can be thought of as the conjunction of the condition of causal locality and the common-cause principle. In this subsection we apply those conditions to the setup of CHSH experiments.

On the assumption that the experimental settings can be treated as free variables, whose choice of setting on one wing is made at spacelike separation from the experiment on the other, a dependence of the probability of the outcome of one experiment on the setting of the other would seem straightforwardly to be an instance of a nonlocal causal influence. The condition that this not occur can be formulated as follows.

(PI) For all parameter settings a, a', b, b', all z, and all $s \in S_a$ and $\tau \in T_b$,

$$\pi^1_{a,b}(s \mid z) = \pi^1_{a,b'}(s \mid z); \tag{11.17}$$

$$\pi^2_{a,b}(\tau \mid z) = \pi^2_{a',b}(\tau \mid z). \tag{11.18}$$

This is the condition that has come to be known as *parameter independence*, following Shimony (1986), Shimony (1990)).

For fixed values of the experimental settings, Bell's Principle of Local Causality entails that the outcomes of the experiments on the two systems be independent, conditional on the specification z of the complete state of the system at the source. This is the *outcome independence* condition.

(OI) For each pair of parameter settings a, b, and all z, and all $s \in S_a$ and $\tau \in T_b$,

$$\pi_{a,b}(s, \tau \mid z) = \pi^1_{a,b}(s \mid z)\, \pi^2_{a,b}(\tau \mid z). \tag{11.19}$$

A violation of PI would seem to permit superluminal signalling, but this requires an additional assumption: that the state of the system be controllable. It would not hold for a theory on which there are principled limitations on the ability of would-be signallers to control the states of the systems they are dealing with; an example of a theory of this sort is Bohmian mechanics, which violates PI but does not permit superluminal signalling. In some of the literature, PI has incorrectly been treated as equivalent to no-signalling.

11.1.3. Explicit experimental assumptions

11.1.3.1. Unique experimental outcomes

Of the potential outcomes of a given experiment, one and only one occurs, and hence it makes sense to speak of *the* outcome of an experiment. However, Everettian, or "many-worlds", interpretations state that all potential outcomes actually occur. Even in these interpretations, though, each individual world has a unique outcome.

11.1.3.2. Experimental settings as free parameters

It is assumed that the complete state z is always sampled from the same probability distribution ρ, no matter what choice of experiment is made, and that, for this reason, the subset of experiments corresponding to any given choice of settings is a fair sample of the distribution of z. This will be discussed in more detail in Chapter 12 .

11.1.3.3. Timing of experimental outcomes

In experiments, care should be taken that, for the two subsystems, the choice of experimental setting and the registration of results take place at spacelike separation. It is assumed that experiments have unique results. The question arises as to *when* the unique result emerges. It is typically assumed that the result is definite once a detector has been triggered or the result has been recorded in a computer memory.

This difficulty will remain unresolved until the quantum mechanical measurement problem is solved.

11.2. Appendix: quantum mechanical calculation of the CHSH inequality

We will calculate the joint measurement probabilities of two entangled photons, find their correlation function, and show how it perfectly violates the limits of classical physics.

11.2.1. 1. The initial entangled state

In an optical setup using spontaneous parametric down-conversion (SPDC), a pair of photons are generated and sent to measurement equipment A and B in a maximally entangled polarisation state described by the symmetric Bell state, Equation 11.13:

$$|\Phi\rangle = \frac{1}{\sqrt{2}}\left(|\chi\rangle_1 |\chi\rangle_2 + |\upsilon\rangle_1 |\upsilon\rangle_2\right)$$

Here, $|\chi\rangle$ and $|\upsilon\rangle$ represent horizontal and vertical linear polarisations. This state means that if settings a at A and b at B both measure in the χ or υ direction, they will always get identical results (either both χ or both υ).

11.2.2. 2. The measurement bases

Settings a and b can be independently rotated.

- the a setting of polariser I is the angle α;
- the b setting of polariser II is the angle β.

A polariser at angle θ measures whether a photon is polarised along that angle (+1 result) or perpendicular to it (−1 result). We can write these measurement states as:

$$|{+\theta}\rangle = \cos\theta\,|\chi\rangle + \sin\theta\,|\upsilon\rangle$$

$$|{-\theta}\rangle = -\sin\theta\,|\chi\rangle + \cos\theta\,|\upsilon\rangle$$

11.2.3. 3. Calculating joint probabilities

We require the probability that the settings a and b both detect a photon passing through their polarisers $(+1, +1)$. According to Born's rule, this is the squared magnitude of the inner product of the measurement state and the initial state:

$$\pi_{++}(\alpha, \beta) = |\langle +\alpha, +\beta \mid \Phi\rangle|^2$$

First, let us find the inner product:

$$\langle +\alpha, +\beta|\Phi\rangle = \frac{1}{\sqrt{2}}\left((\langle\chi|\cos\alpha + \langle\upsilon|\sin\alpha\right)_1 \otimes ((\langle\chi|\cos\beta + \langle\upsilon|\sin\beta)_2\,(|\chi_1, \chi_2\rangle + |$$

Because $\langle\chi|\chi\rangle = 1$ and $\langle\chi|\upsilon\rangle = 0$, this simplifies to:

$$\langle +\alpha, +\beta|\Phi\rangle = \frac{1}{\sqrt{2}}(\cos\alpha\cos\beta + \sin\alpha\sin\beta)$$

Using the trigonometric identity $\cos(\alpha-\beta) = \cos\alpha\cos\beta+\sin\alpha\sin\beta$,

$$\langle +\alpha, +\beta|\Phi\rangle = \frac{1}{\sqrt{2}}\cos(\alpha - \beta)$$

Squaring this gives the joint probability of both getting +1:

$$\pi_{++}(\alpha, \beta) = \frac{1}{2} \cos^2(\alpha - \beta)$$

By doing the same for the other combinations (for example, using $|-\alpha\rangle$ or $|-\beta\rangle$), we obtain the full set of joint probabilities:

- π_{++} (both pass): $\frac{1}{2}\cos^2(\alpha - \beta)$
- π_{--} (both blocked): $\frac{1}{2}\cos^2(\alpha - \beta)$
- π_{+-} (a passes, b blocked): $\frac{1}{2}\sin^2(\alpha - \beta)$
- π_{-+} (a blocked, b passes): $\frac{1}{2}\sin^2(\alpha - \beta)$

11.2.4. 4. The expectation value (correlation function)

The correlation function $E(\alpha, \beta)$ is the average product of their measurement outcomes. We weight the matching results ($+1 \times +1 = 1$ and $-1 \times -1 = 1$) positively, and the opposing results ($+1 \times -1 = -1$) negatively:

$$E(\alpha, \beta) = \pi_{++} + \pi_{--} - \pi_{+-} - \pi_{-+}$$

Substituting the probabilities just calculated:

$$E(\alpha, \beta) = \left(\frac{1}{2}\cos^2(\alpha - \beta) + \frac{1}{2}\cos^2(\alpha - \beta)\right) - \left(\frac{1}{2}\sin^2(\alpha - \beta) + \frac{1}{2}\sin^2(\alpha - \beta)\right)$$

$$E(\alpha, \beta) = \cos^2(\alpha - \beta) - \sin^2(\alpha - \beta)$$

Using the identity $\cos(2\theta) = \cos^2\theta - \sin^2\theta$, we arrive at the quantum correlation function:

$$E(\alpha, \beta) = \cos(2(\alpha - \beta))$$

This is exactly the result we stated in Equation 11.14, with $\alpha - \beta = \theta_{ab}$, confirming that the quantum mechanical predictions for the correlation function violate the CHSH inequality for certain choices of angles α and β.

12. Bell's Theorem, Locality and ontology

12.1. Primary Literature & Philosophical Texts

- **Albert, D. Z.** (1992). *Quantum Mechanics and Experience.* Cambridge, MA: Harvard University Press.
  - *Context:* This is David Albert's foundational text. He uses accessible analogies (such as color and hardness experiments) to break down the EPR paradox, hidden variables, and the philosophical consequences of Bell's Theorem.
- **Bell, J. S.** (1964). "On the Einstein Podolsky Rosen Paradox." *Physics Physique Fizika*, 1(3), 195–200.
  - *Context:* The original paper where John Stewart Bell mathematically proved that local hidden variable theories cannot reproduce all the predictions of quantum mechanics.
- **Bell, J. S.** (1987). *Speakable and Unspeakable in Quantum Mechanics: Collected Papers on Quantum Philosophy.* Cambridge: Cambridge University Press.
  - *Context:* A collection of Bell's papers on quantum philosophy, detailing his views on the "gross nonlocality of nature."

12.2. Historical & Theoretical Context

- **Einstein, A., Podolsky, B., & Rosen, N.** (1935). "Can Quantum-Mechanical Description of Physical Reality be Considered Complete?" *Physical Review*, 47(10), 777–780.
  - *Context:* The historic "EPR paper" that introduced the concepts of local realism and quantum entanglement, which Bell's Theorem later tested.
- **Myrvold, W., Genovese, M., & Shimony, A.** (2020). "Bell's Theorem." *The Stanford Encyclopedia of Philosophy.* Zalta, E. N. (Ed.).
  - *Context:* A peer-reviewed philosophical overview tracking how David Albert and other modern philosophers interpret statistical independence and nonlocality.

Clauser, J.F., 1976, "Experimental investigation of a polarization correlation anomaly," Physical Review Letters, 36: 1223–1226.

——, 2017, "Bell's theorem, Bell inequalities, and the 'probability normalization loophole'," in Bertlmann and Zeilinger (eds.), 451–484.

Clauser, J.F. and Horne, M.A., 1974, "Experimental consequences of objective local theories," Physical Review D, 10: 526–535.

Clauser, J.F., Horne, M.A., Shimony, A., and Holt, R.A., 1969, "Proposed experiment to test local hidden-variable theories," Physical Review Letters, 23: 880–884.

Clauser, J.F. and Shimony, A., 1978, "Bell's theorem: experimental tests and implications," Reports on Progress in Physics, 41: 1881–1927

13. Comparison of quantum physical ontologies

In discussing ontology in physics both the real and ideal spheres, as introduced by Nicolai Hartmann, play a role. The pure mathematics, applied to construct theory, is in the ideal sphere and the physical entities are in the real sphere. For example, self-adjoint operators exist in quantum theory and they represent observables (properties of things) that exist (this is claimed as a truth) in the physical world. However, it must always be kept in mind that truth claims are fallible.

The ontologies discussed in this chapter are proposals for what exists and does not exist in real sphere interpretations of several formulations of quantum mechanics. These proposals are to be contrasted with the aspect of critical ontology that provides the structures for the investigation. A guiding statement, in the spirit of critical ontology, for this investigation, is that by Maudlin (2019):

> A physical theory should clearly and forthrightly address two fundamental questions: what there is, and what it does. This section explores a range of options on 'what there is' in theories that seek to produce identical or very similar empirical results. If the empirical results were not in agreement, then those theories that did not agree with experiment would be eliminated as not being true, although they could still be of theoretical interest. It is because theories with radically different existence claims can have the same empirical consequences that ontology investigation is interesting and relevant.

In what follows familiarity with standard quantum physics, as taught to physics undergraduates, is assumed. If a reminder is needed then Jennan Ismael's contribution to The Stanford Encyclopedia of Philosophy (Ismael 2021) may be helpful.

In the multi-layer model of the real, the discussion that follows deals with physical entities in the inorganic layer. The theories are creations than reside in the objective part of the spiritual layer. There are theories that involve the psychic layer but they will not be considered in this chapter. The psychic layer can only come into play if there is someone there to have perceptions and thoughts.

13.1. Local beables

John Bell (Bell 2004a), uses his concept of local beables (Bell 2004b) to propose that theses random events are what give effect to local beables. Bell's local beables are a significant development as it was a move away from focusing on what is observed to what exists. The locality of the beable, however, is still tied to the observation that particles such as electrons or photons appear locally in experimental situations. *Beable* also implies a process of coming into being. So, a particle may exist in an interaction or detection but what of the existence of the particle as such? The concept of local beables in general leaves such questions unanswered. The application to GRW will now be considered.

13.2. Wavefunction ontology

In standard quantum physics the wavefunction permits a range of possible events. Of the possible events one happens due to a measurement process. But what happens - what is the physics - when there is no measurement? Ghirardi, Rimini and Weber (Ghirardi et al. 1986), (GRW) introduced a supplementary physical effect to standard quantum mechanics that randomly causes the collapse of the wavefunction, creating a way of obtaining macroscopic events from microscopic quantum systems, as will be explained below.

The GRW mechanism is as follows. Let the initial the wave function be

$$\psi(t, q_1, q_2, \dots, q_N)$$

t is time and $q_1, q_2, \dots, q_N$ are position coordinates. The probability per unit time for a GRW spontaneous collapse event is N/τ, where τ is a new constant of nature, chosen to be in the order of 10^{14} seconds. The collapsed wavefunction is

$$\psi' = \frac{j(x - q_n)\psi(t, \dots)}{R_n(x)},$$

where q_n is chosen at random (probability $= 1/N$)

The definition of the weighting function j introduces at least one further constant of nature. This is tuned to be in the region of 10^{-5}cm to preserve the observed microscopic effects while generating the commonly observed macroscopic world. There is nothing but the wavefunction and the collapse is part of its dynamics.

However, the GRW collapse events happen to the wavefunction, not something else. The wavefunction is then well localized in ordinary space at, at most, a mesoscopic scale. Each is centred on a particular space-time point (x, t). So, Bell proposes these events as providing the local beables of the theory. This would make the beables appearances of the object (Sosein) not a representation of the object itself (Dasein). That being so, these local beables are sparse events in a system of particles and would be very rare for a small number of particles. That is, in the time scales typical of experiment nothing would happen outside the detector. Entanglement provides the mechanism for the commonly experienced existence of macroscopic bodies such as detectors. Note that the spontaneous event is a reduction of the wavefunction. This leaves open the relationship to the charge, mass and spin carrying particle. In addition, if this spontaneously reduced wavefunction is what exists then its relation to the actual particle still needs explanation. Implicitly there is a return to Born probability mechanism but without clarity on this the ontology is incomplete.

13.2.1. Distributed charge and mass

An alternative wavefunction ontology, but still within GRW theory, proposes that the particle has only a "fuzzy" position, with "more of it" located in places that correspond to its location in configurations which are assigned high amplitudes by the wave-function. This suggests a picture in which the particle is "smeared out" in space, and the effect of the GRW hit is to concentrate most of the smear within 10^{-5}cm of a particular location. In this ontology the charge and mass of an electron would presumably be distributed across the support of the wavefunction. This theory does propose that a particle exists as such and is extended in space. This therefore provides a more complete ontology.

13.2.2. The multiverse

A third set of theories are a development of Everett's relative state formulation of quantum mechanics (Everett 1957), that are commonly known as many worlds or multiverse interpretations. In this case just the evolving wavefunction features as the candidate entity. The wave function describes all the possibilities for the system and all possibilities happen. This gives rise to many (an infinity of) physical worlds. Wallace (2012) provides a comprehensive discussion of this set of theories.

Everett's criterion for the real existence of a branch is: If the wavefunction ψ of a system is a superposition $a\phi + b\theta$ then the "branches" ϕ and θ exist, and in each branch, everything physically exists that would exist if that were the entire state of the system. However, there are many ways to represent a function as a linear representation of other functions. Some further structure is required to provide a path to candidates for entities. One approach is an appeal to decoherence, achieved through entanglement. Here the quantity capturing possible states of affairs is $|\psi|^2$ and entanglement can pick out specific decomposition that depends on the physics of the interaction bohmhiley2006

$$|\psi|^2 \to |a\phi|^2 + |b\theta|^2$$

That is, there is no interference term. However, the more detailed theory (Wallace 2012) works with approximate decoherence.

Examining critically the ontology of many worlds, it shares some of the issues with the GRW theory, especially in the Bell beables version. The smeared, or extended particle interpretation is not available to it but a complete theory must show how familiar phenomena emerge.

A macro-object is a pattern, and the existence of a pattern as a real thing depends on the usefulness–in particular the explanatory power and predictive reliability–of theories which admit that pattern in their ontology.

What makes a collection of electrons, protons, and neutrons a particle accelerator, rather than something else, has to do with how the microscopic parts are arranged, or structured. A significant part of that structure is the spatial arrangement of the microscopic parts through time. However, a quantum wavefunction contains no microscopic parts localized in space-time, so its behaviour cannot create a macroscopic particle accelerator in anything like the way the behaviour of localized electrons, neutrons, and protons can. The problem remains of how to populate familiar space-time with locally existing entities.

13.3. Bohmian ontology

Bohmian mechanics (Bohm and Hiley 2006), (Dürr and Teufel 2009), is a formulation of quantum mechanics that is constructed to give the same results as that of standard quantum mechanics but in which particles have continuous trajectories. For a discussion of the theory's ontology the mathematics of how this is accomplished is not relevant. The ontology includes:

- Particles with a well-defined position x(t) which varies continuously and is causally determined.
- A guiding field that is derived from the solution of Schrödinger's equation, so that it too changes continuously and is causally determined.
    - The particle is never separate from this guiding field that fundamentally affects it. There is no dynamics without this field.
    - The guiding field is not affected by the particle.

The Bohmian mechanics emphasises that the particle has a continuous trajectory and that these trajectories exist in the physical domain. The particle does not follow the laws of classical dynamics but is guided by a field that is directly derived from the solution to the Schrödinger equation. The claims made about what exists are not made within a wider ontology, but simply posit particles and guiding fIelds. It is tacitly assumed that to exist is to physically exist.

The distinction has been made between existing in the theory (the ideal sphere) and existing physically (the real sphere). There is a version Bohmian mechanics in which the guiding field is not in the real sphere of the ontology but in the ideal sphere. The guiding field is then a law governing the behaviour of the particle but not a physical entity like the particle. Allori (2013) presents this as a primitive ontology. Allori claims that a physical theory

> … will be about a given primitive ontology: entities living in three-dimensional space or in space-time. They are the fundamental building blocks of everything else, and their histories through time provide a picture of the world according to the theory (the scientific image).

These primitive variables describe what exists in the inorganic layer of the real sphere whereas non-primitive variables exist in the physical theories in the ideal sphere. In a physical situation where no person is active there is no mechanism for the ideal sphere to influence the real sphere. The primitive ontology must therefore include the guiding field so that Bohmian mechanics can operate as a fully-fledged physical theory. However, the wavefunction that gives rise to the guiding field is a function on configuration space, rather than the three-dimensional space of natural phenomena.

13.4. Conclusion

Critical ontology, as an investigative method, has now been applied to a range of proposals that claim the physical existence of particle or wavefunctions or both. Of these it is Bohmian mechanics, with particles and guiding fields as physical entities, that provides the most near to complete physical ontology.

A major deficiency in GRW theories is that the ontological status of the spontaneous reduction of the wavefunction. The mathematical description is of a stochastic process but it is not clear what this process is a property of. This will need a discussion of dispositional properties.

14. Ontological Framework: Nicolai Hartmann and Quantum Mechanics

The ontological framework for this blog is from Nicolai Hartmann's *new ontology programme* that was developed in a number of very substantial publications in the first half of the 20th century. In my interpretation of it, it is the open and pluralistic nature of Hartmann's new ontology that is most attractive. It can include categories of substance, process, organism, consciousness and more. Its layer structure allows the discussion of emergence of multiple entities and supervenience, whatever the ultimate ontological status of the layer model. The openness and pluralism is more important to me than the layered ontology. However, many chapters are specifically about quantum mechanics. Quantum mechanics was already making confirmed predictions when Hartmann was writing, and he provided some analysis, but, did noy get much further than identifying that quantum mechanics would provide some ontological challenges. In Hartmann's layered ontological model the foundation is the inorganic stratum, with categories such as space, time, process and substance.

At one stage it was thought that the challenges of quantum mechanics could have a solution in epistemology. However, the problem was that the physics is incomplete and so the problem was not just that of interpretation. The ontological status of the theory can not be fully addressed without repairing the physics. The major flaw in standard quantum mechanics, as discussed in other chapters, is that there is no physical explanation of events.

As a theory that includes a physical explanation of events, consider Ruth Kastner's relativistic transactional interpretation (RTI) (Kastner 2022) as a working theory that plugs the holes in the physics that were the main concern. For ontological analysis, the most significant aspect of RTI is the nature of the quantum substratum and its relationship with the empirical domain. Ruth Kastner does address metaphysical issues but here an indeprendent view will be presented. The analysis of the physics is for a universe that *is* independent of a knowing subject (Popper 1972).

14.1. Ontological status of quantum substratum

The quantum substratum is a domain of potentiality. It is not only not observable; it is not actual. The only indictor of its nature is the mathematical theory that relates it to actual events. This mathematical theory could be treated as a mere prediction and technology instrument. However I want to examine how the universe could be in itself. The potentiality substratum is real in that it is the source of physical events.

A potentiality interpretation of quantum mechanics is not an innovation of RTI. It was proposed by Heisenberg (Heisenberg 1958) and is championed by researchers who do not adopt RTI (Kastner et al. 2018). The three authors, it seems, could not agree with Ruth Kastner on adopting RTI as the best current explanation. The other authors are strongly influenced by A N Whitehead's process metaphysics (Whitehead 2010). On a first pass, a metaphysics with a fundamental event and process ontology, such as Whitehead's, seems consistent with RTI. however, there are many details and interpretations of Whitehead's metaphysics that I have yet to engage with.

In RTI events take place due to a transaction process in the substratum. Processes in the quantum substratum are disposed to create events that manifest in space-time and shape the future potential manifestation of events (objective quantum state reduction). The process can be summarised:

1. The physical system, described by relativistic quantum mechanics, evolves to a state that is favourable to a set of possible transactions.
2. The context of these possible transactions (at a given time in the substratum dynamics) and the state of the system defines a complete set of transition probabilities from a complex of incommensurable possible transitions.
3. One of the possible transitions takes place. This is, as far as the mathematical description guides, a physical stochastic process step.
4. The outcome is a space-time event and a change in the substratum state conditioned on that event.

Why should a quantum substratum of pure potentiality give rise to space-time events? In the mathematical structure describing the physics of the substratum time is a category and, space and momentum are among the potentialities. It would be conceivable that events would appear in a space-time block universe. Alternatively it is the relationship among the events that constitute space-time (Kastner 2022). Space-time is emergent and not ontological fundamental if the events are the elements of its fabric. Fundamentally there are no substances. For example, quantum states describing electrons represent entities that cannot exhibit all their potential attributes at the same times. The picture in which quantum mechanics is a modification of classical mechanics that can keep the substance-particle as the fundamental ontology is no longer defendable. Classically Matter is fundamental, now, in the quantum description adopted here, matter-like entities emerge in a dynamic network of event relationships. This is a profound ontological innovation even greater than the introduction of electromagnetic fields.

Whatever the final status of a layered ontology turns out to be, I think that it will be useful in examining how, in detail, a substratum of pure potentiality can give rise to a rich world of physics, biology, psychology, society and culture.

15. Ontology and the Quantum Object

15.1. Introduction to Physical Ontology

Quantum mechanics is part of physics and experimentation is an activity in physics. Therefore, the entities of interest in physics must be observable to some extent. They must make an appearance. That is, entities of interest must be **objects**; they appear in at least one **field of sense** (Chapter 2).

The ontological framework for the discussion will be that quantum objects are physical objects that have some properties that can appear in one of a collection of fields of sense. To move from ontology to physical theory, an explanatory structure is needed and, when possible, a mathematical formulation is adopted that describes the behavior of the quantum objects in a precise way.

15.2. The Problem of Contextuality

There are puzzling and paradoxical effects of quantum mechanics, be they the thought experiments of Einstein, Podolsky and Rosen (EPR) or Wigner's friend with Schrödinger's cat, or even double-slit experiment. These effects to the need to re-examine what is meant by a **physical property** of an object.

In quantum mechanics, it is known that all properties need a specific context in which to appear as quantified values. This is known to be the case in general from the work of **Kochen and Specker** (Kochen and Specker 1967), but has been a common understanding among physicists based on insights obtained from experiments such as that of **Stern and Gerlach** (Gerlach and Stern 1922).

15.3. Position, Momentum, and the Field of Sense

Before moving on to spin, consider the **Heisenberg Uncertainty Principle**, which is usually taken to mean that an electron, for example, cannot have a precise value of its position and its momentum at the same time.

> **Question:** Does this mean that it sometimes has position as a property and sometimes momentum but never both together?

That understanding of the theory would mean that these properties are of the context rather than the object of the experiment. According to the ontology of objects and their properties being developed here, a particle (such as an electron) always has the properties of position and momentum, represented by an operator, but these properties take values by appearing in distinct **Fields of Sense**. This means that the context in which the particle position takes a value is different from that in which its momentum takes a value.

15.4. Mathematical Formulation of Spin

We now consider a particle with spin 1/2 (an electron) situated and free to move in three dimensions. Following Faddeev and Yakubovskii (2000), the electron combines its spin Hilbert space ($\mathbb{C}^2$) with that associated with the electron's movement in three dimensions ($\mathbb{R}^3$).

The Hilbert space $\mathcal{H}$ for the electron's position and momentum is the space $L^2(\mathbb{R}^3)$ of square-integrable complex-valued functions. In mathematical notation, the state space with spin becomes the enlarged Hilbert space:

$$\mathcal{H}_S = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$$

This captures the intuition that the state function becomes a pair of complex-valued $L^2(\mathbb{R}^3)$ functions. For a pure state $\Psi(x)$ in $\mathcal{H}$, there are two orthogonal states in $\mathcal{H}_S$:

$$\Psi_\uparrow = \begin{pmatrix} \Psi(x) \\ 0 \end{pmatrix}, \; \Psi_\downarrow = \begin{pmatrix} 0 \\ \Psi(x) \end{pmatrix}$$

15.4.1. Spin Operators and Pauli Matrices

Focusing on the spin property $\hat{S}$, a self-adjoint operator S on $\mathbb{C}^2$ can be represented as the linear combination of the identity matrix I and the **Pauli matrices**:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Quantum physics is brought into this mathematics by introducing the Planck constant $\hbar$ and defining the self-adjoint spin-1/2 operators as:

$$S_j = \frac{\hbar}{2}\sigma_j, \quad j = 1, 2, 3 \qquad (1)$$

These follow the angular momentum commutation relations:

$$[S_1, S_2] = i\hbar S_3, \quad [S_2, S_3] = i\hbar S_1, \quad [S_3, S_1] = i\hbar S_2$$

15.5. Spin Eigenvalues and State Vectors

The operators S_j have eigenvalues $\pm\hbar/2$, which are the admissible values of the projections of the spin on some direction. The vectors

$$\Psi_\uparrow = \begin{pmatrix} \Psi(x) \\ 0 \end{pmatrix} \quad \text{and} \quad \Psi_\downarrow = \begin{pmatrix} 0 \\ \Psi(x) \end{pmatrix}$$

are eigenvectors of the operator S_3 with eigenvalues $+\hbar/2$ and $-\hbar/2$.

The eigenvectors of S_1 and S_2 are linear combinations of $\Psi_\uparrow$ and $\Psi_\downarrow$. Therefore, these vectors describe states with a definite value of the third projection of the spin. Similar constructions can be made for the other spin components.

15.6. Non-Commutativity and Interaction

As the components do not commute, they cannot simultaneously take precise values in all components. The spin will appear to interact with some other object, involving a magnetic interaction, to take a value of $+\hbar/2$ or $-\hbar/2$. Independent of the direction defined by the interaction, the possible values the spin property can take are $\pm\hbar/2$, with probabilities determined by the state of the electron. The probabilities of $\pm\hbar/2$ appearing sum to one.

The property values appear when the appropriate **Field of Sense** is available. However, in relation to this point it should be noted that the isolated system has properties represented by Hermitian operators, such as the spin-1/2 S_j, that have eigenvalues $\pm\frac{1}{2}\hbar$ even in the absence of any interaction term coupling it to another object.

The spin values are quantitative characteristics of the spin property, but the values will only actually **appear** once the appropriate Field of

Sense is available to provide a context for an interaction with another object.

The spin will appear to interact with some other object (involving a magnetic interaction) to take a value. A **Field of Sense** affords a context for the appearance of properties of the quantum object. The values only actually appear once the appropriate Field of Sense is available to provide a context for an interaction with another object.

16. The Combination of Quantum Objects

Quantum objects have been introduced in an earlier chapter. These objects, if not the most fundament entities, are quite basic to the constitution of the inorganic layer of reality. In contributing to the complexity of the real world these objects must combine.

Many calculations in quantum mechanics can be carried out by considering a single isolated system and its the operator representing the total energy of the system - the Hamiltonian. That is how the spectrum of the Hydrogen atom is calculated and how tunnelling is analysed. However, it is especially important in examining contextual issues to consider mathematical representation of the combination and interaction of objects. Having introduced the concept of the σ-complex for an isolated system, the σ-complex of two systems S_1 and S_2 will now be constructed.

As is standard in the von Neumann formulation (Neumann 1932), if the Hilbert spaces $\mathcal{H}_1$ associated with object S_1 and $\mathcal{H}_2$ associated with object S_2, the Hilbert space of the combined object $S_1 + S_2$ is the tensor product $\mathcal{H}_1 \otimes \mathcal{H}_2$.

Given the combined system $S_1 + S_2$ with the σ-complex $Q(\mathcal{H}_1) \oplus Q(\mathcal{H}_2)$, there is a unique Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$ such that $Q(\mathcal{H}_1) \oplus Q(\mathcal{H}_2) \cong Q(\mathcal{H}_1 \otimes \mathcal{H}_2)$, where $\cong$ means equivalence unique up to isomorphism. For the proof, in finite dimensional Hilbert spaces, see Kochen (2015).

The combination of two quantum objects gives a quantum object. This would build up a world consisting only of quantum objects. Is this the case or do the quantum charactertics weaken with many combinations? We will return to this question.

17. Quantum chance

In contrast to other formulations of quantum mechanics, the theory proposed here describes a world of objective chance events.

17.1. Quantum States

As presented in an earlier chapter, the properties of a quantum object are represented by self-adjoint operators, and these operators act on the elements of a Hilbert space. The concept of the state of a quantum system or object will now be presented. This concept of state is closely aligned with that in statistical mechanics, but the probabilistic quantum state is a property of an individual object rather than a statistical ensemble. As quantum states generalize classical probabilities, the discussion starts with a brief overview of probability measures.

17.1.1. Probability Measures

Classical probability (Kolmogorov 2018) provides a probability measure—a countably additive, $[0, 1]$-valued measure—that is a function:

$$p : B \to [0, 1]$$

with domain B, a σ-algebra of subsets of the state space Ω, such that $p(\mathbf{1}) = 1$, where $\mathbf{1}$ is the identity in the algebra B, and:

$$p\left(\bigcup_i e_i\right) = \sum_i p(e_i) \quad \text{for pair-wise disjoint elements } e_1, e_2, \ldots \text{ in } B.$$

In the case of a system $\mathbf{S}$, an object more complex than a single particle, the probability function p gives the probabilities over the σ-algebra of events for that system as a whole. A physical theory predicts the probabilities of outcomes of any possible situation given the complete initial state.

States of a system $\mathbf{S}$ in quantum mechanics are, in general, represented by density matrices acting on a Hilbert space, $\mathcal{H}_{\mathbf{S}}$. Density matrices are non-negative, trace-class operators, ρ, of trace 1, that is:

$$\rho = \rho^* \geq 0, \quad \text{with } \mathrm{tr}(\rho) = 1.$$

The expectation of a physical quantity, $\hat{X} \in \mathcal{O}_{\mathbf{S}}$ (the set of properties for the system), at time t in a state given by a density matrix ρ is defined by:

$$p(X(t)) := \mathrm{tr}(\rho X(t))$$

where $X(t)$ is the operator representing the physical property $\hat{X}(t)$. This can be extended to all bounded operators in $\mathcal{A}(\mathcal{H}_{\mathbf{S}})$, the $*$-algebra of all bounded operators acting on $\mathcal{H}_{\mathbf{S}}$.

From the mathematical formulation by Kochen (Kochen 2015), adapted here, there is a one-to-one correspondence between states p on the σ-complex $Q(\mathcal{H})$ and density operators. The use of the same symbol as for the classical event probability $p(x)$ for the quantum state is because this mathematical description of the quantum state is a generalization of the classical probability model from a single σ-algebra to a σ-complex. A density operator ρ defines a probability measure p on $Q(\mathcal{H})$, the σ-complex of projections. The converse, that a state p defines a unique density operator ρ on $\mathcal{H}$, follows from a theorem of Gleason (Gleason 1957). A state on the lattice of projections on $\mathcal{H}$ defines a unique density operator. A lattice consists of a partially ordered set in which every pair of elements has a unique least upper bound and a unique greatest lower bound.

17.1.2. Pure and Mixed States

In Quantum Mechanics, there is a one-to-one correspondence between the pure states of a system and rays of unit vectors ψ in $\mathcal{H}$, such that $p(X) = (\psi, X\psi)$. The pure states correspond to one-dimensional projections π_ψ (with ψ in the image of π_ψ) and $p(X) = \mathrm{tr}(\pi_\psi X) = (\psi, X\psi)$.

That even the pure states predict probabilities that are not 0 or 1 is what is to be expected of projections onto property values. A pure state simply predicts the probabilities of property values that can become actual in interactions. Mixed states are mixtures of the pure states, and in general, there is no unique decomposition of a mixed state into pure states.

17.2. Properties or Observables

An observable is nothing more than a property of a quantum object that has a set of possible values that become actual values in interaction with other objects and can appear in various contexts. It need not actually be "observed" in an experiment to do so. In the theory, each such property is represented by a self-adjoint operator that has a spectrum that corresponds to the set of possible property values.

This concept of property can, as argued above, be treated within the σ-complex formulation (Kochen 2015). Kochen shows that there is a one-to-one correspondence between observables ω such that:

$$\omega : B(\mathbb{R}) \to Q(\mathcal{H}),$$

where $B(\mathbb{R})$ is the σ-algebra of Borel sets (any set in a topological space that can be formed from open sets, or from closed sets, through the operations of countable union, countable intersection, and relative complement is a *Borel set*) generated by the open intervals of the real numbers, $\mathbb{R}$.

Given $\pi_\lambda = \omega((-\infty, \lambda])$, there is a Hermitian operator A such that:

$$A = \int \lambda \, d\pi_\lambda.$$

Conversely, given a Hermitian operator A on $\mathcal{H}$, the spectral decomposition above defines the representation of a property ω as the spectral measure $\omega(s) = \int_s d\pi_\lambda$, for $s \in B(\mathbb{R})$. This establishes the one-to-one correspondence and the equivalence of a key element of the reconstruction to that of standard quantum mechanics.

It follows that if $\omega : B(\mathbb{R}) \to Q(\Omega)$ is a representation of a property with corresponding self-adjoint operator X, then, for the state p with corresponding density operator ρ, the expectation of ω is:

$$\mathrm{Exp}_p(\omega) = \mathrm{tr}(X\rho).$$

The result shows the close connection between the spectrum of an operator and the σ-algebra of property values. For instance, for the case of a discrete operator X, the spectral decomposition $X = \sum a_i \pi_i$ defines the σ-algebra of property values generated by the set $\{\pi_i\}_i$. Conversely, given the σ-algebra of property values, the elements of the set $\{\pi_i\}_i$ allow the definition, for each sequence of real numbers a_i, of the Hermitian operator $\sum a_i \pi_i$.

17.3. Symmetries

In the standard formulation of quantum physics, symmetries appear as unitary transformations of Hilbert space vectors or Hermitian operators. Here they also appear naturally as symmetries of a σ-complex.

An automorphism of a σ-complex Q is a one-to-one transformation $\alpha : Q \to Q$ of Q onto Q such that for every σ-algebra B in Q and all $e, e_1, e_2, \ldots$ in every B:

$$\alpha(e^\perp) = \alpha(e)^\perp \quad \text{and} \quad \alpha\left(\sum_i e_i\right) = \sum_i \alpha(e_i),$$

where $\perp$ denotes the orthogonal complement.

There is a one-to-one correspondence between symmetries $\alpha : Q(\mathscr{H}) \to Q(\mathscr{H})$ and unitary operators u on $\mathscr{H}$ such that $\alpha(X) = uXu^{-1}$, for all $X \in Q(\mathscr{H})$. If a state p corresponds to the density operator ρ, then:

$$p_\alpha(X) = p(\alpha^{-1}(X)) = \mathrm{tr}(\rho u^{-1} X u) = \mathrm{tr}(u \rho u^{-1} X) \qquad (*)$$

so that the state p_α corresponds to the density operator $u\rho u^{-1}$. This shows the mathematical equivalence between the σ-complex and the density matrix formulations under symmetry transformation.

Now that the mathematical formalism in terms of σ-complexes has been presented, it is possible to develop a theory that describes the micro-physical world as governed by quantum chance. By chance is meant objective probability that is a physical property of objects in the real sphere rather than a description of uncertainty in the epistemic sphere. This will, among other attributes, allow the derivation of the quantum conditional probability that provides an explanation of quantum state reduction.

In standard quantum mechanics, the term "reduction" usually means the reduction of the wavefunction and so is conceptually tied to the Schrödinger picture. Although there is a mathematical equivalence between the Schrödinger picture and the formalism developed in this paper, the state as a set of probability measures on a σ-complex of projection operators describes a physics of dispositions rather than waves in either real space or some more abstract space. As such, the position is closer to that of matrix mechanics of early quantum physics, but with more ontological detail on the role of probabilities.

17.3.1. Conditional States

State reduction is a phenomenon that may seem unique to quantum mechanics and has no counterpart in classical mechanics, but this is not the case. Conditional probability plays that role classically, but the objective ensemble or relative frequency interpretations of probability are available, which means that no conceptual problem is posed by the "reduction" of the probability distribution. With this in mind, conditional probability will now be generalized to quantum mechanics.

If p is a state on $Q(\mathcal{H})$ and Y a projection operator in $Q(\mathcal{H})$ such that $p(Y) \neq 0$, then it will be shown that there exists a unique state $p(\,\cdot \mid Y)$ conditional on Y. If ρ is the density operator corresponding to p, then to see that the operator $Y\rho Y/\mathrm{tr}(Y\rho Y)$ corresponds to the state $p(\,\cdot \mid Y)$, note that if X lies in the same σ-algebra as Y, then X and Y commute, so:

$$\begin{aligned}\mathrm{tr}(Y\rho Y X)/\mathrm{tr}(Y\rho Y) &= \mathrm{tr}(\rho X Y)/\mathrm{tr}(\rho Y)\\ &= p(X\cdot Y)/p(Y)\\ &= p(X \mid Y),\end{aligned}$$

which is the standard form for conditional probability. It is proposed that the expression:

$$p(X \mid Y) = \operatorname{tr}(Y\rho YX)/\operatorname{tr}(Y\rho Y) \qquad (**)$$

generalizes the conditional probability state to the case when X and Y do not commute.

The mapping of ρ to $Y\rho Y/\operatorname{tr}(\rho Y)$ is the formula for the reduction of state given by the von Neumann-Lüders Projection Rule (Lüders 1951). In standard quantum mechanics, this rule is an additional principle appended to quantum mechanics. Here it appears as the unique answer to conditioning a state to a specific projection operator.

From the symmetry transformation of the state, equation (*):

$$\begin{aligned} p_\alpha(X \mid Y) &= \frac{p_\alpha(XY)}{p_\alpha(Y)} \\ &= \frac{p(\alpha^{-1}(XY))}{p(\alpha^{-1}(Y))} \\ &= \frac{p(\alpha^{-1}(X)\alpha^{-1}(Y))}{p(\alpha^{-1}(Y))} \\ &= p(\alpha^{-1}(X) \mid \alpha^{-1}(Y)) \end{aligned}$$

The results in this subsection hold when X and Y are in the same commutative von Neumann sub-algebra corresponding to a sub-σ-algebra of $Q(\mathcal{H})$. The next subsection covers the more interesting case of when they are in different sub-algebras and there is a correction to the form of the conditional probability that is well known from classical probability.

17.3.2. Total Probability in Classical and Quantum Conditional Probability

The Law of Total Probability (or Law of Alternatives in the discrete case, as here) provides a useful method to partition and analyze conditional probabilities. Firstly, in the classical case, let $Y_1, Y_2, \ldots$ lie in a $*$-algebra of commuting projection operators with $Y_i Y_j = 0$ for $i \neq j$, and let $Y = \sum_i Y_i$, then:

$$\begin{aligned}
p(X \mid Y) &= p\left(\sum_i (XY_i)\right) / p(Y) \\
&= \sum_i (p(XY_i)/p(Y_i))(p(Y_i)/p(Y)) \\
&= \sum_i p(X \mid Y_i) p(Y_i \mid Y)
\end{aligned}$$

Not only is this partition useful, but it also displays clearly how the quantum case differs from the classical.

From equation (**), above:

$$\begin{aligned}
p(X \mid Y) &= \mathrm{tr}(Y\rho Y X)/\mathrm{tr}(\rho Y) \\
&= \mathrm{tr}\left(\sum_{i,j} Y_i \rho Y_j X\right) / \mathrm{tr}(\rho Y) \\
&= \sum_i \mathrm{tr}(Y_i \rho Y_i X)/\mathrm{tr}(\rho Y) + \sum_{i \neq j} \mathrm{tr}(Y_i \rho Y_j X)/\mathrm{tr}(\rho Y) \\
&= \sum_i p(X \mid Y_i) p(Y_i \mid Y) + \sum_{i \neq j} \mathrm{tr}(Y_i \rho Y_j X)/\mathrm{tr}(\rho Y).
\end{aligned}$$

This shows that for conditional properties, an interference term must be added to the classical law of total probability. When X commutes with each Y_i, the interference term is zero.

A measurement of a property is the most familiar example of conditioning with respect to several properties. If a property represented by an operator A has a spectral decomposition $\sum_i a_i \pi_i$, then measuring the observable amounts to registering the values of the properties given by the π_i. The interaction algebra B_A is that generated by the π_i.

Conditioning can be used to define the context for the dynamics of a quantum object. In general, this requires the notion of conditioning with respect to several conditions.

Given a system with σ-complex Q and disjoint elements $Y_1, Y_2, \ldots$ in a common σ-algebra in Q with $\sum_i Y_i = \mathbf{1}$, and a state p, the state conditional on $Y_1, Y_2, \ldots$ is defined to be $p(\,\cdot \mid Y_1, Y_2, \ldots) = \sum p(Y_i) p(\,\cdot \mid Y_i)$. This can be written more compactly as $p(\,\cdot \mid B_A)$, the state conditional on the interaction algebra B_A.

For a quantum system, with σ-complex $Q(\mathcal{H})$ and if ρ is the density operator corresponding to the state p:

$$p(\cdot \mid B_A) = \sum \operatorname{tr}(\rho Y_i)(Y_i \rho Y_i / \operatorname{tr}(\rho Y_i)) = \sum Y_i \rho Y_i,$$

so that for each X the probability $p(X \mid B_A) = \sum_i \operatorname{tr}(Y_i \rho Y_i X)$.

The natural definition for applying a symmetry to the conditioned state is given by:

$$p_\alpha(X \mid B_A) = p(\alpha^{-1}(X) \mid \alpha^{-1}(B_A)).$$

The σ-algebra B_A generated by the $Y_1, Y_2, \ldots$ is simply a sub-algebra of the σ-complex.

17.4. Summary

There is an equivalence between quantum states and the probabilities describing objective chance. That we are dealing with a generalization of probability theory is made evident in the equation for the conditional probability that introduces extra terms due to quantum interference.

18. Modal categories and quantum chance

18.1. Introduction

The motivation for examining modal categories is to provide a more refined account of the ontology of dispositions and chance in quantum mechanics. Therefore, the emphasis here will be on the *real sphere*, although the modes apply also in the *ideal sphere* but in a way appropriate to that sphere.

By analyzing how different interpretations of quantum theory handle the transitions between possibility, contingency, necessity, and actuality, we can uncover the underlying implicit ontologies that distinguish these frameworks.

18.2. The Hierarchy of Modal Categories

Following the ontological framework laid out by Nicolai Hartmann (1937), we distinguish between positive and negative modes of being. Intuitively, necessity is more than actuality, and actuality is more than possibility. This provides a clear sense of direction because the lower positive mode is contained within the higher:

$$\text{Possibility} \subset \text{Actuality} \subset \text{Necessity}.$$

What is actual must at least be possible, and what is necessary must at least be actual.

18.2.1. Positive Modes of Being

- **Necessity:** Not being able to be otherwise.
- **Actuality:** Being this way and not otherwise.
- **Possibility:** Being able to be one way or not.

18.2.2. Negative Modes of Being

Conversely, the negative modes offer descending degrees of definiteness and being: * **Contingency:** Not being necessary (also being able to be different). * **Non-actuality:** Not being so. * **Impossibility:** Not able to be so.

With the negative modes, **impossibility** is a minimum of being, representing extreme non-being. **Contingency** is fundamentally the non-existence of necessity. The definiteness of the way of being is less even in non-actuality than in contingency, and even less in impossibility than in non-actuality.

Of the three negative modes, it is **contingency** that will be the most significant for our needs in quantum mechanics because it is a border case with a trace of positivity, providing a minimum opening to *being-so*.

18.3. Modal Profiles of Quantum Interpretations

To illustrate how these modes apply, consider the measurement of a spin component of a quantum object. Empirically, it can take only one of two values: $\pm\hbar/2$. There are at least five descriptions of the world which are modally distinct in the real sphere.

18.3.1. Standard Quantum Mechanics

In standard quantum mechanics (Ismael 2021), it is contingent (not necessary) for a spin component to take only one of these two values. The language of “uncertainty” is inappropriate in a world described by standard quantum mechanics. The Heisenberg Uncertainy relationship, $\Delta x \Delta p \geq \hbar$, still hold of course but should be interpreted as a relationship between ranges of possibility for physical values. The values are not merely hidden; they have no *actual* being until they come into existence.

Standard quantum mechanics does not provide a mechanism to describe the transition from contingency to actuality. It goes no further than to describe possibilities and the probability of them becoming actual in a context.

18.3.2. Bohmian Mechanics

In a world described by Bohmian Mechanics (Dürr and Teufel 2009), the spin wavefunction guides the electron to take one of two values after following a determined trajectory. For every exact starting point of the electron, the end spin state is fully determined.

It is only due to the practical impossibility of determining the starting point that makes the spin value seem contingent. Therefore, it can be said to be **epistemically contingent** but not so in reality. In the exactly specified situation, the value that becomes actual does so **necessarily**.

18.3.3. Ghirardi–Rimini–Weber (GRW) Theory

18.3.3.1. Probabilistic Interpretation

If the world is as described by the Ghirardi, Rimini, and Weber (GRW) theory (Ghirardi et al. 1986) with the wavefunction still providing probabilities of outcomes, the ontological situation is structurally similar to standard quantum mechanics. This may seem puzzling because this theory is explicitly intended to dispel the mystery of quantum measurement (or, more generally, actuality). However, even after the spontaneous wavepacket reduction given by the GRW mechanism, some contingency remains; or, if the spin component value is actualized due to the spontaneous reduction, nothing is explained by it—it is merely posited.

18.3.3.1.1. Mass Density Interpretation

There is a distinct version of the GRW theory (Egg and Esfeld 2014) in which the wavefunction provides a **mass density** of the particle rather than a probability density. In typical situations, the mass density will spontaneously reduce to a physical configuration consistent with either $+\hbar/2$ or $-\hbar/2$.

18.3.4. The "Many-Worlds" Description

In the "Many-Worlds" description Wallace (2012), the world is infinitely larger than can be observed. Only one of an infinite set of sub-worlds is observable to us. The spin component takes both values on two sub-worlds that are empirically separate. Through correlation, it is known that if in one sub-world the spin takes the value $+\hbar/2$, then in the other it takes $-\hbar/2$. In this description of the world, all values are **actualized by necessity**.

18.3.5. Summary Matrix

Interpretation	Status of Outcomes	Ontological Category	Role of Chance
Standard QM	Contingent	Objective Real	Fundamental transition without physical mechanism
Bohmian Mechanics	Necessary	Epistemic	Practical ignorance of exact starting trajectories
GRW (Probabilistic)	Contingent / Posited	Objective Real	Spontaneous reduction mechanism
GRW (Mass Density)	Physically Reduced	Objective Real	Continuous physical transformation
Many-Worlds	Necessary	Objective Real	Multi-verse branching across sub-worlds

18.4. 3. Quantum Chance and the Problem of Actualization

When the mathematical formulation of quantum theory brings probabilities rather than wavefunctions to the fore, the term **chance** is used to mean objective random events rather than a state of uncertainty or partial belief. Before an event occurs, the complete description of the system is given by a probability measure.

Using these ontological terms, the probability measure over the σ-complex of the quantum system—an object in the *inorganic level of real being*—describes the objective chance that its properties (represented by self-adjoint operators) will take one of a spectrum of possibilities that will become actual in that same inorganic level of reality.

The probability measure describes a contingent mode of being for the quantum system with a spectrum of values that are possible and become actual. What remains missing is an understanding of the timing of the actualization.

$$\text{Contingent Spectrum} \xrightarrow{\text{? Timing \& Physical Cause ?}} \text{Actualized State}$$

In all the versions of quantum theory considered so far, time behaves the same way as in classical physics. The timing of the actualization of the possible values of the observable and the physical cause (at the inorganic level of reality) of this actualization remains a major open problem.

The analysis of events in **quantum tunneling** may provide critical insight into this problem.

19. The Quantum Mechanics of Events, Histories and Trees

19.1. *Gemini feedback*

19.2. 1. Foundations vs. Interpretation (Credo 1 & 2)

Your alignment here is crucial. Fröhlich's insistence on "completing the foundations" rather than adding a loose philosophical gloss ("interpretation") matches your goal of a generalized probability theory.

By demanding a **"structure-preserving map"** between mathematical symbols and concrete physical phenomena, Fröhlich is advocating for a strict mathematical realism. In your framework, this means that the complex of probability spaces cannot just be a computational trick; it must structurally map directly to how physical possibilities manifest.

19.3. 2. The Ontological Friction: What Constitutes "What Is"? (Credo 3)

Your critique of paragraph (3) hits on a fundamental divide in modern quantum foundations: **Event/Process Ontology vs. Substance/State Ontology.**

- **Fröhlich's View (Event-Ontology):** For the ETH school, the ultimate building blocks of reality are *events* and the *histories* they form. Objects like electrons, protons, or fields are secondary; they are essentially the structural rules or "bonds" that constrain which histories are possible. Reality is a verb, not a noun.
- **Your View (Substance/State-Ontology):** You argue that events cannot exist in a vacuum—there must be *things* (particles, fields, physical quantities) to which things happen. The atoms, fields, and their possible states must be granted ontological status.

19.3.1. Bridging the Gap

To reconcile this for your upcoming posts, consider this perspective: **Can your physical quantities be defined by the structure of the probability spaces themselves?** If an atom is "real," its reality in an event-based framework might be represented by a invariant structural feature of the complex of probability spaces. The "thing" is the persistent topology of possibilities, while the "event" is the actualization of a point within that topology.

19.4. 3. Mathematical Rigor as an Epistemic Guardrail (Credo 4 & 5)

Your defense of mathematical rigor against the "rigor for rigor's sake" critique is vital. As you noted, Feynman warned us about fooling ourselves; mathematical rigor is the ultimate tool for exposing implicit, lazy assumptions.

In quantum mechanics, loose mathematics often hides the measurement problem behind vague terms like "environment" or "decoherence" without specifying *when* or *how* a definite outcome actually emerges. Fröhlich uses rigorous operator algebra (von Neumann algebras, specifically type III_1 factors in relativistic settings) to force the mathematics to tell us exactly when a possibility becomes an event.

19.5. Moving Forward: The Horizon of ETH

The six topics you listed at the end are the core battlegrounds of quantum foundations. As you dive into them, keep these specific questions in mind for your generalized probability theory:

- **On Loss of Predictability:** How does Fröhlich's view of the future branching into trees match your network of probability spaces?
- **On Locality:** Does your complex of probability spaces naturally prevent faster-than-light signaling while maintaining quantum non-locality?
- **On the Nature of Events:** In ETH, an event occurs when a mathematical condition called the **"Information Loss Constraint"** is met. You will want to look at how this constraint can be translated into a property of your probability spaces.

Your approach of using ETH to patch the gaps in your own theory is an excellent methodology. It ensures you aren't building in a vacuum, but rather testing your generalized probability theory against one of the most mathematically rigorous frameworks available today.

Which of the six topics from Fröhlich's list do you plan to tackle first in your next post?

It sounds like you are laying the groundwork for a deeply rigorous and conceptually clean exploration of the **ETH Approach (Events, Trees, and Histories)**. Bridging your own work—where the quantum state is a complex of probability spaces—with Fröhlich's framework is an excellent conceptual exercise.

Let's break down your reflections on Fröhlich's "credo," particularly the areas of alignment and the friction point regarding ontology.

19.6. End of feedback

It is time to return to quantum mechanics. The approach I have been developing is a generalised probability theory were the quantum state sits on a complex of probability spaces. In my review post of October 2022 I referred to the work of Fröhlich and colleagues and their search for a fundamental theory of quantum mechanics. They call it ETH (Events, Trees, and Histories) (Fröhlich 2021). Theirs is also an approach that proposes that quantum mechanics is fundamentally probabilistic and that it describes events and not just measurements. So, I will, over several posts, work through their theory to learn how some of the gaps in my own approach may be addressed. The picture below gives an early indiction how the concept of possibilities fit into the ETH scheme. An event is identified with the realisation of a possibility.

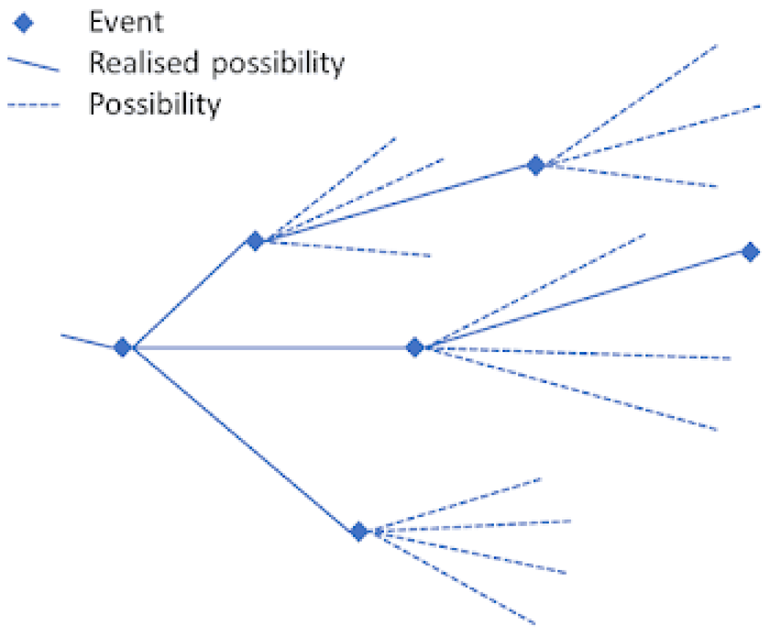


Figure 19.1.: Illustration of ETH - Events, trees, and histories

It has been a theme of my posts to try and clarify the philosophical fundamentals, especially ontology, associated with a physical theory. So, I will start the review of ETH with the introduction to a paper in which Fröhlich sets out his "credo" for his endeavour (Fröhlich 2021). His credo is:

! Fröhlich's Credo

1. Talking of the "interpretation" of a physical theory presupposes implicitly that the theory has reached its final form, but that it is not completely clear, yet, what it tells us about natural phenomena. Otherwise, we had better speak of the "foundations" of the theory. Quantum Mechanics

> has apparently not reached its final form, yet. Thus, it is not really just a matter of interpreting it, but of completing its foundations.
>
> 2. The only form of "interpretation" of a physical theory that I find legitimate and useful is to delineate approximately the ensemble of natural phenomena the theory is supposed to describe and to construct something resembling a "structure-preserving map" from a subset of mathematical symbols used in the theory that are supposed to represent physical quantities to concrete physical objects and phenomena (or events) to be described by the theory. Once these items are clarified the theory is supposed to provide its own "interpretation". (A good example is Maxwell's electrodynamics, augmented by the special theory of relativity.)
> 3. The ontology a physical theory is supposed to capture lies in sequences of events, sometimes called "histories", which form the objects of series of observations extending over possibly long stretches of time and which the theory is supposed to describe.
> 4. In discussing a physical theory and mathematical challenges it raises it is useful to introduce clear concepts and basic principles to start from and then use precise and – if necessary – quite sophisticated mathematical tools to formulate the theory and to cope with those challenges.
> 5. To emphasize this last point very explicitly, I am against denigrating mathematical precision and ignoring or neglecting precise mathematical tools in the search for physical theories and in attempts to understand them, derive consequences from them and apply them to solve concrete problems.

where I have added the numbering for easy reference. Let's take them one by one.

I agree completely with this comment although I may have lapsed occasionally into using the term “interpretation” loosely. So, a possibility to be investigated is that in addition to the standard formulation of quantum mechanics there may be an additional stochastic process that describes event histories.

The use of “interpreted” in the second paragraph has to do with the scope and meaning of the theory. We have the physical quantities that are to be described or explained, their mathematical representation and then the various theoretical structures that can make use of these quantities in their mathematical representation. In this way it should be clear from the outset what the intended theory is about. It about things and their mathematical representation.

This statement poses more of a problem. For me, the ontology has more to do with the concerns in the previous paragraph. While I agree that there are events, there must be physical quantities that participate in these events. These quantities must also form part of the ontology. For example, atoms may be made up of more fundamental particles. The atoms and the more fundamental particle are part of the ontology, and it is part of the structure of the ontology that atoms are made up of electrons, protons, and neutrons. Neutrons and protons are made up of still more fundamental particles. I would also include fields and possible states of the physical objects in the ontology.

Again, I agree. Feynman is known to have said that doing science is to stop us fooling ourselves. He was thinking primarily of comparing predictions with the outcome of experiments. However, mathematics also plays this role. By formulating rigorously the mathematics of a theory and following strictly the consequences we can avoid introducing implicit assumptions that make thing work out “all right” when they should not. When we get disagreement with experiment then we can be sure that it is the initial assumptions about objects or their mathematical representation that is at fault.

Frölich’s reputation is as an especially rigorous mathematical physicist and not only philosophers, but many physicists take such a rigorous approach to the mathematics to be rigour for rigour’s sake. While I do not claim his skills, I am more than happy to try ans learn form an approach that emphasises precise mathematics.

Within this “credo” Fröhlich and collaborators address:

- Why it is fundamentally impossible to use a physical theory to predict the future.
- Why quantum mechanics is probabilistic.
- The clarification of “locality” and “causality” in quantum mechanics.
- The nature of events.
- The evolution of states in space-time.
- The nature of space-time in quantum mechanics.

We will work our way through these topics in upcoming posts.

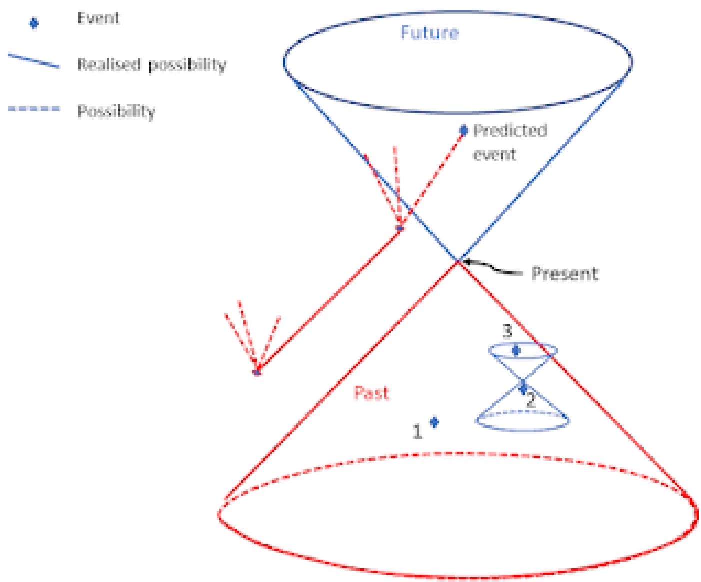
20. The Impossibility of Prediction and Quantum Indeterminacy

The previous post introduced the ETH (Events, Trees and Histories) approach to the foundations and completion of quantum theory. This post addresses why it is impossible to use a physical theory to predict the future and why quantum mechanics is probabilistic although the Schrödinger equation is deterministic.

Prediction uses available information to get knowledge about what will happen. It is an epistemic concept. The impossibility of prediction does not mean that the world is not deterministic, not governed by probabilistic law or even not governed by any laws at all. However, if we could predict the future reliably then this would be evidence that the world is deterministic. We will examine Fröhlich's (2021) arguments.

20.1. Impossibility of Prediction

It is always possible for someone to make a series of successful guesses about the future but what we are considering here is prediction based on available information and physical theory. To predict an event, we must know of everything that can affect that event. That is what the event is and when and where it takes place. Fröhlich's [1] argument is simple and illustrated by the diagram below.



The diagram uses the standard light-cone representation of space-time. The Future is at the top and the Past at the bottom. The predictor sits at the Present and has in principle access to all information in their Past light-cone. That is, they can access information on all past events but no information on what happens outside their Past light-cone.

There is causal structure within the Past light-cone. Events 1 and 2 are space-like separated. They cannot influence each other. Event 3 is in the future of 2, so event 2 can influence 3. Of significance for the argument are the events outside the predictor's light-cones (Future and Past) that are in the past light-cone of the predicted event. These events can influence what happens at the space-time point of the predicted event, but the predictor can have no knowledge of them. Therefore, the predictor cannot predict in principle what will happen at a future space-time point. In practice it is common knowledge that prediction is seen to work in many situations. These situations are controlled to isolate the predicted phenomenon from the influence outside the predictor's control.

20.2. Indeterminacy of Quantum Mechanics

The impossibility of reliable prediction does not imply indeterminacy.

Consider an isolated system. That is, over a period of time its evolution is independent of the rest of the universe. It is only for isolated physical systems that we know how to describe the time evolution of operators representing physical quantities in the Heisenberg picture (in terms of the unitary propagation of the system).

In the Heisenberg picture states of S are given by density matrices, $ ho$, acting on a separable Hilbert space, $\mathcal{H}$, of pure state vectors of S as in the mathematical formulation presented previously. Let $\hat{X}$ be a physical property of S, and let $X(t) = X(t)^*$ be the self-adjoint linear operator on $\mathcal{H}$ representing $\hat{X}$ at time t. Then the operators $X(t)$ and $X(t')$ representing $\hat{X}$ at two different times t and t', respectively, are related by a unitary transformation:

$$X(t) = U(t', t)X(t')U(t, t')$$

where, for each pair of times t, t', $U(t, t')$ is the propagator (from t' to t) of the system S, which is a unitary operator acting on $\mathcal{H}$, and $\{U(t, t')\}_{t,t' \in \mathbb{R}}$ satisfy

$$U(t, t') \cdot U(t', t'') = U(t, t''), \quad \forall t, t', t''$$
$$U(t, t) = 1, \quad \forall t$$

However, in the Copenhagen interpretation, whenever a measurement is made, at some time t, say the deterministic unitary evolution of the state of S in the Schrödinger picture is interrupted, and the state collapses into an eigenspace of the self-adjoint operator representing the physical quantity that is measured and over that eigenspace the probabilities are given by Born's Rule. This is what we have previously called the selection of a σ-algebra from the σ-complex of S.

As I am working towards a formulation of quantum mechanics that does not give a special status to measurement or observers. In the post *Modal categories and quantum chance* Born's rule is invoked as follows:

> The probability measure describes a contingent mode of being for the quantum system with a spectrum of values that are possible and become actual. What is missing is an understanding of the timing of the actualisation. In all the versions of quantum theory considered so far time behaves the same way as in classical physics.

While the question of timing is still to be resolved, quantum mechanics should be a theory that incorporates random events that are not derived from the deterministic evolution of the state. However, that evolution does govern the probabilities of becoming actual of property values associated with the spectrum of the self-adjoint operators representing the physical properties of S.

It should be noted that Bohmian mechanics provides an alternative model in which the uncertainty of the outcome is due to uncertainty in the initial conditions for the dynamical evolution. In the Bohmian theory the uncertainty is epistemic, and the dynamics is deterministic. The Bohmian theory needs to introduce this uncertainty into an otherwise deterministic theory to obtain empirical equivalence to standard quantum mechanics.

The formulation of the quantum mechanics presented in this blog is, so far, consistent with the ETH approach [1]. Some aspects of special relativity have now been introduced even though a relativistic formulation of quantum mechanics has not been presented yet. This will be part of the formulation of a mathematical description of the quantum “event”.

21. Events in quantum mechanics: a simple proposal

In standard quantum mechanics, **events** are observations. The occurrence of an event typically requires an experimental setup; an actual observer appears only in thought experiments such as Wigner's friend or Schrödinger's Cat. I do not deny that observing experimenters exist, but I claim they are not essential to the workings of the physical world. This note outlines a simple, local account of how events arise.

The quantum pointer behind the two slits in the interference experiment and the Stern–Gerlach setup with a magnetic field indicate that an interaction can select one σ-algebra from a σ-complex. Once a σ-algebra is selected, the postulate is that the associated probability distribution is sampled and that sample is an *event*.

This description looks classical in form even though it is expressed in quantum-state language. The magnetic field is treated classically as a parameter in the quantum Hamiltonian.

Experiments examined

- Double slit interference
- Stern–Gerlach

These examples motivate the algebra-selection plus sampling picture described below.

21.1. Proposal

A local interaction between a quantum system and an apparatus (pointer, Stern–Gerlach device, etc.) **selects a single σ-algebra** from a prior complex of possible algebras that describe the system's properties. A **σ-algebra** denotes the set of measurable properties and their logical combinations that can be assigned probabilities. This selection occurs at the interaction and precedes macroscopic detection.

Once a single σ-algebra is selected, the situation admits a classical probability description. At that point an event may be modelled as a sample drawn from the resulting probability distribution, analogous to sampling in classical statistics. The actualisation of a potential value is caused by the interaction and is the immediate next step after the complex reduces to one algebra.

After this actualisation, the particle may proceed to a detector and its properties recorded in accord with the experiment's purpose.

Toy model (two-level system coupled to a pointer)

A short, standard toy model makes the idea concrete.

- **System**: two-level system with orthonormal basis $\{\,|a_1\rangle, |a_2\rangle\,\}$.
- **Apparatus (pointer)**: large Hilbert space with pointer states $|\phi\rangle$.
- **Interaction Hamiltonian (idealised von Neumann coupling)**:

$$H_{\text{int}} = g\,\hat{A} \otimes \hat{P},$$

where $\hat{A}$ has eigenstates $|a_i\rangle$ with eigenvalues a_i, and $\hat{P}$ is the pointer momentum conjugate to the pointer position.

- **Initial state**:

$$|\Psi(0)\rangle = (c_1|a_1\rangle + c_2|a_2\rangle) \otimes |\phi_0\rangle.$$

- **After interaction** (unitary $U = \exp(-\frac{i}{\hbar}H_{\text{int}}t)$) the joint state becomes entangled:

$$|\Psi\rangle = c_1|a_1\rangle \otimes |\phi_1\rangle + c_2|a_2\rangle \otimes |\phi_2\rangle,$$

where $|\phi_i\rangle$ are pointer states correlated with system outcomes.

- **Reduced density matrix of the system**:

$$\rho_s = \mathrm{Tr}_{\text{ptr}}\,|\Psi\rangle\langle\Psi| = \sum_{i,j} c_i c_j^*\,\langle\phi_j|\phi_i\rangle\,|a_i\rangle\langle a_j|.$$

If the pointer states become effectively orthogonal, $\langle\phi_j|\phi_i\rangle \approx \delta_{ij}$, then ρ_s is approximately diagonal in the $\{|a_i\rangle\}$ basis and the probabilities for outcomes are $p_i \approx |c_i|^2$.

Interpretation in the proposal. The interaction has singled out the algebra generated by the projectors $\{\,|a_i\rangle\langle a_i|\,\}$ (a σ-algebra of events for this measurement context). Once that algebra is selected, the classical probability distribution $\{p_i\}$ is available and the event is modelled as sampling that distribution: outcome i occurs with probability p_i.

This toy model shows how a local interaction can (i) pick a preferred algebra/basis and (ii) produce a classical probability distribution from which an event can be sampled.

21.2. Remarks and context

- **Relation to decoherence**
  The algebra-selection step is closely related to decoherence: the apparatus (and environment) cause pointer states to become orthogonal, effectively selecting a preferred algebra or basis. The present outline emphasises that once such a basis is selected, one may treat the next step as classical sampling.

- **Measurement problem**
  This account reframes measurement as (i) local algebra selection via interaction and decoherence, followed by (ii) sampling from the resulting classical distribution. It avoids invoking conscious observers while acknowledging that further work is needed to explain why a single sample is realised rather than only an effectively diagonal density matrix.

- **Wigner's friend and observer-relativity**
  If different observers assign different algebras, the proposal ties the algebra-selection to the local interaction that couples system and apparatus. Reconciling observer-relative descriptions with a single objective event requires further clarification.

- **Entanglement and Bell correlations**
  A local algebra-selection mechanism must still reproduce Bell-type correlations. The global quantum state and the local selection rules together must yield the correct joint probabilities without superluminal causal influences; this is a key constraint on any full theory.

21.3. Development avenues

1. **Relativistic quantum theory**
   Investigate Fröhlich's relativistic quantum theory (Fröhlich 2021) to embed algebra selection and actualisation in a relativistic framework.

2. **Branching mathematics**
   Study the mathematics of branching in Everettian (many-worlds) formulations. The multiverse formulation (Wallace 2012) has an event problem—how to interpret branching as events—and understanding branching may allow replacing universe splitting with sampling of a probability distribution over possibilities.

21.4. Suggested additions for a fuller manuscript

- Define terms briefly (e.g., σ-algebra, σ-complex, selection).
- Provide a worked toy model with explicit parameter choices and timescales.
- Compare explicitly with decoherence, objective collapse, and Everett interpretations.
- Identify any empirical differences or tests that could distinguish this outline from other accounts.

Part III.

Organic Layer

22. Natural selection

22.1. Literature on Spontaneous Organisation and Natural Selection

Here is a curated list of key literature exploring the interplay between spontaneous self-organisation and natural selection, complete with BibTeX citations.

22.1.1. 1. Foundational Monographs

22.1.1.1. The Origins of Order: Self-Organization and Selection in Evolution

- **Authors:** Stuart A. Kauffman
- **Year:** 1993
- **Summary:** This seminal book argues that natural selection is not the sole architect of biological complexity. Kauffman uses mathematical models to show how spontaneous, physics-driven self-organisation creates the stable, ordered structures (like genetic regulatory networks) upon which Darwinian selection acts.

```
@book{kauffman1993origins,
  title={The Origins of Order: Self-Organization and Selection in Ev
  author={Kauffman, Stuart A.},
  year={1993},
```

```
  publisher={Oxford University Press},
  address={Oxford, UK}
}
```

22.1.2. 2. Theoretical and Philosophical Biology

22.1.2.1. Self-Organization Proposes what Natural Selection Disposes

- **Authors:** Fabio Boschetti and Brad Batten
- **Journal:** *Biological Theory* (2008)
- **Summary:** This paper clarifies the division of labour between the two forces. It conceptualises self-organisation as the mechanism that generates innovative biological forms, while natural selection acts as the environmental filter determining which forms survive.

```
@article{boschetti2008self,
  title={Self-organization Proposes what Natural Selection Disp
  author={Boschetti, Fabio and Batten, Brad},
  journal={Biological Theory},
  volume={3},
  number={4},
  pages={308--321},
  year={2008},
  publisher={Springer}
}
```

22.1.2.2. Self-Other Organization: Why Early Life Did Not Evolve Through Natural Selection

- **Authors:** Liane Gabora
- **Journal:** *Biology and Philosophy* (2006)
- **Summary:** Gabora investigates abiogenesis and early protocells. She argues that because the earliest forms of life lacked genetic code and replicated through messy, uncoded chemical interactions, their evolution was driven entirely by self-organising principles rather than Darwinian selection.

```
@article{gabora2006self,
  title={Self-other organization: Why early life did not evolve thro
  author={Gabora, Liane},
  journal={Biology and Philosophy},
  volume={21},
  number={3},
  pages={353--377},
  year={2006},
  publisher={Springer}
}
```

22.1.3. 3. Computational and Recent Advances

22.1.3.1. Natural Induction: Spontaneous Adaptive Organisation without Natural Selection

- **Authors:** Christopher L. Buckley, Christopher S. Adams, Richard A. Watson, and Michael Levin
- **Journal:** *Entropy* (2024)

- **Summary:** This recent paper explores how physical systems (like viscoelastic networks) can spontaneously learn and optimise their behaviour through recurrent interactions. It provides a concrete mechanism for adaptive organisation that completely bypasses the need for natural selection or external training.

```
@article{buckley2024natural,
  title={Natural Induction: Spontaneous Adaptive Organisation w
  author={Buckley, Christopher L. and Adams, Christopher S. and
  journal={Entropy},
  volume={26},
  number={6},
  pages={499},
  year={2024},
  publisher={MDPI}
}
```

23. The genetics game

Instead of requiring rational thought, the “payoffs” in genetic game theory are biological fitness (survival and reproduction rates). The foundational components include:

- Evolutionarily Stable Strategy (ESS): Formalized by John Maynard Smith and George R. Price in 1973, an ESS is a strategy that, if adopted by all members of a population, cannot be invaded by any alternative strategy. If everyone adopts an ESS, natural selection cannot improve upon it.
- The Prisoner’s Dilemma & Cooperation: Game theory explains how cooperation and altruism can evolve. In the classic Prisoner’s Dilemma, defecting yields the highest individual reward. However, through mechanisms like kin selection (where you share genes with relatives), a “cooperative” gene can ensure its own propagation across the population, even at the expense of the individual.
- Signaling Games: Researchers have found that genes within the genome essentially participate in signaling games. Genes act as senders and receivers of signals, adopting strategies of deception, imitation, cooperation, and competition to influence cellular biochemistry and gene expression.
- The Gene’s Eye View: Popularized by Richard Dawkins, this perspective frames the individual organism merely as a “vehicle” or “survival machine” that genes use to play the game of life and maximize their representation in future generations.

Maynard Smith and Price (1973), Maynard Smith (1982), Dawkins

(1976), Massey and Mishra (2018), Archetti and Schuster (2014), McGlothlin et al. (2022)

Part IV.

Psychic Layer

24. Hartmann and Hayek: A Comparative Ontology

24.1. Introduction

This section examines the ontological parallels between **Nicolai Hartmann** and **Friedrich A. Hayek**.
Although there is no historical evidence of influence either way, their philosophies converge on three major themes:

- a **layered** or **stratified** (the section titled 1.4) structure of reality
- **emergence** and the irreducibility of higher levels
- **anti-reductionism** as a metaphysical position

The critical ontology of Hartmann was developed with the ambition to be a complete ontology that would set the the way of being for all entites. Hayek is known as an economist. But, in addition to his technical work in economics he had wider interest in socail science and psycnology. Hayek did not declare that his work in these areas is explicitly ontological and it s certainly not narrowly so. His work often has an epistemological flavour but his his emergent orders in what Harmann calls the *Psychical* and *Spiritual* layers provide a mechanism to explain how new entites come into existence. The goal is to show how Hartmann's systematic ontology can illuminate and strengthen Hayek's emergentist psychological and social ontologies.

Key sources include Hartmann's *Der Aufbau der realen Welt* Hartmann (1940) and Hayek's *The Sensory Order* Hayek (1952), as well as recent secondary literature such as Lewis's analysis of Hayek's ontology (Lewis 2017).

24.2. Hartmann's Stratified Ontology

24.2.1. Levels of Reality

As previously discussed Hartmann distinguishes four major strata (the section titled 1.4):

1. **Inorganic**
2. **Organic**
3. **Psychic**
4. **Spiritual**

Each level has its own **categories**, **laws**, and **forms of determination**.
Higher levels depend on lower ones but are not reducible to them Hartmann (1953).

24.2.1.1. The Categorial *Novum*

Hartmann's doctrine of the *novum* states that higher levels introduce **genuinely new properties**.
This emergence is **ontological**, not merely epistemic.

24.3. Dependence Relations

Hartmann articulates several forms of dependence:

- **Foundation (Fundierung)**
- **Superstructure**
- **Autonomy of levels**

These relations allow for **asymmetric dependence**: the higher requires the lower, but the lower is not determined by the higher.

24.4. Hayek's Emergentist Ontology

24.4.1. The Sensory Order

In *The Sensory Order* Hayek (1952), Hayek argues that mental phenomena form a **classification system** that emerges from neural events but cannot be reduced to them.

This is Hayek's most explicit ontological work.

24.4.2. Complex Phenomena and Spontaneous Order

Hayek's social theory—developed in *Studies in Philosophy, Politics and Economics* Hayek (1967) and *Law, Legislation and Liberty* Hayek (1973)—treats markets, legal systems, and traditions as **higher-level orders**.

These orders:

- arise from individual interactions
- exhibit properties not predictable from micro-data
- constrain and guide individual behaviour

24.4.3. Anti-Reductionism

Hayek rejects both:

- **physicalist reduction** of mind
- **individualist reduction** of social institutions

His anti-reductionism is grounded in the **limits of knowledge** and the **structure of complex orders**.

24.5. Structural Parallels Between Hartmann and Hayek

24.5.1. Levels vs. Orders

Hartmann's *Schichtenbau der Welt* and Hayek's theory of complex orders both posit:

- hierarchical structures
- emergent properties
- irreducible higher-level regularities

24.5.2. Novum vs. Spontaneous Order

Hartmann's *categorial novum* parallels Hayek's idea that higher-level orders exhibit **new patterns** not derivable from lower-level data.

24.5.3. Dependence and Constraint

Both philosophers endorse **asymmetric dependence**:

- higher levels/orders depend on lower ones
- once formed, they **constrain** lower-level processes

24.5.4. Realism About Higher-Level Entities

Hartmann is a robust realist about higher-level entities.
Hayek is realist about social orders, rules, and institutions.

Both reject eliminativism.

24.6. Differences and Tensions

24.6.1. Systematic vs. Fragmentary Ontology

Hartmann offers a systematic metaphysics; Hayek's ontology is dispersed across psychology and social theory.

24.6.2. Normativity

Hayek's ontology is tied to liberal political theory (rule of law, limits of constructivism).
Hartmann's ontology is descriptive and non-political.

24.6.3. Methodological Divergence

- Hartmann: phenomenological realism, categorial analysis
- Hayek: theory of complex phenomena, epistemic limits, evolutionary accounts

24.7. A Hartmannian Reconstruction of Hayek

24.7.1. Mapping Hayek into Hartmann's Framework

Hartmann's categories can clarify:

- the ontological status of rules
- the nature of social institutions
- the structure of emergent orders

24.7.2. Strengthening Hayek's Anti-Reductionism

Hartmann provides a **metaphysical justification** for Hayek's methodological anti-reductionism.

24.7.3. Implications for Social Ontology

A combined Hartmann–Hayek framework yields:

- a robust realist account of institutions
- a layered model of social reality
- a metaphysics of emergent social order

24.8. Conclusion

Hartmann and Hayek, though intellectually distant, converge on a shared metaphysical vision of **layered reality**, **emergence**, and **irreducibility**.
A systematic comparison reveals that Hartmann's ontology can deepen and stabilise the metaphysical foundations of Hayek psychological and social theories. In turn Hayek provides detail on the ppsychic and spiritual consequences of an ontology of emergence.

25. Sensory order

This chapter will examine how the work of Hayek on emergent order can help develop the ontoology of the Psychic level and clarify the realtionship to the lower biological level and the higherSpiritual level.

- Hartmann develops a stratified ontology of reality (Seinsschichten), where higher levels emerge from but are irreducible to lower ones. As is explaind in chapter **Strata of being** (the section titled 1.4).
- Hayek develops a theory of spontaneous orders and complex phenomena, also emphasizing irreducibility and emergent structures.

This comparitive study focuses on:

- Emergence: Hartmann's categorial Novum vs. Hayek's complex phenomena.
- Layered ontology: Hartmann's strata vs. Hayek's multi-level social orders.
- Anti-reductionism: Both reject physicalist reductionism.
- Rules and categories: Hartmann's categories vs. Hayek's rules-based order.
- Realism: Both defend the reality of higher-order structures.

25.1. Emergence and layers

Part V.

Spiritual Layer

26. Ontological Critiques of the Notion of Government

26.1. Introduction

This overview surveys several major **ontological critiques of the concept of government**.
Unlike political or moral critiques, ontological critiques question the *being*, *existence*, or *metaphysical status* of government as such.
These critiques arise from different philosophical traditions, each challenging the assumption that "government" is a coherent or real entity.

The four strongest traditions are:

1. **Anti-entity / nominalist critiques**
2. **Process-ontology critiques**
3. **Social-constructivist critiques**
4. **Normative-ontological (anarchist metaphysics) critiques**

Each section below outlines the core argument and cites representative authors.

26.2. Anti-Entity Critiques: Government as a Reified Abstraction

A classical ontological critique holds that "government" is not a real entity but a **reified abstraction**.
On this view, only individuals exist; collective entities such as "the state" or "the government" are conceptual fictions.

Max Stirner famously described such abstractions as *"spooks"*—phantoms that dominate human thought despite lacking real existence (Stirner 1844).
Similarly, methodological individualists argue that social wholes cannot possess agency or intention independent of the individuals composing them (Hayek 1948a).

This critique denies the ontological status of government altogether: if only individuals exist, then attributing will, authority, or moral standing to "government" is a category mistake.

26.3. Process-Ontology Critiques: Government as a Set of Practices

A second critique arises from **process philosophy** and **social ontology**.
Here, government is not a substance or entity but a **dynamic ensemble of practices, relations, and performances**.

Michel Foucault's work on *governmentality* treats governing as a historically contingent set of techniques rather than the action of a unified entity (Foucault 2007).
Bruno Latour's actor-network theory similarly dissolves institutions into networks of heterogeneous actors (Latour 2005).
Judith Butler extends this to political structures more broadly, arguing that political forms are performative and contingent (Butler 2015).

If government is a process rather than a thing, then it cannot serve as the bearer of sovereignty or authority.
Its ontological instability undermines claims to inherent legitimacy.

26.4. Social-Constructivist Critiques: Government as a Status Function

A third critique accepts that government exists, but only as a **socially constructed status function**.
John Searle argues that institutions exist because people collectively accept certain rules and assignments of function (Searle 1995).
However, this dependence on collective intentionality makes government **ontologically fragile**.

Charles Tilly's historical work shows that states emerged from contingent patterns of coercion and capital accumulation, not from any essential or natural form (Tilly 1992).
Benedict Anderson's analysis of nations as *imagined communities* extends this fragility to the political identities that sustain governments (Anderson 1983).

On this view, government has no mind-independent existence; its being is parasitic on shared belief and recognition.

26.5. Normative-Ontological Critiques: The Incoherence of Political Authority

The strongest analytic critique comes from **normative ontology**: government claims rights (to coerce, command, tax, punish) that no individual possesses.
If individuals lack these rights, then aggregating them cannot create such rights.

Robert Paul Wolff argues that political authority is incompatible with individual autonomy (Wolff 1970).
A. John Simmons shows that no existing theory successfully grounds a duty to obey the state (Simmons 1979).
Michael Huemer provides the most systematic contemporary version, arguing that the state's special moral status is a metaphysical fiction (Huemer 2013).

This critique concludes that government is **ontologically impossible** as a bearer of legitimate authority.

26.6. Conclusion

These four traditions—nominalist, process-ontological, constructivist, and normative-ontological—offer powerful challenges to the metaphysical coherence of "government."
They differ in method and emphasis, but all converge on a central insight:

> The concept of government is far less ontologically secure than political theory typically assumes.

Whether government is a fiction, a process, a fragile construct, or a metaphysical impossibility, each critique destabilises the idea that government is a natural or necessary feature of social reality.

27. Libertarian Foundations

27.1. Robert Nozick and Michael Huemer: A Comparative Analysis of Libertarian Foundations

Robert Nozick and Michael Huemer are both central figures in contemporary libertarian political philosophy, yet their approaches diverge sharply in foundations, method, and normative structure. Nozick represents the classical, rights-absolutist tradition inaugurated in *Anarchy, State, and Utopia*, whereas Huemer exemplifies the post-Nozick turn toward common-sense morality and non-ideal political reasoning.

27.1.1. Foundations: Self-Ownership vs. Common-Sense Morality

Nozick grounds libertarianism in the principle of **self-ownership** and the associated system of **side-constraints**. Rights are treated as near-absolute limits on permissible action, and any state activity beyond the minimal state is ruled out because it violates these constraints. His theory is metaphysically ambitious, deriving political conclusions from strong foundational axioms.

Huemer rejects this approach. He argues that political philosophy should begin not with metaphysical claims about ownership but with **ordinary moral intuitions**. His libertarianism is built from widely shared judgments about coercion, harm, and fairness. Rights are important but **not inviolable**; they may be overridden in exceptional circumstances. This makes Huemer's framework more flexible and less dependent on controversial metaphysical premises.

27.1.2. Method: Ideal Theory vs. Non-Ideal Reasoning

Nozick's method is architectonic and ideal-theoretic. He constructs a systematic philosophical model and uses abstract thought experiments—such as the Wilt Chamberlain argument—to show that patterned theories of justice inevitably conflict with liberty.

Huemer's method is deliberately non-ideal. He focuses on real-world institutional failures, epistemic limitations, and the moral presumptions ordinary agents rely on. His work aligns with the contemporary PPE (Philosophy, Politics, and Economics) style: empirically sensitive, anti-systematic, and grounded in practical reasoning rather than metaphysical theory-building.

27.1.3. Political Authority and Coercion

Both philosophers challenge the moral legitimacy of political authority, but they do so in different ways. Nozick argues that the minimal state is justified because it protects rights without violating them; anything more extensive is morally impermissible. His "Tale of the Slave" illustrates the continuity between political coercion and morally problematic domination.

Huemer's critique is broader. He maintains that political authority is a **moral illusion**: states possess no special moral permission to coerce beyond what individuals have. Coercion is presumptively wrong, though not absolutely forbidden. This position is more radical in scope but less rigid in its foundations.

27.1.4. Redistribution and the Scope of the State

For Nozick, redistribution is categorically unjust because it violates self-ownership. The minimal state is the only morally legitimate political structure.

Huemer also opposes most forms of redistribution, but his opposition is grounded in common-sense moral presumptions rather than rights absolutism. He allows that extreme circumstances may justify coercive redistribution, though such cases are rare.

27.1.5. Summary Table

Dimension	Robert Nozick	Michael Huemer
Foundation	Self-ownership; rights absolutism	Common-sense morality; rejects absolutism
Method	Ideal theory; abstract system	Non-ideal; empirical; intuitive
State	Minimal state justified by rights	State authority morally unjustified
Coercion	Nearly always wrong	Presumptively wrong but overridable
Redistribution	Rights violation	Usually wrong, sometimes justified
Tone	Architectonic, metaphysical	Practical, intuitive, epistemic

27.1.6. Concluding Contrast

Huemer's project can be understood as a post-Nozick development: he preserves many libertarian conclusions but replaces Nozick's controversial metaphysical foundations with moral intuitions that ordinary agents already accept. In doing so, Huemer offers a more accessible and empirically grounded libertarianism, while Nozick remains the paradigmatic architect of rights-based minimal-state theory.

28. Eucken's liberal order

Times of crises trigger authoritarian solutions. On the very short time scale this may be but liberal democracies should be resilient enough to return freedoms and within a framework of law and regulation. It's when freedoms are put on hold that the robust legal framework ensuring the rule of law enables a return to decent liberal democracy. A liberal order is one to which personal freedom is fundamental but recognised the constraints and opportunities afforded by the interaction with others. These interactions will give rise to conflicts and tradeoffs. The freedom to enter into an exchange must be balanced with the freedom not to enter an exchange. Should a shop keeper have the unreserved right not to sell to an individual or should an individual have the right to purchase from an individual goods or services on offer without prejudice? To deal with type of situation rules of interaction are required. Rules will range from international treaties and conventions, national to local laws, regulations and customs. To put this in place there needs to be an interplay of politics, economics and values. The commercial world cannot merely request that the political get out of its way nor that values can be ignored in the pursuit of profit. Where does this leave personal freedom? It cannot be allowed to degenerate into a mere freedom to obey the rules.

It is a great misfortune that "Principles of Political Economy" by Walter Eucken ((**eucken2004grundzuge?**)) was never translated into English. His model of a responsible and open liberal market order that addresses the deficiencies of laissez-fair or market fundamentalism as its modern version is sometimes known could form the foundation for addressing the current ills in western economies. This is not just a matter of economic theory. This model provided the blueprint for the new West German economy and social compact after the Second World War. It also needed politicians with the will to implement it, in this case Ludwig Erhard. The astute branding of the model by Alfred Muller-Armack as the Social Market Economy, who became head of policy for Erhard helped form a cross-party consensus in favour of the model but with liberals and conservatives emphasising more the free market aspect and social democrats the social. While politically astute in the short to medium term the initial integrity of the model became diluted as more regulation and control was added during the 1960s.

This chapter will present Euckens model and variants in an attempt to interpret it in the context of current challenges, of which the most significant are growth, fairness, climate change and preclusive action. Climate change and its consequences is an example of an externality that Eucken did not anticipate1 but solutions can be found and should be built into the measures that his framework provides.

28.1. Introduction

Liberal economics is taken to start with the work of Adam Smith . As this was developed it became known as classical economics. But this term can only apply to a family of attitudes that recognise the strength of markets in allocating scare resources and generating fair prices. Its first phase was largely descriptive and could be thought of as a branch of social philosophy. As its findings started to be recognised for their applicability there was a move to start to treat the subject matter of economics as physicist treat mater, leading to an ever increasing use of mathematical techniques and models. This process has culminated in the mathematical models of neo-classical economics becoming an academic discipline that becomes more and more concerned with its internal system of rewards for research that hold only a tenuous link to practical problems. It is not our purpose here to provide a detailed history of economics but mention the academically dominant neo-classical school to emphasise that the thinkers covered here are distinct and separate from that orthodoxy. The goal here is to describe economic thinking developed st time to deep crisis at any given time, in relation to the thought and contributions of a key economic thinker, is indeed liberal and how far it fall short.

Neoclassical economics is the economic theory that appears to be most widely in use by experts and is remarkably like physics, in that it is based on a few simple principles, which lead to a rigorous mathematical formulation. This mathematical formulation leads to some basic theorems, which are taken to be an approximate description of a well functioning economy. A large body of results and models has been built up around this theory, and there is a community of experts that express confidence in its basic correctness. However this approach cannot take account of human aspects such as presferences, inventiveness and philanthropy, although narrow self interest is built in.

At the very least, the neoclassical theory of general equilibrium establishes that economics is one of the mathematical sciences. It sets a standard for clarity, rigour and generality that alternative approaches to economics must aim to live up to. This theory is a model that mathematical scientists in other disciplines should be interested to study and understand, because there is a serious claim it has succeeded in capturing something true in a simple mathematical structure about human beings. But if the evolution of the other mathematical sciences is any guide, one cannot expect that a successful theory in a domain is the last word. It is because of its success that we should expect that eventually any mathematical theory of real phenomena is to be replaced by a deeper theory that explains more and corrects misunderstandings.

There are reasons to think that the present is a good time to seek to improve economic theory. There are prominent voices criticizing the neoclassical model, or its offshoots in theories of pricing and risk in financial markets, as having badly failed as a basis of policy advice in the last decade leading to the 2008 economic crisis. Nonetheless it appears to be the case that there is a view among some economists that the model of an economy expressed by the theory of general equilibrium does teach lessons that have implications for policy questions. Among such views is the claim that markets do not require more than minimal regulation because they naturally evolve to a state of stable equilibrium, in which every participant is more satisfied and better off than they would be were the market constrained by external regulation2.

There are also long standing critiques of the neoclassical notion of general equilibrium, some coming from academic economists, others coming from scientists in other disciplines. In real markets there must be dynamics, and an understanding of how equilibrium arises – if it indeed does – from a more general non-equilibrium dynamics is a necessary step to answering many key questions about market behaviour.

Related to this are other fundamental issues about prices. Is a price an absolute quantity, or is it a ratio? In physics we have a basic rule which is that observables are always expressible as ratios of two quantities expressed in the same units, so they are a pure number. Applied to economics, this rule suggests that decisions should not be based on the units or currencies prices are expressed in. Is there a way this can be enforced in the formal treatment? A better observable might be constructed from measuring the result of performing a cycle of trades which begin and end in the same currency. Here one can take the ratio of two prices in the same currency. However, there is a problem, which is that all such quantities are supposed to vanish in equilibrium. But perhaps this is what we want for the non-equilibrium theory. Perhaps these are quantities whose dynamics is responsible for the evolution from non-equilibrium to equilibrium.

28.1.1. The liberal family

Liberalism does not have its Karl Marx. It evolved through times of trouble due to wars of religion while looking back to classical Greece and Rome for some clues as to how a more stable and tolerable order could be created. This meant abstracting Athenian democracy from the reality of slavery and patriarchy. Add to this the institutionalised privilege in the Roman republic. Many groups and societies have had their councils and while it is difficult to make a case from any one being an ancestor of an existing national assembly many of them a prototype; initial attempts at parliaments. Common to the liberal family is some form of representative parliament. This is fundamental to the political aspect of liberal order. In different places different detailed models have been developed.

Since prehistoric times there have been local market and then there evolved trading rounds between them. These developed in a context of a political order, often closely aligned with an enforced religion, but was tolerated along side these authoritarian structures because of their use in distributing goods and generating revenue. That they also had a side effect of stimulating innovation (technical, ethical and social). Authoritarian regimes can tolerate and even be supportive of technical innovation but ethical and social change needed policing followed by suppression.

28.2. Liberal Principles

28.2.1. Dimensions of liberalism

One way to try identify something systematic about the wide array of views espoused by liberals depending on their understanding of their principles is to look at how views range along a number of dimensions. Liberals generally support limited government, individual rights (including civil rights and human rights), a markets economy, democracy, secularism, gender equality, racial equality, internationalism, freedom of speech, freedom of the press and freedom of religion. To help evaluation to what extend a particular political and economic order is liberal we will introduce four dimensions.

- Individual: personal liberty is the fundamental concept although constrained by the freedoms of others.
- Political : this derives form the primacy but need to compromise of individual liberty to develop a workable accommodation that sacrifices as little individual freedom as possible.
- Social: the enlightened liberal will be socially tolerant and concerned to address suffering and hardship while insisting on the need to personal responsibility. In this dimension common values that sit constructively with the other dimensions will be values while those who do not share them will be tolerated if at all possible.
- Economic : at a fundamental level this is the freedom to enter or not into an exchange of goods or services.

These values can be treated a dimensions as a range of positions can be taken in them.

28.2.2. Market fundamentalism

This is a return to laissez-faire. But perhaps it goes further. The expression was popularized by business magnate and philanthropist George Soros in his book The Crisis of Global Capitalism (1998)

28.2.3. Reformed liberal economics

In this chapter we will look at combining to notions of reform. The first is to change something for the better and the second is to regroup as sports team or a military company would regroup following a setback.

28.2.4. What should we expect from theories of markets?

Economics provides an approximate and idealised description of a certain aspects of the world. An economic theory is intended to capture a few aspects of very complex and dynamical emergent phenomena, which occur in large markets in human societies. As such, there is no perfect model, the goal is to enunciate principles, which are general enough, but also powerful enough, to underlie the construction of a variety of models adopted to specific circumstances and questions, and to be the basis of explanation and, to a limited degree, prediction. Because no theory or model can perfectly capture the complexity of an actual economy, all principles, theories and models, must be considered provisional and phenomenological, and continually tested against data3.

To the extent that there is a connection to other areas of science it comes about because the economy is a subset of human activity, which forms a subset of processes in the biosphere. The biosphere is a far from equilibrium system, driven to an approximate, slowly evolving, steady state by the flow of energy through it. It is characterized by cycles of material and energy whose rates of flow are governed by feedback processes acting on diverse scales4. The economy of the world is a subset of that system of cycles and flows which is interesting to us because it is subject to peculiar kinds of decision making, about which we keep records.

Furthermore, an economy is subject to external events and conditions, which being human events are not subject to simple modelling or reliable predictions. So there is a strong limitation in what we can expect from a formal economic theory. But we still can investigate the possibility that there are some principles that lead to models, which can make some useful and testable predictions-at least for the behaviour of markets between their disruptions by unpredictable events and innovations. To the extent that these are robust and independent of the fine-grained details of how human actors make decisions our models will be reliable. This is exactly where behavioural and experimental economics pose challenges to the neoclassical model.

Thus, we are not inclined to side with who suggest that because human behaviour is involved there are no useful regularities that can be captured by mathematical models of a market. Even as we realise that there can be no perfect model of an economy, we seek to replace the Arrow-Debreu model with an alternative that is an example of a useful model, that applies to real people in interaction, and we take its shortcomings as starting points to attempt to improve it rather than as indications that there cannot be a useful general theory of economies and markets.

28.3. The liberal market order

The Keynesian revolution was a response to the realisation that *laissez-faire* was not sufficient to deal with complex national and international trading. In addition the social pressures due to economic dynamics and adjustment could not just be left to play themselves out. An alternative response to this situation was developed in Germany during the 1930s. Evidently a liberal reform program had to be clandestine and camouflaged in abstract academic terms. The intellectual leadership of this activity took place at the University Freiburg, which is still a centre for robust liberalism. The key players were Walter Eucken, Franz Böhm and Hans Großmann-Doerth. The main character in the argument here is Eucken who presented the detailed description of the political and economic order that would permit the operation of a free and open society in "Grundsätze der Wirtschaftspolitik" ("The Principles of Political Economy") .

Eucken's basic idea is the following: "The competitive order is able to properly coordinate investments over the long term. Because with its price mechanism it has the instrument to detect disproportionality and finally to correct them. In this it is superior to all other orders ...

Without a competitive order, no state capable of action can emerge and without a competitive state, no competitive order."5

The political order proposed by Eucken, that was labelled the social market economy only later by the politically more pragmatic Alfred Müller-Armack, can be represented as a diagram, Figure [fig:order]. Central to the proposed model is a functioning price system. Prices provide the efficient signals that allows the market to coordinate the allocation of goods and services without a central coordinator. However the whole order needs to be considered as a mutually enforcing system. Indeed the key concept of competition is an aspect of the system as a whole rather than a constituent. An in depth overview has been provided by Karen Horn ((**Horn?**)).

28.3.1. Constituent principles of the market order

The basic principle of the competitive order is a functioning price system. As Eucken was completing his work, the information signalling function of the price system and the impossibility of effective and efficient centralisation was explained by Hayek in *The Use of Knowledge in Society* (Hayek (1948b)). Flexible prices inform sellers about changing shortages and preferences and they also limit the power of all actors through implicit reports of profits and losses and divert scarce resources into efficient uses.

The second principle is the primacy of monetary policy. Around Eucken's basic principle of a effective price system, the further constituent principles are grouped as inseparable, to be considered necessary and acting simultaneously. The primacy of monetary policy refers to the priority of currency stability. What is required is a stable domestic currency and its external value. The authorities should refrain from deliberately influencing the external value of the domestic currency, for example to promote exports - especially since this leads rapidly to a completely unfruitful devaluation race between countries. A stable domestic and external currency value is required in order not to distort natural relative assessments of current and future domestic and foreign consumption.

The third constituent principle is free market access, both nationally and internationally. That is an essential prerequisite for competition. Walter Eucken formulates his postulate as follows: "… the principle of open markets must apply to the economic policy, , because if they are closed there is the acute danger of obstruction of complete competition." 6 If markets are not freely accessible and if therefore the"opening up of supply and demand" has not occurred, then the established producers can rest in monopoly-like positions and gradually lose sight of the service to the consumer.

The fourth principle is the right to private property. Philosophically, the right to private property derives from natural law; every human being is considered the owner of himself. The power of disposition over the fruits of their labour is derived from this too. The protection of private property is also considered an essential pillar of the competitive order: only those who have the power to dispose of their possessions can fully devote themselves to competition and exercise their power of disposition in a responsible manner. This principle requires the participation of all citizens; it must also be flanked by effective competition policy.

The fifth principle is freedom of contract or the voluntary nature of all economic decisions. The competitive order is based on the fact that economic decisions are decentralised and made voluntarily. Individual companies and individuals conclude agreements on their transactions and negotiate their terms between themselves - nothing and no one may hinder this. Only when the legal power over the respective property underlying the transactions remains undistorted can the competitive process be such that the economic resources actually flow to the location of their best use. What Eucken wrote on freedom of contract is clear: "First, it is indispensable. Second, freedom of contract can not be granted for the purpose of concluding contracts that restrict or eliminate the freedom of contract."7

The sixth constituent principle is liability. This principle is particularly important, as has become very clear in the wake of the international financial and economic crisis. Its neglect in many areas of the economy and wider society has given us the 2008 crisis in the first place. The principle of liability is so crucial because only those who are fully liable for the consequences of their actions are really responsible. Only liability ensures that all those involved in the competition are behaving in a fair rather than exploitative manner, even though they may not have a corresponding moral compass. Liability thus ensures that competition is not unfair, but can be relied on to ensure performance and delivery.

The last of the seven constituent principles, Consistency of Economic Policy, concerns politics. One should not be tempted to dismiss this as a less important for this reason. On the contrary, on the question of the role of the state an essential minimum condition is defined. The principle of the consistency of economic policy demands that the political conditions will be reliably predictable, so that the economic actors can plan. This requirement applies in general.

28.3.2. Regulatory principles of the market order

The first principle is competition policy. Concentrations of power must be prevented. Competition policy in the narrower sense deals with targeted interventions against cartels and monopolies. The justification for such interventions is that the competitive order must prevent the formation of permanent positions of power, no matter what their nature. However, since Eucken, economic theory has undergone a development that verifiably shows that it does not depend so much on the actual absence of monopolies. It is more important that the markets are open to competitors and not isolated. A monopolist who can not rest will ultimately provide just as good a supply to consumers as a competitive company.For foreign markets, this idea is even contained in the third constituent principle of Walter Eucken, the principle of open markets;

but the application to domestic markets can also be subsumed under it. This competition policy must also act with general principles and avoid discrimination.

The second principle is income redistribution. A certain degree of redistribution as a correction of the distribution of income as it arises spontaneously in the market is quite reasonable and permissible in the rules of competition. Eucken makes it his second regulating principle. The second principle, however, is subordinate to the first principle; it does not stand for itself. It therefore does not include any - possibly socio-politically motivated - correction of the market result, which makes it imperative to redeem the social market economy's promise of social compensation. On the contrary, the income redistribution always has the task of strengthening the price mechanism and thus the efficiency of the market. That makes them part of the regulatory policy.By contrast, an independently operated, self-reliant distribution policy would reach deep into the realms of process policy. Distributional corrections run the risk of weakening them. They are always a balancing act.

As the third regulating principle Walter Eucken called the improvement of the "economic calculation". With this he expanded the constituent principle of liability. By "economic accounting" is meant entrepreneurs completed, not necessarily includes all macroeconomic costs. Today, economists speak of "external effects" in this context. External effects are costs or benefits of those who are not involved, which are caused as side effects in connection with a specific microeconomic decision, but are not taken into account. Such side effects are common. Eucken remarked: "The system works very well, but it does not take into account the repercussions which the individual economic plans and their execution exert on the macroeconomic data - if these repercussions are not felt in the own planning area of the individual management. [...] Consider the destruction of forests in America. "

Walter Eucken's fourth regulating principle addresses a topic that affects the labour market. This topic has recently rekindled public discussion in Germany in connection with the introduction of minimum wages. It's about the possibility that falling wages do not lead to a fall off of work but on the contrary to an increase. However, Walter Eucken considers this effect to be a rare special case. Because he assumed that a worker in a flexible competitive order with falling wages simply moves to another job or another region: "In free movement and professional mobility of work, the solution is facilitated through other occupations." If this possibility, however is de facto limited, for example, by local demand monopolies for workers, then, according to Eucken, the state with minimum wages to counteract to thwart exploitation - but only then.

So that was the system of "constituent and regulating principles" that the Freiburg economist Walter Eucken designed more than sixty years ago. These principles of competitive order are timeless. Even today, they can be used as an objective yardstick for politics, even and especially in times of crisis. The "Freiburg Imperative" contains the core of the neoliberal project. It is the basis of the social market economy.

28.4. Externalities

28.5. Preclusive action

28.6. Conclusions

Climate change was anticipated early in the 20th century but was not widely recognised as a priority in the first half of the century. In 1896, a Swedish chemist, Svante Arrhenius noticed that the gas carbon dioxide (CO_2) was especially good at trapping heat radiation. He realised that this meant that the massive increases in atmospheric CO_2, caused by coal burning during the industrial revolution, would fuel Fourier's greenhouse effect and lead to global warming of the planet.

To anticipate one conclusion of the following, it does not appear to be the case that an objective reading of the mathematical theory of neoclassical economics provides an argument for laissez-faire economics over a mixed economy or for limiting regulation of financial markets. The case for free markets is more subtle and depends on valuing individual freedom and showing that free markets facilitate this freedom.

This is generally the case for scientific theories.

Neglecting to take account of undesirable effects of economic activity, such as global warming, can be brought into the model and through the price mechanism leading to a number of proposals for carbon pricing .

„Die Wettbewerbsordnung ist imstande, die Investitionen auf die Dauer richtig aufeinander abzustimmen. Denn sie besitzt mit ihrer Preismechanik das Instrument, um Disproportionalitäten festzustellen und um sie schließlich zu korrigieren. Darin ist sie allen anderen Ordnungen überlegen ... Ohne eine Wettbewerbsordnung kann kein aktionsfähiger Staat entstehen und ohne einen aktionsfähigen Staat keine Wettbewerbsordnung."

„ ... muss für die Wirtschaftspolitik der Grundsatz gelten, die Öffnung der Märkte durchzuführen, weil bei ihrer Schließung die akute Gefahr der Behinderung der vollständigen Konkurrenz gegeben ist."

„Erstens: Sie ist unentbehrlich. Zweitens: Vertragsfreiheit darf nicht zu dem Zwecke gewährt werden, um Verträge zu schließen, welche die Vertragsfreiheit beschränken oder beseitigen."

29. Hartmann and Hayek on layered reality

Both Nicolai Hartmann and Friedrich von Hayek defend a **stratified, anti-reductionist picture of reality**, but they do so from very different traditions. Hartmann works within realist phenomenology and critical ontology; Hayek comes from Austrian economics, theoretical psychology, and liberal social theory. Conceptually, however, they converge on three core ideas: **levels**, **emergence**, and **irreducibility**.

29.1. Ontological levels

- **Hartmann:** Reality is articulated into ontological layers (inorganic, organic, psychic, spiritual). Each level has its own categories and laws, and higher levels depend on but are not reducible to lower ones. The doctrine of *Schichtenbau der Welt* is explicitly ontological and systematic.

- **Hayek:** Reality—especially social and mental reality—is structured in levels of complexity. Markets, legal orders, and mental phenomena are higher-level "orders" generated by interactions at a lower level (individual actions, neural events), but they cannot be exhaustively described in the language of those lower levels.

In both, **levels are not mere heuristic abstractions**; they mark genuine ontological differences.

29.2. Emergence and the "novum"

- **Hartmann:** Higher levels exhibit a *categorial novum*: new determinations and laws that do not appear at lower levels. Emergence is ontological, not just epistemic.

- **Hayek:** Complex phenomena (mind, market, law) display patterns that cannot be predicted or derived from micro-data alone. His notion of "spontaneous order" and his theory of complex phenomena imply that higher-level structures have properties not reducible to the properties of their elements.

Hartmann's *novum* and Hayek's "spontaneous order" are structurally similar: both mark **genuinely new properties** at higher levels.

29.3. Dependence and constraint

- **Hartmann:** Higher levels are **founded on** lower ones (Abhängigkeitsverhältnis): they require the lower level as a basis, yet possess their own autonomous laws. Dependence is asymmetric: the higher cannot exist without the lower, but the lower is not determined by the higher.

- **Hayek:** Higher-level orders **emerge from** lower-level interactions, but once in place they **constrain** and **channel** those interactions (e.g. rule-governed behaviour, price signals). The dependence is again asymmetric: no market without individuals, but individual actions are shaped by the market order.

Both thus combine **ontological dependence** with **relative autonomy** of higher levels.

29.4. Anti-reductionism

- **Hartmann:** Explicitly rejects both upward and downward reduction. Physicalism fails because it cannot capture the categorial novelties of life, consciousness, and spirit.

- **Hayek:** Rejects reduction of mental and social phenomena to physical or individual-psychological descriptions. His work in *The Sensory Order* and later social theory argues that different levels require **different conceptual frameworks**.

For both, anti-reductionism is not just methodological; it is grounded in a **metaphysics of levels**.

29.5. Rules, categories, and order

- **Hartmann:** Builds a systematic theory of categories: each level has its own categorial structure, and there are also "modal" and "formal" categories that cut across levels.

- **Hayek:** Develops a theory of **rules** (of just conduct, of social interaction) and **orders** (cosmos vs. taxis). Rules are not reducible to individual intentions; they form part of the structure of a higher-level order.

A fruitful analogy: Hartmann's **categories** and Hayek's **rules** both function as **structuring principles** of levels/orders.

29.6. Realism about higher-level entities

- **Hartmann:** Robust ontological realism: higher-level entities (organisms, persons, cultural formations) are real, not mere constructs.

- **Hayek:** Realist about social and mental orders: markets, legal systems, and traditions are not fictions; they are objective orders with their own regularities, even though they are unintended.

Both oppose any view that treats higher-level entities as merely "illusory" or "nothing but" lower-level facts.

29.7. Key differences

- **Systematicity:** Hartmann offers a fully systematic ontology; Hayek's ontology is fragmentary and embedded in psychology and social theory.
- **Normativity:** Hartmann's ontology is largely descriptive; Hayek's social ontology is tightly connected to normative liberalism (rule of law, limits of constructivist rationalism).
- **Method:** Hartmann proceeds via categorial analysis and phenomenological realism; Hayek via theory of complex phenomena, epistemic limits, and evolutionary accounts of institutions.

A comparative project can treat Hartmann as providing a **general metaphysical framework** within which Hayek's emergentist social ontology can be located and clarified.

30. Economic Entities

An economic ontology typically identifies a handful of core entity types that everything else is built from. The usual breakdown:

Agents (who acts) - Households/individuals — consumers, laborers, savers - Firms/enterprises — producers, employers - Government/state — regulator, taxer, spender - Financial institutions — banks, intermediaries - Foreign sector — other economies, trade partners

Resources/factors - Land (natural resources) - Labor - Capital (physical and financial) - Entrepreneurship/organization (sometimes treated as a fourth factor)

Objects of exchange - Goods (durable, non-durable) - Services - Money/currency - Financial instruments (bonds, equities, derivatives) - Property rights/claims

Relations/processes - Markets (the venues/mechanisms where agents interact) - Transactions (exchange events) - Contracts (formalized obligations) - Prices (the valuation relation between agents and goods) - Production (transformation of inputs to outputs) - Distribution (allocation of output/income)

Institutions (the rules layer) - Property rights - Legal/regulatory frameworks - Norms and conventions - Organizations (firms, central banks, regulatory bodies) as durable structures rather than momentary actors

Flows vs. stocks — a cross-cutting distinction: - Stocks: wealth, capital assets, inventory, debt outstanding - Flows: income, expenditure, investment, output over a period

If you're building this for a formal ontology (e.g., OWL/RDF for economic data, or a conceptual model for a course/paper), it helps to separate these into three layers: **actors** (who), **artifacts** (what is exchanged/held), and **events/relations** (transactions, contracts, production) — with institutions sitting alongside as constraints on how actors and artifacts can relate.

Is there a particular use case — academic ontology design, a data model, or general conceptual understanding? That'd help me tailor the depth and structure.

Public domain status depends on original publication date and jurisdiction, but here's how the list breaks down:

Clearly public domain - Marx, *Capital* Vol. I (1867) — original German text is public domain. The classic English translation by Samuel Moore and Edward Aveling (1887) is also public domain, and is the version widely available on Marxists Internet Archive, Project Gutenberg, etc. - **Menger, *Grundsätze der Volkswirtschaftslehre* (1871)** — original German is public domain (author died 1921, well past life+70).

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If you want, I can point you to specific open-access repositories (Marxists.org, Mises.org, archive.org, SEP bibliography links) for the PD/freely-available ones rather than just flagging status.

Part VI.

End Matter

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